The Ternary Operating Point: Threshold Control and Annealing in 1.58-Bit Language Models

Haydar Kilic[a,1,*]

[a]*Department of Artificial Intelligence Engineering, Ostim Technical University, Ankara 06374, Turkey*

Abstract

Ternary quantization-aware training compresses language-model weights to a three-level codebook, but the fraction of weights surviving quantization, the network's operating point, is treated as an implementation detail rather than a design variable, and, to our knowledge, no prior work establishes whether it can be controlled or whether any control persists through training. We prove that under the absmean rule this fraction is fixed at 0.6899 for every zero-mean initializer, independently of initialization scale, and derive the closed-form bijection between the threshold constant and the operating point it induces. Two mechanisms for unlocking it are then compared under a pre-registered protocol: Threshold-Aware Initialization, which reshapes the weight-magnitude distribution, and a mechanism-matched threshold offset, evaluated over 3,016 runs spanning 1.4 M–352 M non-embedding parameters, two architecture families and fifteen seeds at the confirmatory scales. Introducing a retention coefficient, the slope of final sparsity on imposed initial sparsity, separates interventions that determine outcomes from those that merely order them. The two diverge: a fifty-point manipulation of initial sparsity leaves fifty points at the threshold (retention 1.003–1.006) and two to four at the initializer (0.052–0.069), although both yield near-perfect monotone trends. The initializer is outperformed by its own control and carries an almost identical penalty in full precision, indicating a generic initialization cost. Annealing the threshold early improves held-out loss by 0.060–0.201 nats at all eleven scales. All quality claims are token-equal at a fixed 20 M-token budget, at which no run had converged. Control resides in the threshold, not the weight distribution, and durability must be measured rather than inferred.

Keywords: ternary quantization, quantization-aware training, 1.58-bit language models, weight initialization, sparsity control, threshold annealing, pre-registered experimental design

1. Introduction

The deployment cost of large language models is now dominated not by arithmetic but by memory. In the small-batch, latency-sensitive regime typical of interactive serving, autoregressive inference is bandwidth-bound: each generated token requires the model's weights to be streamed from memory to the compute units, so decoding latency tracks the number of bytes moved far more closely than the number of arithmetic operations performed [1, 2]. Reducing the precision of the weights therefore attacks the binding constraint directly, which adding parallel compute cannot do.

Among the available formats, the ternary codebook $\{-1, 0, +1\}$ occupies a distinctive position. Encoding a ternary weight requires at most $\log_2 3 = 1.585$ bits against the 16 of half precision (roughly a tenfold reduction), while retaining what a strictly binary code cannot express: the capacity to represent irrelevance. A binary network must assign every connection a non-zero magnitude, whereas a ternary network can set a connection to zero, so the codebook records which connections are absent as well as the sign of those that remain. Transformers can be trained under such a constraint from scratch rather than compressed after the fact: BitNet [3] demonstrated this for binary weight matrices at transformer scale, and BitNet b1.58 [4] established the ternary variant. Quantization-aware ternary pretraining has since become an active line of work, on the practical promise that a model may be trained once and deployed in a fraction of the memory a half-precision model of the same size would require.

Training under a discrete constraint, however, introduces a design variable that full-precision training does not possess. At every optimizer step, each weight must be judged large enough to survive quantization or small enough to be set to zero, and that judgement requires a threshold. The fraction of weights that survive it, which we term the operating point of the network, determines how much of the model is active at any moment, and therefore governs both the density of the gradient signal reaching the weights and the sparsity of the deployed artefact. On its face it is among the most consequential quantities in the training procedure.

Despite its centrality, the operating point is not treated as a design variable in any of the three literatures that bear on it. The omission takes a different form in each.

Quantization research fixes the threshold without stating what it implies. The dominant conventions set the threshold as a constant multiple of the mean absolute weight: $0.7\,\mathbb{E}|w|$ in the ternary weight networks of Li et al. [5], and $1.0\,\mathbb{E}|w|$ (the

*Corresponding author.
Email address: haydar.kilic@ostimteknik.edu.tr (Haydar Kilic)
[1]ORCID: 0000-0002-2551-3772

absmean rule) in BitNet b1.58 [4] and the work that follows it. These constants are reported as implementation details. They are in fact statements about the operating point: for a zero-mean Gaussian tensor, a threshold constant and a surviving fraction determine one another exactly. The choice between the two conventions is thus a choice between two substantially different initial sparsities, made in the literature without either operating point being named. A reader is consequently unable to tell whether reported differences between ternary recipes reflect the recipes or merely the operating points they silently select. Where the quantizer's free parameters are learned rather than fixed, it is the non-zero levels that are optimized [6], or, in the pruning literature, the threshold itself [7]; neither line characterizes the operating point it induces, so the same question remains unasked.

Initialization theory is provably silent in this regime. The canonical prescriptions (Xavier [8], Kaiming [9], orthogonal [10] and LSUV [11]) form a theory of second moments, and differ from one another chiefly by a choice of scale, orthogonal initialization adding a correlation structure that leaves the marginal magnitude distribution effectively unchanged after normalization. The absmean rule normalizes each weight by a statistic of its own tensor, so the resulting ternary configuration is invariant to precisely such rescalings. It follows, though we have not seen it stated, that every standard initializer places a ternary network at the same operating point, and that a practitioner attempting to trade quality against density by adjusting the initialization scale cannot move the density at all.

Sparsity research measures ordering, not durability. The pruning literature has repeatedly found that interventions applied at initialization are less influential than they appear [12, 13], but its instruments were developed for thresholds supplied externally by a schedule. The absmean threshold is structurally different: it is a functional of the weight tensor itself, and therefore moves as the weights move. Whether an operating point imposed at initialization survives such a self-referential threshold is a question the existing durability results do not answer, and the customary evidence, a monotone relationship between an imposed value and an observed outcome, cannot distinguish an intervention that orders final states from one that determines them.

The gap that these three omissions jointly leave is specific and testable: the ternary operating point is never established as a controllable variable, and the persistence of any control over it is never measured.

We close this gap in three steps.

First, we establish the invariance analytically. We prove that for any zero-mean initializer the initial active fraction under an absmean threshold is $\rho_0 = 0.6899$ independently of the initialization scale, and we derive the closed-form correspondence between the threshold constant and the operating point together with its inverse. This converts an implementation detail into a design variable with a stated meaning, and the prediction is verified empirically across two orders of magnitude in initialization scale, over which the active fraction varies by 6×10^{-4}, and across four initializer families, each within 10^{-3} of the prediction.

Second, we construct two mechanisms that move the operating point, and set them against each other. Threshold-Aware Initialization (TAI) shapes the weight-magnitude distribution so that a chosen fraction of weights lies above the threshold that the distribution itself induces, leaving the absmean rule untouched. The threshold-offset control does the converse: it retains the standard initializer and scales the threshold by the analytically derived constant that realizes the same operating point. Because both arms reach an identical target by different routes, the comparison isolates whether the operating point or the weight distribution carries the effect, a question that a design containing only the proposed method could not have asked. We additionally introduce a γ-warmup schedule that anneals the threshold during early training rather than displacing the initial distribution.

Third, and this is the methodological core of the study, we measure persistence rather than inferring it. Alongside the conventional rank statistic, we report a retention coefficient: the slope of final sparsity on imposed initial sparsity, which separates an intervention that orders outcomes from one that determines them. The analysis plan, the primary baseline and the equivalence margin were frozen before execution, and the design includes a placebo arm in which the operating-point label varies while the realized intervention is held constant, so that an artefact of the analysis would be visible as one.

We state the outcome plainly here, because the paper's value depends on the reader trusting its reporting, and because one of its central results is a refutation of the hypothesis it was designed to confirm.

The retention measurement separates the two mechanisms decisively and in a direction we did not anticipate. A fifty-percentage-point manipulation of initial sparsity leaves fifty points of final sparsity when applied at the threshold (retention 1.003–1.006, $R^2 = 1.000$), and between two and four points when applied at the initializer (retention 0.052–0.069). Training returns the network to a final sparsity near 0.30 from wherever the initializer places it, so the sparsity that our own proposition fixes at initialization proves to be approximately where the optimizer goes unaided. The rank association between imposed and final operating point is meanwhile near-perfect, which is to say that the customary evidence would have supported a strong causal claim that the retention coefficient refutes.

Consistently with this, the proposed initializer is outperformed by its own mechanism-matched control at matched operating points, and a 2×2 ablation shows it to carry an almost identical quality penalty in full precision and under quantization, the signature of a generically weaker initializer rather than of a failure of operating-point control. Both of these findings come from arms constructed so that the proposed method could lose.

What survives, and what we regard as the paper's positive contribution, is the schedule. Annealing the threshold briefly at the start of training lowers held-out loss and reduces steps-to-target at **all eleven scales** tested, spanning 1.4 M to 352 M non-embedding parameters, the loss improvement ranging from 0.060 to 0.201 nats. The same advantage holds at all thirteen architecture $\times$ scale cells against the pre-registered baseline,

at twelve of thirteen against the data-selected minimum, on a depth-controlled ladder, and at the upper four of five learning rates. The locus of the effect is thus the timing of the threshold, not the initial weight distribution.

The main contributions of this paper are as follows:

An analytical characterization of the ternary operating point: We prove that the initial active fraction under an absmean threshold is independent of the initialization scale, fixing it at $\rho_0 = 0.6899$ for every zero-mean initializer, and we derive the closed-form bijection between the threshold constant and the operating point together with its inverse. This gives the quantization literature a stated meaning for a constant it has treated as an implementation detail, and it reveals that the rule-of-thumb threshold of Li et al. [5] and the absmean convention of Ma et al. [4] correspond to operating points of 0.780 and 0.690 respectively. To our knowledge this correspondence has not been reported.

Two algorithmically distinct mechanisms for controlling the operating point, with a separation guarantee: We introduce Threshold-Aware Initialization, a closed-form two-mode magnitude construction whose realized operating point is provably exact under a stated separation condition and is verified per invocation against a binomial tolerance, together with a γ-warmup schedule that anneals the threshold during early training. We pair these with a mechanism-matched threshold-offset control derived from the same analysis, so that the operating point and the weight distribution can be varied independently.

Retention as a measured quantity: We introduce the retention coefficient as the statistic that distinguishes an intervention which orders final states from one which determines them, and demonstrate that the two diverge sharply in this setting: a near-perfect rank association coexists with the erasure of 93–95% of the intervention. We further identify the structural cause (a threshold defined as a functional of the weights it judges) and derive from it a falsifiable prediction for quantizer families outside the one studied here.

A pre-registered empirical validation at scale: We report 3,016 persisted runs consuming 50.9 A100-hours across an eleven-point ladder from 1.4 M to 352 M non-embedding parameters, two architecture families, a depth-controlled ladder, and fifteen seeds at the confirmatory scales, with the hypothesis family, primary baseline and equivalence margin frozen before execution and no run excluded for divergence.

A negative result of independent interest, established under falsification controls: We show that shaping the initial weight distribution is not an effective route to operating-point control, and that its apparent cost is a generic initialization penalty rather than a quantization-specific one. This conclusion rests on a mechanism-matched control and a label-decoupled placebo, both of which were pre-registered and both of which performed the function for which they were built.

Methodological instruments transferable beyond this problem: We report variance over the pooled seed $\times$ evaluation-fold matrix and find the evaluation fold to be the larger component, which bears on the near-universal convention of reporting seed variance alone. We derive the equivalence margin from the measured quantization cost rather than fixing it by fiat. And we exploit the presence of two algebraically equivalent implementations in the design to obtain an empirical bound on the harness's own reproducibility floor, against which our smallest effects are explicitly assessed.

A complete, re-analysable research artefact: The run store, provenance manifest, analysis pipeline and figure-generation code are released in full, so that every table and figure in this paper can be regenerated from the raw records without a GPU.

The remainder of this paper is organized as follows. Section 2 reviews the three literatures on which this work draws (low-bit quantization, weight initialization, and sparsity scheduling), together with the methodological literature on evaluating training interventions, and states the specific gap each leaves. Section 3 presents the formalism and the experimental protocol: the quantization model and the proof of scale invariance, the TAI construction and its separation guarantee, the γ-warmup schedule, the mechanism-matched and placebo controls, the model family and scale ladder, the leakage-free corpus partition, the optimization and calibration procedure, the convergence and divergence diagnostics, the arms under comparison, the reported metrics, and the frozen statistical analysis plan. It closes with the ablations and confound controls, the reproducibility protocol, the computational cost of each intervention, and the deviations from the pre-registered plan. Section 4 reports the empirical outcome in the order in which the hypotheses were registered, beginning with the validity of the executed grid, the empirical verification of the locked operating point, and the convergence status of the token budget; that last determination governs the wording of every subsequent claim. Section 5 supplies the mechanisms behind those findings, relates them to prior work, quantifies the principal threats to validity, and states the practical recommendations and the structural limitations of the design. Section 6 concludes and identifies the experiments that would most directly strengthen or overturn the results reported here.

2. Related Work

This work sits at the intersection of three literatures that have developed largely independently: low-bit weight quantization, weight initialization theory, and the study of sparsity schedules and their durability. Each has converged on conventions that the others do not examine, and the object of this paper (the

ternary operating point, i.e. the fraction of weights that survive quantization) is a quantity that all three touch and none isolates. We review them in turn, closing each review with a statement of what remains unaddressed and how the present study addresses it. A fourth subsection situates our experimental protocol within the methodological literature on evaluating training interventions, since several of this paper's conclusions are negative and their credibility rests on that protocol. A fifth draws the three literatures together and states how this paper addresses them jointly.

2.1. Low-bit weight quantization and the status of the threshold

The modern line begins with the observation that weights can be constrained to a small discrete alphabet during training provided gradients are propagated through a surrogate. BinaryConnect [14] constrained weights to $\{-1, +1\}$ while retaining full-precision accumulators; binarized neural networks [15] extended the treatment to activations, binarizing both with the sign function alone; and XNOR-Net [16] added per-channel scaling factors to recover representational range. The common enabling device across this family is the straight-through estimator [17], whose theoretical status remains only partially understood but whose empirical adequacy for coarse-grained quantization is by now well established.

Ternary codebooks $\{-1, 0, +1\}$ emerged as the natural refinement, on the argument that a zero level costs little in storage ($\log_2 3 = 1.585$ bits against 1) while restoring the capacity to express irrelevance, which a strictly binary code cannot. Ternary weight networks [5] formalized this as an approximation problem, seeking the ternary tensor minimizing ℓ_2 distance to the full-precision weights, and derived the widely reused symmetric threshold $\Delta = 0.7\,\mathbb{E}|w|$ as a tractable surrogate for the exact solution. Trained ternary quantization [6] relaxed the symmetry assumption by learning the two non-zero levels, and dynamic network surgery [18] allowed pruned connections to be reinstated. Each relaxes one part of a hand-set quantization rule, but neither revisits the threshold itself. Loss-aware and non-uniform schemes [19, 20] pursued the complementary strategy of optimizing the codebook against the task objective rather than against reconstruction error.

Two branches diverge in the large-model era. Post-training quantization methods compress an already-trained model: GPTQ [21] and AWQ [22] reconstruct a fixed weight matrix under a bit budget, while SmoothQuant [23] and outlier-aware integer decomposition [24] target instead the activation outliers that obstruct low-bit inference. Because training dynamics play no role in either case, the questions this paper asks do not arise: there is no trajectory along which an imposed operating point could be retained or erased. The second branch trains under the constraint from the outset. BitNet [3] demonstrated that binary weight matrices can be trained from scratch at transformer scale, and BitNet b1.58 [4] established the ternary variant, adopting the absmean rule in which the quantization threshold is set from the mean absolute weight of each tensor. The absmean convention, together with the accompanying claim that 1.58-bit models can approach full-precision quality at sufficient scale, has since been widely inherited.

Across this literature the threshold is treated in one of two ways, and neither exposes what we take to be the operative quantity. It is either fixed by a heuristic constant ($0.7\,\mathbb{E}|w|$ in Li et al. [5], $1.0\,\mathbb{E}|w|$ under the absmean convention of Ma et al. [4]) or it is allowed to move with the weights during training [6, 18], in which case it is never characterized in isolation. In both cases the constant is reported as an implementation detail rather than as a design variable with an interpretable consequence. What is missing is the observation that a threshold constant is a statement about the fraction of weights that will survive: for a zero-mean Gaussian tensor the two are related in closed form, so the choice between 0.7 and 1.0 is a choice between roughly 22% and 31% initial sparsity made without either paper saying so. To our knowledge no prior study states this correspondence, and consequently none compares the two conventions on the axis that actually separates them.

A second and more consequential omission concerns persistence. The literature reports the sparsity a ternary network exhibits at convergence, and it reports the threshold rule that was applied; it does not report whether the threshold rule is what determined the observed sparsity, or whether training would have arrived at a similar density from a different starting configuration. The distinction is invisible when only one threshold convention is run, which is the norm.

We supply the missing correspondence analytically and exploit it experimentally. Proposition 1 and Eq. (9) give the exact map between the threshold constant c and the initial active fraction ρ_0 under a zero-mean initializer, and Eq. (19) inverts it, so that any desired operating point can be realized by a closed-form constant. Evaluating the map at the classical value of Li et al. [5] places that heuristic at $\rho_0 = 0.780$, and the absmean convention of Ma et al. [4] at $\rho_0 = 0.690$; our two measured optima bracket the classical value, which we read as convergent evidence rather than as a claim of priority. More importantly, we make the operating point itself a swept variable, over a range that brackets both conventions, which permits the persistence question to be asked directly rather than inferred.

2.2. Weight initialization: a variance theory in a scale-invariant regime

Initialization theory for deep networks is, with few exceptions, a theory of second moments. Glorot and Bengio [8] derived a scale that preserves activation and gradient variance across layers under linear assumptions, and He et al. [9] corrected it for rectified nonlinearities. Orthogonal initialization [10] strengthened the guarantee by preserving norms exactly in the linear regime, and layer-sequential unit-variance initialization [11] replaced analytical variance propagation with a data-driven calibration pass, rescaling each layer in sequence until its measured output variance reaches unity. Later work extended the programme to architectures where normalization layers are absent or undesirable [25, 26] and to identity-preserving schemes [27].

A parallel development partially displaced the question. Batch [28] and layer [29] normalization, and their computationally lighter successor RMSNorm [30], render networks substantially insensitive to initialization scale by renormalizing activa-

tions at every layer. Xiong et al. [31] showed that placing the normalization inside the residual block leaves gradients well behaved at initialization, and the pre-normalized arrangement they analysed is now the default in language modelling. The prevailing view is therefore that initialization matters chiefly in the absence of normalization.

Comparatively little work addresses initialization when the forward pass is quantized. Where the topic is raised, it is generally in the service of training stability, through devices such as bounding the range of the latent full-precision weights [15, 32], rather than as a control variable in its own right. The implicit assumption, inherited from the full-precision setting, is that a well-conditioned initializer in the variance sense remains well-conditioned after quantization.

The assumption fails in a specific way. The absmean rule normalizes each weight by a statistic of the tensor to which it belongs, so the resulting ternary code is invariant to any positive rescaling of the tensor (our Lemma 1). The prescriptions in the variance-propagation programme differ from one another chiefly by such a rescaling. It follows that Xavier, Kaiming, orthogonal and LSUV initialization, schemes that differ materially in the full-precision setting, place a ternary network at the same operating point, and that the entire theoretical apparatus is silent on the quantity that determines how a ternary network begins. The consequence for practice is that a researcher attempting to trade quality against density by varying the initialization scale cannot move the density at all, and no prior work states this.

Our own attempt to exploit this observation failed, and the initialization literature is partly vindicated by that failure. Constructing a bimodal magnitude distribution to place the network at a chosen operating point does move ρ_0, but the resulting initializer is measurably worse in ordinary full-precision terms as well, by an amount indistinguishable from its cost under quantization (Section 4.10). The variance-based programme is silent about the operating point, but it is not silent about what makes a good initializer, and our results are consistent with its verdict.

We formalize the silence as a proposition rather than an observation: Proposition 1 establishes that the initial active fraction of an absmean ternary layer is independent of the initialization scale σ, and we verify it empirically across two orders of magnitude in σ and across four standard initializer families. We further show that initialization nonetheless retains influence in this regime through channels the operating point does not capture: a properly completed LSUV calibration, which reaches the same operating point as a standard Gaussian to within 10^{-3}, nevertheless attains a lower held-out loss at nine of the eleven scales of our ladder. The useful statement is therefore not that initialization ceases to matter under ternary quantization, but that it ceases to matter through the operating point, and that the operating point must consequently be controlled elsewhere.

2.3. Sparsity schedules, learned thresholds, and the durability of early interventions

The pruning literature reached early consensus that sparsity is best imposed gradually. Gradual magnitude pruning [33] established the now-standard practice of raising a sparsity target along a schedule while training continues, and comparative studies [34] confirmed its robustness against more elaborate alternatives. Dynamic sparse training methods [35, 36] maintain a fixed sparsity budget while permitting the support to change, and movement pruning [37] selects weights by the direction of their motion rather than their magnitude, explicitly acknowledging that a weight's importance is a property of the trajectory and not of any single point on it.

A distinct strand makes the threshold itself a learned quantity. Soft threshold reparameterization [7] replaces a hard, hand-set cutoff with a learned per-layer one and reports that the resulting sparsity budget is markedly non-uniform across layers; continuous sparsification [38] reaches a sparse network instead through a continuous relaxation of an ℓ_0 penalty, so that the retained support emerges from a differentiable objective rather than from a cutoff set by hand. These works are the closest antecedents of our threshold-offset control, and they establish that thresholds are consequential; they do not, however, characterize the threshold in terms of the operating point it induces, nor do they compare a threshold intervention against a distributional one.

The question of whether an intervention applied at initialization survives training has been asked most sharply in the lottery-ticket literature. The original hypothesis [39] held that a sparse subnetwork identified after training could be retrained from its original initialization to comparable quality, which was widely read as evidence that the surviving support is determined early. Subsequent work substantially complicated the picture: weight rewinding to an early but non-zero iterate proved necessary at scale [13], and a systematic evaluation of pruning-at-initialization criteria [12] found that several published methods are insensitive to the very information they purport to use: randomly shuffling which weights a layer prunes left accuracy unchanged or improved it, so that only the per-layer pruning fraction carries information. The broader methodological survey of Blalock et al. [40] reached a compatible conclusion from the other direction, reporting that the absence of standardized benchmarks and metrics leaves most published pruning results difficult to compare at all.

Two things are notable about this literature from the vantage of the present study. First, its instrument for testing durability is essentially always correlational: an intervention is applied, an outcome is measured, and a monotone relationship between the two is taken as evidence of control. The falsification devices introduced by Frankle et al. [12], shuffling the pruning mask within a layer while holding its sparsity fixed, are the exception, and their value is precisely that they broke claims which favourable accuracy comparisons had appeared to support. Second, the thresholds this literature studies are external to the weights they judge: a pruning criterion compares weights against a cutoff supplied by a schedule or, where the cutoff is learned, against a separate parameter optimized alongside them. Absmean ternary quantization is structurally different, because the threshold is a functional of the weight tensor itself and therefore moves as the weights move. Whether an initialization-time operating point survives such a self-referential threshold is not a question the pruning literature has had occasion to ask, and its

answers do not transfer by analogy.

We make durability a measured quantity rather than an inference. Alongside the conventional rank statistic, we report a retention coefficient (Eq. (32)), the slope of final sparsity on imposed initial sparsity, which distinguishes an intervention that orders outcomes from one that determines them. The distinction is not academic in our data: the rank association between the imposed operating point and the final one is nearly perfect, yet the retention coefficient of the initialization mechanism is approximately 0.05–0.07, so the large majority of a 50-percentage-point intervention is erased during training, while the mechanism-matched threshold control retains it essentially in full (Section 4.8). Had we followed the customary practice of reporting the monotone trend alone, we would have reported a strongly confirmed causal claim. We additionally adopt the falsification strategy of Frankle et al. [12] in a form suited to this setting, an arm in which the operating-point label varies while the realized partition is held fixed, so that an artefact of the analysis would be visible as such. Finally, Section 5.2 offers a mechanism for the asymmetry that is specific to self-referential thresholds and yields a testable prediction for quantizers outside the family we study.

2.4. Evaluating training interventions: variance, pre-registration and equivalence

Because several of this paper's principal findings are negative, and because one of them contradicts the hypothesis the study was designed to confirm, the protocol carries an unusual share of the evidential weight. We therefore situate it explicitly.

A now-substantial body of work documents that small reported improvements in machine learning frequently fail to survive controlled re-evaluation. Melis et al. [41] showed that architectural gains in language modelling largely dissolve under equalized hyperparameter search; Lucic et al. [42] reached the same conclusion for generative adversarial networks; and Blalock et al. [40] report the related finding for pruning, that the absence of standardized benchmarks and metrics leaves the field unable to judge how much progress it has made. Dodge et al. [43] traced much of the problem to under-reported search budgets, and Bouthillier et al. [44] to the practice of estimating variance from a single source, typically the seed, when several sources contribute comparably. On the inferential side, Lakens [45] argues that a claim of non-inferiority requires an equivalence test rather than a failure to reject a null, and Holm [46] supplies the standard step-down correction for a pre-specified family of tests.

Quantization and sparsity research has adopted these practices unevenly. Seed counts are often small or unreported, the baseline against which a method is compared is frequently selected after the results are known, and a claim to match full precision is typically supported by the absence of a significant difference rather than by an equivalence procedure with a stated margin. Falsification controls of the kind introduced by Frankle et al. [12] remain rare.

Our design imports these practices in full and adds two elements that we have not seen combined in this literature. The hypothesis family, the primary baseline, the equivalence margin and the analysis plan were frozen before the confirmatory grid was executed, and the margin is derived from the data as a fraction of the measured quantization cost rather than fixed by fiat. Variance is estimated over the pooled seed $\times$ evaluation-fold matrix, which in our measurements reveals the evaluation fold to be the larger of the two components, a result with implications beyond this paper, since seed-only variance is the near-universal convention. Beyond this, the design includes two arms constructed so that the proposed method could lose: a mechanism-matched control that reaches the same operating point by a different route, and a placebo whose label varies while its realized intervention does not. Both performed their function; the first defeated the proposed method and the second confirmed that the sparsity effect is not an analysis artefact. We report these outcomes as the principal evidence that the study's positive findings are not selection effects.

2.5. Positioning

In summary, the quantization literature fixes a threshold constant without stating the operating point it implies and without testing whether that operating point persists; the initialization literature supplies a theory of second moments that is provably silent once the quantizer is scale-invariant; and the sparsity literature establishes that early interventions are often less durable than they appear, but its instruments were developed for external thresholds and do not transfer to a threshold defined by the weights themselves. This paper addresses the three jointly: it gives the closed-form correspondence between threshold constant and operating point, shows that the operating point is consequently locked under every standard initializer, and then measures, rather than assumes, which of two mechanisms for unlocking it survives training. The methodology of Section 3 is what allows the resulting answer, which is not the one we expected, to be reported with confidence.

3. Methodology and Experimental Setup

This section specifies the quantization formalism under study (Section 3.1), the analytical result that motivates the proposed method (Section 3.2), the Threshold-Aware Initialization (TAI) algorithm and its schedule variant (Sections 3.3–3.4), the mechanism-matched and placebo controls that render the central hypothesis falsifiable (Sections 3.5–3.6), the model family and scale ladder (Section 3.7), the corpus and leakage-free partition (Section 3.8), the optimization protocol and per-arm learning-rate calibration (Section 3.9), the convergence and divergence diagnostics (Section 3.10), the arms under comparison (Section 3.11), the evaluation metrics (Section 3.12), the pre-registered hypotheses and the statistical analysis plan (Section 3.13), the ablation and control suite (Section 3.14), the reproducibility protocol (Section 3.15), and the computational cost of the interventions (Section 3.16). Every quantity reported in Section 4 is produced by the procedure described here. The design was not revised after the confirmatory grid was launched; where measurement forced a departure from the registered protocol, it is recorded in Section 3.17.

3.1. Problem formulation

3.1.1. Ternary linear layers

We consider decoder-only transformer language models in which every linear projection inside the transformer blocks is replaced by a quantization-aware BitLinear layer [3]; the tied input embedding and output head are retained in half precision and are not quantized. Let $\mathbf{W} \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ denote the weight matrix of such a layer, with fan-in $f = d_{\text{in}}$, and let $\mathbf{x} \in \mathbb{R}^{d_{\text{in}}}$ denote its input. The layer computes

$$\mathbf{y} \;=\; \widehat{\mathbf{W}}\,\hat{\mathbf{x}}, \qquad \hat{\mathbf{x}} \;=\; \mathrm{STE}(\tilde{\mathbf{x}},\, Q_a), \qquad \widehat{\mathbf{W}} \;=\; \mathrm{STE}(\mathbf{W},\, Q_w), \tag{1}$$

where $\tilde{\mathbf{x}} = \mathrm{RMSNorm}(\mathbf{x})$ is the root-mean-square normalized input [30],

$$\mathrm{RMSNorm}(\mathbf{x}) \;=\; \frac{\mathbf{x}}{\sqrt{\frac{1}{d_{\text{in}}}\sum_j x_j^2 + \varepsilon}} \odot \mathbf{g}, \qquad \varepsilon = 10^{-5}, \tag{2}$$

with a learned per-channel gain $\mathbf{g}$, and STE denotes the straight-through estimator [17]

$$\mathrm{STE}(u, Q) \;=\; u + \mathrm{sg}[\, Q(u) - u \,], \tag{3}$$

in which $\mathrm{sg}[\cdot]$ is the stop-gradient operator. Consequently the forward pass is quantized while the backward pass propagates $\partial\mathcal{L}/\partial u$ unmodified, with the optimizer updating the latent `bfloat16` parameters directly.

Activations are quantized to signed 8-bit integers with a per-token dynamic scale, following the symmetric per-token convention of [3, 4],

$$\begin{aligned} Q_a(\mathbf{x}) \;&=\; \frac{1}{s}\,\mathrm{clip}(\,\mathrm{round}(s\,\mathbf{x}),\, -128,\, 127), \\ s \;&=\; \frac{127}{\max\,(\|\mathbf{x}\|_\infty,\, 10^{-5})}, \end{aligned} \tag{4}$$

where $\|\cdot\|_\infty$ is taken over the last (feature) axis. The scale places $s\mathbf{x}$ in $[-127, 127]$, so the lower clip bound is never attained; it is retained only to keep the output within the signed 8-bit range under floating-point rounding.

3.1.2. Absmean ternary weight quantization

Weights are ternarized with the absmean rule. Given a threshold scale $c > 0$, define

$$\gamma(\mathbf{W}; c) \;=\; \max\!\left(c \cdot \frac{1}{n_{\mathbf{W}}}\sum_{i,j}|w_{ij}|,\; 10^{-5}\right) \;=\; \max\,(c\,\mathbb{E}|w|,\; 10^{-5}), \tag{5}$$

where $n_{\mathbf{W}} = d_{\text{out}} d_{\text{in}}$ is the number of entries in $\mathbf{W}$, and

$$\begin{aligned} Q_w(\mathbf{W}; c) \;&=\; \gamma(\mathbf{W}; c) \cdot \mathrm{clip}\!\left(\mathrm{round}\!\left(\frac{\mathbf{W}}{\gamma(\mathbf{W}; c)}\right),\, -1,\, +1\right) \\ &\in\; \gamma \cdot \{-1, 0, +1\}^{d_{\text{out}} \times d_{\text{in}}}. \end{aligned} \tag{6}$$

The nominal recipe uses $c = 1$, i.e. $\gamma = \mathbb{E}|w|$. Because $\mathrm{round}(\cdot)$ maps to zero on $(-\frac{1}{2}, \frac{1}{2})$,[2] an individual weight survives quantization, i.e. is assigned a non-zero ternary code, if and only if

$$|w_{ij}| \;\geq\; \tfrac{1}{2}\gamma(\mathbf{W}; c). \tag{7}$$

Definition 1 (Ternary operating point)**.** *For a weight matrix* $\mathbf{W}$ *and threshold scale* c*, the active fraction is*

$$\rho(\mathbf{W}; c) \;=\; \frac{1}{n_{\mathbf{W}}}\sum_{i,j}\mathbf{1}\Big[\,|w_{ij}| \geq \tfrac{1}{2}\gamma(\mathbf{W}; c)\,\Big] \;=\; 1 - \mathrm{sp}(\mathbf{W}; c), \tag{8}$$

where sp *denotes the induced ternary sparsity. Evaluated at initialization,* $\rho_0 \triangleq \rho(\mathbf{W}^{(0)}; c)$ *is termed the initial operating point; evaluated at the end of training,* ρ_T *is the final operating point. Model-level quantities are the unweighted mean over all BitLinear layers.*

Throughout, ρ is measured under the same quantizer and the same live threshold scale c that the forward pass employs at the time of measurement, so that the reported sparsity characterizes the network as executed rather than a counterfactual re-quantization of it. For the full-precision control arm, in which no weight is ternarized, sparsity is defined to be exactly zero. A separate accessor evaluates ρ at the canonical threshold $c = 1$ and is used only in the analytical verification of Section 3.2, where σ-independence at the canonical threshold is precisely the quantity of interest.

3.2. The locked operating point of zero-mean initializers

The premise of this work is that the initial operating point of an absmean ternary layer is not a free design variable under any standard initialization scheme. This is a consequence of the scale invariance of (6).

Lemma 1 (Scale invariance of the ternary code)**.** *For every* $\lambda > 0$ *and every* $\mathbf{W}$*, the ternary code pattern induced by* (6) *satisfies*

$$\begin{aligned} &\mathrm{clip}(\,\mathrm{round}(\lambda\mathbf{W}/\gamma(\lambda\mathbf{W}; c)), -1, 1) \\ &\qquad =\; \mathrm{clip}(\,\mathrm{round}(\mathbf{W}/\gamma(\mathbf{W}; c)), -1, 1), \end{aligned}$$

and hence $\rho(\lambda\mathbf{W}; c) = \rho(\mathbf{W}; c)$*.*

Proof. γ is positively homogeneous of degree one in $\mathbf{W}$ by (5) (away from the 10^{-5} floor), so the ratio $\lambda w_{ij}/\gamma(\lambda\mathbf{W}; c) = w_{ij}/\gamma(\mathbf{W}; c)$ is invariant, and therefore so are round, clip and the indicator in (8). □

Proposition 1 (σ-independence of the initial operating point)**.** *Let* $w_{ij} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$ *with* $\sigma > 0$*. Then, in the large-*$n_{\mathbf{W}}$ *limit and at threshold scale* c*,*

$$\rho(c) \;=\; 2\left[1 - \Phi\!\left(\tfrac{c}{2}\sqrt{\tfrac{2}{\pi}}\right)\right], \tag{9}$$

[2] The implementation rounds halves to even, so a weight with $|w_{ij}|$ exactly equal to $\frac{1}{2}\gamma$ is annihilated rather than retained. The boundary carries zero probability under any continuous weight density, and ρ is measured from the quantizer output rather than from the comparison in (7), so neither the analysis nor the measurement is affected.

where Φ is the standard normal cumulative distribution function. In particular ρ does not depend on σ, and at the canonical threshold $c = 1$,

$$\rho_0 \;=\; 2\,[1 - \Phi(0.39894\ldots)] \;=\; 0.6899\ldots \tag{10}$$

Proof. For a half-normal magnitude, $\mathbb{E}|w| = \sigma\sqrt{2/\pi}$, so $\gamma = c\,\sigma\sqrt{2/\pi}$ and the survival condition (7) reads $|w_{ij}| \geq \frac{c}{2}\sigma\sqrt{2/\pi}$. Dividing by σ gives the σ-free condition $|z| \geq \frac{c}{2}\sqrt{2/\pi}$ with $z \sim \mathcal{N}(0,1)$, whence (9). Setting $c = 1$ yields (10). □

Proposition 1 applies directly to Gaussian, Xavier and Kaiming initialization, which differ from one another only in the choice of σ. Orthogonal initialization imposes a correlation structure that the proposition does not assume, but one which leaves the marginal magnitude distribution effectively unchanged after normalization, so it reaches the same locked operating point. All four therefore begin at $\rho_0 \approx 0.69$, i.e. at approximately 31% initial ternary sparsity. The claim is verified empirically in two independent ways: (i) distributionally, by sampling 10^6 Gaussian variates at $\sigma \in \{0.02, 0.1, 0.5, 1.0, 2.0\}$ and measuring ρ directly; and (ii) architecturally, by instantiating the full model under each candidate initializer and averaging ρ over all BitLinear layers, with a pre-registered tolerance of $|\rho - 0.69| < 0.02$ for the locked verdict.

Remark 1 (LSUV in a pre-normalized architecture: measured, not assumed). Layer-Sequential Unit-Variance initialization (LSUV) is included as a baseline and is implemented in full, comprising both the orthogonal pre-initialization and the data-driven layer-sequential variance calibration pass; the calibration batch is drawn from a generator distinct from the training-order generator so that no arm receives a different data sequence as a side effect of being calibrated. Because BitLinear applies RMSNorm to its own input (2), one might expect the calibration to be inert in this architecture: the input to every projection is unit-variance by construction, and an orthogonal weight matrix normalized to $1/\sqrt{f}$ maps unit-variance input to unit-variance output analytically. We do not assume this; we instrument it. Two diagnostics are recorded at every calibration (the number of rescaling operations performed, and the maximum absolute deviation $\max|\mathrm{Var}(\mathbf{y}) - 1|$ encountered at initialization) and both are written to the run manifest and reported in Section 4.2. The measurement decides which of two statements the paper is entitled to make: if the calibration is inert, `standard` and `lsuv` are one locked-ρ_0 family under two labels; if it is not, they share an operating point but are distinct interventions, and the difference between them is informative about what initialization can still do once the operating point is fixed. In either case no best-of-two contrast is taken over them, and the mechanism-matched second family remains the threshold-offset control of Section 3.5, which differs at the operating point rather than in the weight distribution.

3.3. Threshold-Aware Initialization

TAI removes the constraint identified in Proposition 1 by constructing an initial weight distribution whose magnitude law is explicitly bimodal with respect to the absmean threshold, so that ρ_0 becomes a directly specifiable hyperparameter.

3.3.1. Construction

Fix a target $\rho_0 \in (0,1)$, an active magnitude α, a dormant magnitude δ with $0 < \delta < \alpha$, and a jitter half-width $a \in [0,1)$. For each of the $n \triangleq n_{\mathbf{W}}$ entries, draw independently

$$\begin{aligned} u_i, v_i, s_i &\overset{\text{i.i.d.}}{\sim} \mathcal{U}(0,1), \\ J_i = (1-a) + 2a\,v_i &\sim \mathcal{U}(1-a,\,1+a), \\ \varsigma_i &= \begin{cases} -1, & s_i < \tfrac{1}{2}, \\ +1, & \text{otherwise}, \end{cases} \end{aligned} \tag{11}$$

and set the pre-normalized weight

$$\tilde{w}_i \;=\; \varsigma_i\, J_i\,[\alpha\,\mathbf{1}\{u_i < \rho_0\} \;+\; \delta\,\mathbf{1}\{u_i \geq \rho_0\}]\,. \tag{12}$$

Finally, rescale to the fan-in target $\tau = 1/\sqrt{f}$,

$$w_i \;=\; \frac{\tau}{\max\left(\mathrm{std}(\tilde{w}),\; 10^{-9}\right)}\,\tilde{w}_i, \tag{13}$$

which fixes the second moment of the layer to the same value as the standard initializer without altering the ternary code, by Lemma 1.

3.3.2. Closed-form realization of the target

Because $\mathbb{E}[J] = 1$ and the sign is independent of the magnitude, the absmean threshold of the construction (12) is available in closed form,

$$\gamma_{\text{pred}} \;=\; \mathbb{E}|\tilde{w}| \;=\; \rho_0\,\alpha + (1-\rho_0)\,\delta. \tag{14}$$

The construction realizes its target exactly (in the sense that the surviving set is precisely the set $\{i : u_i < \rho_0\}$, whose expected cardinality is $\rho_0 n$) if and only if the two magnitude modes lie on opposite sides of $\frac{1}{2}\gamma_{\text{pred}}$ even in the worst case of the jitter.

Lemma 2 (Separation condition)**.** *Under* (11)–(12)*, and in the large-n limit in which the realized threshold $\gamma(\mathbf{W}^{(0)}; 1)$ converges to γ_{pred}, every active entry survives and every dormant entry is annihilated by* (6) *at $c = 1$ if and only if*

$$\alpha(1-a) \;\geq\; \tfrac{1}{2}\gamma_{\text{pred}} \qquad \textit{and} \qquad \delta(1+a) \;<\; \tfrac{1}{2}\gamma_{\text{pred}}. \tag{15}$$

In that case $\rho(\mathbf{W}^{(0)}; 1) = \frac{1}{n}\big|\{i : u_i < \rho_0\}\big|$, which is $\mathrm{Binomial}(n, \rho_0)/n$ *and hence concentrates at ρ_0.*

Both inequalities in (15) are asserted at every invocation of the initializer rather than assumed, so that a subsequent change to (δ, a) cannot silently invalidate the guarantee. Because (13) removes the absolute scale of (α, δ), only their ratio is operative; we therefore fix $\alpha = 1$ throughout and vary δ. With the operating defaults $\alpha = 1$, $\delta = 0.15$, $a = 0.15$, the conditions hold with margin across the entire ρ_0 grid; the sensitivity analysis of Section 3.14 sweeps (δ, a) and identifies the boundary of the admissible region.

3.3.3. Post-hoc verification

Independently of Lemma 2, the realized operating point is measured after every initializer call by applying (6) to the generated tensor, and is required to satisfy

$$\left|\rho_{\text{real}} - \rho_0\right| \;\le\; \max\!\left(0.02,\; 5\sqrt{\frac{\rho_0(1-\rho_0)}{n}}\right), \qquad (16)$$

i.e. five binomial standard deviations, floored at two percentage points. The run aborts if (16) is violated. This converts the realization of ρ_0 from an asserted property of the method into a per-call measurement, and the realized value is written to the run record.

Remark 2 (Deterministic causal intervention). When a `torch.Generator` is supplied, all three uniform streams in (11) are drawn from it on the host, so that the entire model-level initialization is a deterministic function of the pair (generator state, ρ_0). In the sweep of Section 3.13, the generator is seeded identically across the ρ_0 grid within each seed; consequently, moving ρ_0 changes only the threshold partition of a fixed base noise realization, holding both the underlying random draws and the data order constant. The sweep is therefore a ceteris-paribus intervention rather than an observational comparison, and each seed constitutes an independent replication of that intervention.

Algorithm 1 states the complete procedure.

Remark 3 (On the absence of an iterative search). An earlier formulation of TAI performed a fixed-point search over the active magnitude α. That search was inert for three independent reasons: the candidate tensor was rebuilt from scratch at the start of every iteration, so the updated α was never applied to the returned weights; the realized active set is determined entirely by the mask $\mathbf{1}\{u_i < \rho_0\}$, with the dormant mode below threshold for every iterate in the admissible range; and the terminal normalization (13) renders the absolute scale of (α, δ) irrelevant, only their ratio entering. The first of these is what makes removing the search leave the output bit-identical; the other two make the search pointless even had it been applied. We therefore report the closed-form construction that the search was approximating, together with the explicit verification (16), rather than describing an optimization procedure that does not affect the result.

Algorithm 1: Threshold-Aware Initialization (TAI) for one BitLinear tensor

Input : Weight tensor $\mathbf{W} \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$; target operating point $\rho_0 \in (0, 1)$; active magnitude α; dormant magnitude δ; jitter half-width a; optional generator g.

Output : $\mathbf{W}$ initialized such that $\rho(\mathbf{W}; 1) \approx \rho_0$ and $\text{std}(\mathbf{W}) = 1/\sqrt{d_{\text{in}}}$.

Require: $0 < \delta < \alpha$, $0 \le a < 1$.

$n \leftarrow d_{\text{out}}\, d_{\text{in}}$; $\quad f \leftarrow d_{\text{in}}$; $\quad \tau \leftarrow 1/\sqrt{f}$

$\gamma_{\text{pred}} \leftarrow \rho_0\,\alpha + (1-\rho_0)\,\delta$ `// closed-form absmean threshold, Eq. (14)`

if $\alpha(1-a) < \frac{1}{2}\gamma_{\text{pred}}$ **then abort**: active mode may fall below threshold

if $\delta(1+a) \ge \frac{1}{2}\gamma_{\text{pred}}$ **then abort**: dormant mode may survive threshold

$u, v, s \leftarrow$ three independent $\mathcal{U}(0,1)^n$ draws from g `// shared base noise across` ρ_0

$J \leftarrow (1-a) + 2a\,v$; $\quad \varsigma \leftarrow \mathbf{1}[s \ge \frac{1}{2}] - \mathbf{1}[s < \frac{1}{2}]$

$m \leftarrow \mathbf{1}[u < \rho_0]$ `// active mask;` $\mathbb{E}[m] = \rho_0$ `by construction`

$\tilde{w} \leftarrow \varsigma \odot J \odot (\alpha\, m + \delta\,(1-m))$

$\tilde{w} \leftarrow \tilde{w} \cdot \tau / \max(\text{std}(\tilde{w}), 10^{-9})$ `// fan-in normalization;` ρ `invariant (Lemma 1)`

$\gamma \leftarrow \max(\text{mean}|\tilde{w}|, 10^{-9})$;
$\rho_{\text{real}} \leftarrow \frac{1}{n}\left|\{i : \text{round}(\tilde{w}_i/\gamma) \ne 0\}\right|$

$\text{tol} \leftarrow \max\left(0.02,\; 5\sqrt{\rho_0(1-\rho_0)/n}\right)$

if $|\rho_{\text{real}} - \rho_0| > \text{tol}$ **then abort**: target operating point not realized

$\mathbf{W} \leftarrow \text{reshape}(\tilde{w})$; $\quad$ **return** $\mathbf{W}, \rho_{\text{real}}$

3.4. γ-warmup: annealing the operating point

The initial operating point is, by construction, a property of step 0 only; gradient descent immediately begins to redistribute weight magnitudes. To decouple the choice of the initial point from the abruptness of its imposition, we introduce an optional threshold anneal. Let $g(t)$ denote the threshold scale c used at optimizer step t. The γ-warmup schedule is the linear ramp

$$g_{\text{warm}}(t;\, s_0, \tau_w) \;=\; \begin{cases} s_0 + (1-s_0)\dfrac{t}{\tau_w}, & t < \tau_w, \\ 1, & t \ge \tau_w, \end{cases} \qquad (17)$$

with warmup start $s_0 \in (0, 1]$ and duration τ_w in optimizer steps. Since ρ is monotonically decreasing in c by (9), $s_0 < 1$ lowers the effective threshold and therefore increases the active fraction during the early phase, relaxing toward the canonical $\gamma = \mathbb{E}|w|$ over τ_w steps. The operating configuration is $(s_0, \tau_w) = (0.4, 240)$; the ablation of Section 3.14 sweeps $s_0 \in \{0.3, 0.4, 0.5\}$ and $\tau_w \in \{120, 240, 480\}$.

For completeness, the gradual-pruning baseline uses the complementary rising schedule

$$g_{\text{sched}}(t;\, T) \;=\; 1 + (T-1)\min\!\left(1, \frac{t}{\lfloor 0.7\,S \rfloor}\right), \qquad T = 1.6, \quad (18)$$

where S is the total number of optimizer steps; the threshold rises over the first 70% of training, driving forward sparsity upward monotonically.

All writes to the threshold scale pass through a single sanctioned accessor which updates both the host-side scalar and a persistent zero-dimensional device tensor in place. The device mirror exists so that the compiled training step is guarded on tensor shape and dtype rather than on a Python scalar value: the two arms whose thresholds move every step, γ-warmup and the gradual-pruning baseline, would otherwise trigger a graph recompilation per step. The host scalar remains the single source

of truth, and the two representations are cross-checked against a relative tolerance of 10^{-6} at every run boundary.

3.5. Mechanism-matched control: the γ-offset arm

A design in which TAI is the only arm capable of moving ρ_0 cannot falsify the claim that controlling ρ_0 is beneficial: the hypothesis could fail only through a defect of the apparatus. We therefore include an arm that reaches the same operating point by an entirely different mechanism.

The γ-offset arm retains the standard zero-mean Gaussian initializer and instead fixes the threshold scale to a constant $c^\star(\rho_0)$ chosen so that the initial active fraction equals the target. Because Φ is strictly increasing, $\rho(c)$ in (9) is continuous and strictly decreasing on $(0, \infty)$; hence $c^\star$ is the unique root of

$$\begin{aligned} \rho(c^\star) &= \rho_0, \qquad \text{equivalently} \\ c^\star(\rho_0) &= \frac{2}{\sqrt{2/\pi}}\,\Phi^{-1}\!\left(1 - \frac{\rho_0}{2}\right), \end{aligned} \tag{19}$$

and is obtained numerically by 200 steps of bisection on $[10^{-4}, 20]$, which agrees with the closed form to machine precision. The calibration is validated by the sanity condition $c^\star(0.69) = 1.000 \pm 0.02$, i.e. the locked baseline must be recovered at the locked operating point; the assertion is enforced at definition time.

The two arms therefore differ only in where the operating point is set:

Arm	Mechanism
`tai`	shape the weight-magnitude law; keep $\gamma = \mathbb{E}\lvert w\rvert$
`gamma_offset`	keep the standard magnitude law; scale γ by $c^\star$

The pre-registered readings are stated in advance and are symmetric with respect to outcome. If the two arms are equivalent in quality, the contribution is attributable to control of the operating point rather than to the TAI initializer specifically, and the simpler threshold-offset implementation should be recommended. If TAI dominates, the magnitude law carries value beyond the operating point, plausibly because a permanent rescaling of γ permanently rescales every surviving ternary weight, and with it the activations each layer emits and the gradients they induce, whereas the TAI perturbation decays as weights move. If γ-offset dominates, that is reported as such. Note that $c^\star$ is held fixed for the entire run; this permanence is the mechanism under test and is not an incidental implementation choice.

3.6. Placebo arm for the causal sparsity hypothesis

The causal claim, that ρ_0 shifts the final learned sparsity, is paired with a pre-registered falsification instrument. The placebo arm `tai_negctrl` invokes the identical initializer machinery but decouples the ρ_0 label from the threshold partition it induces: the realized split is always generated at the fixed reference $\rho_0^{\text{ref}} = 0.69$, while the label recorded in the run store varies over the sweep grid. If the apparent relationship between ρ_0 and final sparsity were an artefact of the analysis rather than a causal consequence of the initial partition, the placebo would exhibit the same trend. The frozen prediction is a null trend (Spearman ≈ 0).

Because the placebo shares a name prefix with the treatment arms, its name is matched by exact equality before any prefix test is applied, and the initializer is then retrieved from a table keyed by exact name. Independently of that dispatch, every placebo run asserts $|\rho_{\text{real}} - 0.69| < 0.03$ whatever label it was issued under, so a regression in the matching order would halt the run rather than corrupt the control. A placebo that silently receives the genuine initializer does not merely weaken the primary claim; it inverts it, since the falsification instrument would then report the true effect as an artefact.

3.7. Model family and scale ladder

3.7.1. Block families

All models are pre-norm decoder-only transformers with rotary position embeddings (RoPE, $\theta = 10^4$) [47], tied input/output embeddings, and BitLinear projections throughout the attention and feed-forward blocks. Two block families are studied.

Family A (`decoder_v1`) uses multi-head self-attention with H heads of dimension d/H and a squared-ReLU [48] feed-forward network,

$$\mathrm{FFN}_A(\mathbf{x}) = \mathbf{W}_{\text{down}}\big[\max(0, \mathbf{W}_{\text{up}}\mathbf{x})\big]^{\odot 2}. \tag{20}$$

Family B (`decoder_v2`) uses grouped-query attention [49] with H_{kv} key/value heads and a SwiGLU [50] feed-forward network,

$$\mathrm{FFN}_B(\mathbf{x}) = \mathbf{W}_{\text{down}}[\mathrm{SiLU}(\mathbf{W}_{\text{gate}}\mathbf{x}) \odot \mathbf{W}_{\text{up}}\mathbf{x}]. \tag{21}$$

Because grouped-query attention requires $H \bmod H_{kv} = 0$, the number of key/value heads is set to the largest proper divisor of H, which for prime head counts is $H_{kv} = 1$, recovering multi-query attention [1]. This is well defined for every ladder point, including those with odd head counts. Both families use the identical BitLinear projection, so the quantization pathway is held constant and only the block topology varies.

Residual connections are applied around each sub-block, $\mathbf{x} \leftarrow \mathbf{x} + \mathrm{Attn}(\mathbf{x})$ and $\mathbf{x} \leftarrow \mathbf{x} + \mathrm{FFN}(\mathbf{x})$, with RMSNorm applied inside each BitLinear rather than as a separate pre-norm module.

3.7.2. Parameter accounting

Non-embedding parameter counts are computed analytically as

$$N_{\text{non-emb}} = L\left(2d^2\left(1 + \frac{H_{kv}}{H}\right) + 2\,d\,d_{\text{ff}}\right) + \mathbf{1}[\text{Family B}] \cdot L\,d\,d_{\text{ff}}, \tag{22}$$

where the final term accounts for the SwiGLU gate projection. The attention term counts the query and output projections at d^2 each and the key and value projections at $d\,H_{kv}(d/H)$ each; under multi-head attention $H_{kv} = H$ and it reduces to the familiar $4d^2$, while grouped-query attention shrinks the key and value projections in proportion to H_{kv}/H. Embedding and output-head parameters are excluded throughout, as they are tied and are not quantized.

3.7.3. Ladder definition and confirmatory/exploratory partition

Table 1 defines the scale ladder. Under a fixed single-GPU budget it is not possible to run the full method × seed grid at every ladder point; rather than reduce the seed count uniformly, which would weaken every claim, we partition the ladder explicitly. Confirmatory scales carry the full method set at $n = 15$ seeds and support hypotheses H1–H5, together with the cross-architecture and feedback checks (H7, H8); exploratory scales run a reduced set of 5 seeds and support only the scale-consistency analysis (H6). Every claim in the results states which set it rests on, and no analysis silently iterates over the full ladder.

The span from the smallest confirmatory scale to the largest executed scale exceeds two orders of magnitude in non-embedding parameters. The opt-in `dim2048` point was declared but not enabled under the available budget; it contributes to no result reported in this paper, and the executed ladder therefore comprises eleven scales. We note explicitly that the larger ladder points cannot be trained compute-optimally under a fixed token budget; their role is therefore restricted to establishing the sign consistency of the effect across scale, which a fixed budget can measure honestly, and not to a compute-optimal loss comparison.

3.7.4. Depth-controlled ladder

The principal ladder co-varies width and depth, so an apparent dependence of the effect on scale could reflect a depth effect. We therefore define a second, depth-controlled ladder in which $L = 6$ is held fixed and only width varies, with $H = \lfloor d/64 \rfloor$ (head dimension held at 64) and $d_{\text{ff}} = 64\ \text{round}(8d/(3{\cdot}64))$ (Table 2). The scale trend is refitted on this ladder as a confound-controlled second slope.

3.7.5. Embedding initialization

The tied token embedding is initialized as $\mathcal{N}(0, 0.02^2)$, following the GPT-2 convention [51]. This is stated explicitly because it is consequential in an initialization study: the framework default of $\mathcal{N}(0, 1)$ for an embedding table, left in place on a tied embedding, yields logits whose standard deviation grows as $\sqrt{d}$, producing an initial cross-entropy that is both far above the uniform-prediction baseline $\log|\mathcal{V}|$ and strongly scale-dependent. Such a transient would consume a scale-dependent fraction of a fixed token budget, bias the scale-consistency analysis in a direction unrelated to ρ_0, and dilute the very initialization effect under measurement. With the stated value, the initial loss sits at the uniform-prediction baseline at every scale of the ladder. The setting is identical for every arm and touches no BitLinear tensor, so the mechanism under test is unaffected.

3.8. Corpus, tokenization and leakage-free partition

3.8.1. Corpus

Training data are drawn from the FineWeb-EDU `sample-10BT` stream [52] and tokenized once with the pretrained GPT-2 byte-pair encoder [51] ($|\mathcal{V}| = 50{,}257$), with an end-of-text token appended to each document. The tokenized corpus is truncated to $N_{\text{corpus}} = 48\,\text{M}$ tokens (at least 2× the per-run token budget divided by the training fraction, so that no run exhausts its split) and is cached as a `uint16` memory-mapped array, ensuring that the identical token stream is reused across all sessions and all arms. A real BPE vocabulary is used (rather than a byte-level or reduced vocabulary) so that the held-out multiple-choice benchmarks of Section 3.12 are scored on exactly the vocabulary the models were trained on. Should the corpus stream be unavailable, a deterministic synthetic fallback over the same vocabulary is generated and flagged in the manifest; downstream benchmark evaluation is then automatically disabled rather than reported on non-linguistic data.

3.8.2. Interleaved held-out partition with guard bands

Batches are formed by sampling a uniformly random window start $i \sim \mathcal{U}\{0, \ldots, |D| - \ell - 2\}$ and reading the contiguous span $[i, i + \ell]$, which supplies ℓ input tokens and their shifted targets, with $\ell = 512$. A naive contiguous train/validation cut is therefore insufficient: any training window beginning in $[p - \ell, p)$, where p is the cut point, overlaps validation tokens. Moreover, a single tail block confounds evaluation-data variance with document-distribution shift, since the stream is ordered.

The partition is constructed as follows. The corpus is divided into contiguous chunks of $C = 50{,}000$ tokens. A held-out set of chunks is selected at stride $\lfloor N_{\text{chunks}}/N_{\text{hold}} \rfloor$, where $N_{\text{hold}} = \max(K, \text{round}(\phi N_{\text{chunks}}))$ and $\phi = 0.06$, so that held-out material is interleaved across the entire document distribution rather than concentrated in a tail. The training set is the complement of the union of the held-out chunks dilated by a guard band of ℓ tokens on each side, so that no training window can reach into held-out material by construction. Finally, the held-out region is partitioned into $K = 5$ sub-folds by assigning chunk j to fold $j \bmod K$.

Critically, all K folds share one and the same training set. This is enforced by an identity assertion at construction time. The alternative (deriving a training set per fold, then training on fold 0's complement while scoring all folds) places folds $1, \ldots, K - 1$ inside the training data and would report training loss as held-out variance. Fold 0 serves as the primary validation split; the remaining folds supply the data-sampling variance component of the stability analysis (Section 3.13) without any retraining, since each model is scored on all K folds while still resident in memory.

3.9. Optimization protocol

3.9.1. Token-equal budget

All arms are trained under an identical budget of $B = 2.0 \times 10^7$ tokens at batch size $b = 32$ and sequence length $\ell = 512$, giving

$$S = \left\lceil \frac{B}{b\,\ell} \right\rceil = \left\lceil \frac{2.0 \times 10^7}{16{,}384} \right\rceil = 1{,}221 \quad \text{optimizer steps per run.} \tag{23}$$

Table 1: Scale ladder. d: model width; L: depth; H: attention heads; d_{ff}: feed-forward width; N: non-embedding parameters from Eq. (22), Family A. Confirmatory scales run the full method set at $n = 15$ seeds; exploratory scales run $n = 5$ seeds and support the scale-consistency analysis only. The largest point is opt-in and is enabled only if wall-clock permits.

Tag	d	L	H	d_{ff}	N	Set
`dim192`	192	4	4	512	1.38 M	exploratory
`dim256`	256	4	4	688	2.46 M	**confirmatory**
`dim320`	320	5	5	853	4.78 M	exploratory
`dim384`	384	6	6	1024	8.26 M	**confirmatory (primary)**
`dim512`	512	8	8	1365	19.57 M	**confirmatory**
`dim640`	640	9	10	1706	34.40 M	exploratory
`dim768`	768	10	12	2048	55.05 M	**confirmatory**
`dim896`	896	11	14	2389	82.42 M	exploratory
`dim1024`	1024	12	16	2730	117.42 M	exploratory
`dim1280`	1280	14	16	3413	214.07 M	exploratory
`dim1536`	1536	16	16	4096	352.32 M	exploratory
`dim2048`	2048	20	16	5461	782.91 M	opt-in

Table 2: Depth-controlled ladder used to separate width from depth in the scale analysis. Depth is fixed at $L = 6$ while width varies; $H = \lfloor d/64 \rfloor$ holds head dimension at 64, and d_{ff} follows the SwiGLU-style ratio $8d/3$ rounded to a multiple of 64. N is non-embedding parameters from Eq. (22), Family A. All points run at $n = 5$ seeds and feed the confound-controlled slope of Table 15.

Tag	d	L	H	d_{ff}	N
`wc384`	384	6	6	1024	8.26 M
`wc512`	512	6	8	1344	14.55 M
`wc768`	768	6	12	2048	33.03 M
`wc1024`	1024	6	16	2752	58.98 M

3.9.2. Optimizer and schedule

Optimization uses AdamW [53] with $\beta = (0.9, 0.95)$, weight decay 0.1, and global gradient-norm clipping at 1.0, in fused form where available. The learning rate follows linear warmup over $W = 120$ steps followed by cosine decay to $\eta_{\min} = 10^{-4}$:

$$\eta(t) = \begin{cases} \eta_{\text{peak}} \dfrac{t+1}{W}, & t < W, \\ \eta_{\min} + \dfrac{1}{2}(\eta_{\text{peak}} - \eta_{\min})\left[1 + \cos\left(\pi \dfrac{t-W}{S-W}\right)\right], & t \geq W. \end{cases} \tag{24}$$

Parameters are held in `bfloat16` and optimized in that precision; no `float32` master copy is maintained, and forward passes run under `bfloat16` autocast. The cross-entropy objective is evaluated in a chunked form: the flattened batch is partitioned into 4 pieces; per-piece sums are accumulated and divided by the total token count. This is algebraically identical to the batch mean (each token contributes exactly once with weight $1/n$), but only one piece's `float32` logit expansion is resident at a time, reducing the transient peak by a factor of approximately the chunk count. The reduction matters because the logit expansion cost scales with $|\mathcal{V}|$ and not with d, so the smallest ladder point would otherwise pay the same memory cost as the largest.

3.9.3. Per-arm learning-rate calibration

A single shared learning rate would penalize whichever arm is furthest from its optimum, confounding the method contrast with a tuning artefact. Each (scale, method) cell therefore receives its own $\eta^\star$, selected by minimizing validation loss over the grid $\eta \in \{5\times10^{-4}, 10^{-3}, 2\times10^{-3}, 3\times10^{-3}, 5\times10^{-3}\}$, averaged over three calibration seeds $\{100001, 100002, 100003\}$ that are disjoint from the fifteen reporting seeds. Disjointness ensures that no reported metric is obtained on a seed that participated in its own hyperparameter selection.

Two refinements are applied. First, calibration runs are truncated to 300 steps: the selection is an argmin over the grid, and the ranking separates long before convergence. Second, if the argmin falls on a grid endpoint, the grid is extended once in that direction (doubling upward, halving downward), up to two extensions. This converts the possibility that the optimum lies outside the grid from an untestable limitation into a measured fact: either the curve turns over, and $\eta^\star$ can be reported as interior, or it continues to descend, which is itself reportable. The short-calibration proxy is validated directly at the primary scale by refitting the calibration curve at full length on one seed and confirming that the argmin agrees; this check is necessary because high learning rates look better early than they prove to be, and $\eta^\star$ is selected from only 180 post-warmup steps but applied to 1,221-step runs.

3.9.4. Global quantization-state hygiene

The precision mode, the quantizer recipe and the threshold scale are process-level state read inside the forward pass. Because the harness deliberately uses a clean interpreter exit at session boundaries as a control-flow mechanism, any non-exception-safe manipulation of that state can leak across runs. All three are therefore established by a context manager with entry assertions and guaranteed restoration on every exit path, including exceptions, interrupts and interpreter exit. A post-hoc guard additionally verifies that any run declared ternary reports non-zero sparsity, and aborts without persisting the record if it does not. The failure mode this prevents (a full-precision run

saved under a valid ternary run identifier and skipped forever by the resume logic) is silent, plausible-looking, and would invalidate every downstream statistic.

3.10. Convergence and divergence diagnostics

3.10.1. Validation trace

Held-out loss on fold 0 is probed every $\lfloor 0.08\,S \rfloor$ steps (13 probes per run at the full budget), each probe averaging 12 evaluation batches drawn from a fixed generator so that all arms and seeds are scored on an identical evaluation sequence. The probe costs approximately 2% of run wall-clock and is what makes the convergence claim checkable.

3.10.2. Plateau test

Convergence is assessed by a slope test on the held-out trace rather than by a level comparison on the training loss. Let $\{(t_k, v_k)\}_{k=1}^{K_v}$ be the validation trace and let $\mathcal{T}$ index the final $\max(5, \lceil 0.40 K_v \rceil)$ probes. Fit $v = a + b\,t$ over $\mathcal{T}$ by ordinary least squares, with

$$
\begin{aligned}
b &= \frac{\sum_{k\in\mathcal{T}} (t_k - \bar{t})(v_k - \bar{v})}{\sum_{k\in\mathcal{T}} (t_k - \bar{t})^2}, \\
\mathrm{se}(b) &= \sqrt{\frac{\sum_{k\in\mathcal{T}} r_k^2}{(|\mathcal{T}| - 2)\sum_{k\in\mathcal{T}} (t_k - \bar{t})^2}},
\end{aligned}
\tag{25}
$$

where r_k are residuals, and define the fitted tail descent $\Delta = b\,(t_{\max} - t_{\min})$ and the tolerance band $\beta_p = \varepsilon_p |v_{K_v}|$ with $\varepsilon_p = 0.01$. A run is certified plateaued if and only if all three conditions hold:

$$
\begin{aligned}
&\text{(i)} && |b| - 1.96\,\mathrm{se}(b) \le 0 \ \text{ or } \ |\Delta| \le \beta_p && \text{(no resolvable trend);} \\
&\text{(ii)} && \Delta \ge -\beta_p && \text{(not still descending);} \\
&\text{(iii)} && \Delta \le +\beta_p && \text{(not ascending).}
\end{aligned}
\tag{26}
$$

Conditions (ii) and (iii) jointly imply the second disjunct of (i), so the verdict reduces to $|\Delta| \le \beta_p$: the slope confidence interval is evaluated but cannot by itself change the outcome. We state the criterion in the form in which it is implemented. Fewer than five tail probes yield a negative verdict: inability to certify, rather than certification by absence of evidence. Condition (iii) is not pedantry: a destabilizing run also has a tail that is flat in level terms and would otherwise be admitted into an equivalence claim, silently averaging instability into a statement about quality.

This formulation replaces a level test on single-batch training loss, which is uninformative for two independent reasons. First, single-batch training loss carries per-step sampling noise on the order of 0.05–0.15 nats against a tolerance band of order 0.04 nats, so the verdict is dominated by measurement noise; simulated perfectly flat traces pass only 63–100% of the time depending solely on the noise scale. Second, |mean(tail) − final| is satisfied mechanically by cosine decay, since the loss flattens as $\eta \to \eta_{\min}$ irrespective of convergence. The quantity of interest is whether the loss is still descending, which is the slope.

3.10.3. Divergence detection

The plateau test constrains the trend of the validation trace, not its level: a run that blew up early and then settled at a catastrophic loss has also stopped descending, so it satisfies (26) exactly as a converged run does. Since every claim is a seed-paired contrast, a single undetected divergence in one arm can manufacture a large, significant and entirely spurious effect. We therefore flag a run as diverged if

$$
\begin{aligned}
v_{\text{final}} \text{ is not finite} \quad \text{or} \quad v_{\text{final}} > \kappa \log|\mathcal{V}|, \\
\kappa = 1.25, \qquad \log|\mathcal{V}| = \log 50{,}257 = 10.825,
\end{aligned}
\tag{27}
$$

or if the terminal third of the validation trace rises by more than 50% above its own start while exceeding $\log|\mathcal{V}|$. The bound is principled rather than arbitrary: a model that has learned nothing still predicts the uniform distribution and attains exactly $\log|\mathcal{V}|$, so a materially larger loss indicates confidently wrong predictions. Diverged runs are recorded (so that the resume logic does not repeatedly re-attempt them) and excluded from all analyses; the count of exclusions is reported, since a high rate would itself constitute evidence that the learning-rate grid is too aggressive at the larger scales.

3.10.4. Budget-sufficiency check

A slope test can establish only that no trend is resolvable at the observed noise level; it cannot exclude a slow real descent that additional tokens would reveal. Since the equivalence hypothesis rests entirely on the adequacy of the budget, that adequacy is tested directly rather than assumed. At the primary scale, the pre-registered baseline and the γ-warmup arm are re-run at 2× the token budget on three seeds, holding all else fixed. The loss at B is then read off each long run's own validation trace at the probe nearest the short-budget step count, so the comparison with its loss at $2B$ is within-run and carries no cross-run seed noise. The resulting gap is tested against the equivalence margin of Section 3.13. The decision rule is fixed in advance: if the gap is within the margin for both arms, the equivalence claim retains convergence wording; otherwise it is restated as a claim about a fixed 20 M-token budget and the manifest records the licensed phrasing.

3.11. Methods under comparison

Table 3 enumerates the arms of the confirmatory grid. The set is deliberately non-degenerate: it contains an upper bound, a locked-ρ_0 baseline family, an explicit sparsity-scheduling method, a mechanism-matched operating-point control, and the proposed method with its schedule variant and its swept-optimal variant. The locked-ρ_0 family holds three arms. Two of them, `standard` and `bitnet158`, are the same configuration up to floating-point rounding (Remark 4), so the pre-registered primary baseline is drawn from that family for the reason given there rather than because it contributes a distinct mechanism; `lsuv` shares the operating point but is a distinct intervention, a question Remark 1 leaves to measurement and Section 4.2 settles.

Table 3: Methods in the confirmatory grid. All arms share the identical architecture, corpus, data order per seed, token budget and schedule shape. The peak learning rate is calibrated separately for every (scale, arm) cell on disjoint seeds (Section 3.9.3); apart from that, the arms differ only in the listed mechanism. c is the absmean threshold scale of Eq. (5).

Arm	Initializer	Quantization mechanism
`fp16_control`	standard	No quantization; `bfloat16` throughout. Full-precision reference; sparsity is 0 by definition. The arm name is historical: no IEEE half-precision arithmetic is used anywhere in this work, and this arm differs from the ternary arms only in that Q_w is bypassed.
`standard`	$\mathcal{N}(0, 1/f)$	Absmean ternary, $c \equiv 1$. Locked at $\rho_0 \approx 0.69$ (Proposition 1).
`lsuv`	orthogonal + data-driven variance pass	Absmean ternary, $c \equiv 1$. Same locked operating point; whether the calibration pass is inert here is measured, not assumed (Remark 1).
`bitnet158`	standard	BitNet b1.58 formulation: $\widehat{\mathbf{W}} = \text{clip}(\text{round}(\beta\mathbf{W}), -1, 1)/\beta$ with $\beta = 1/(c\,\mathbb{E}\lvert w\rvert)$. Pre-registered primary baseline; algebraically equivalent to `standard`; see Remark 4.
`sparse_sched`	standard	Gradual pruning: rising threshold $g_{\text{sched}}(t)$ of Eq. (18) with target $T = 1.6$.
`gamma_offset`	standard	Mechanism-matched control: $c \equiv c^\star(\rho_0)$ from Eq. (19), held fixed.
`tai`	TAI, $\rho_0 = 0.75$	Absmean ternary, $c \equiv 1$. Proposed method.
`tai_warmup`	TAI, $\rho_0 = 0.75$	γ-warmup schedule $g_{\text{warm}}(t; 0.4, 240)$ of Eq. (17).
`tai_star`	TAI, $\rho_0 = \rho_0^\star$	As `tai`, with $\rho_0^\star$ fed back from the sweep (Section 3.14).
`tai_negctrl`	TAI at fixed 0.69, variable label	Placebo arm for the causal hypothesis (Section 3.6); sweep only.

Remark 4 (Relationship between the two ternary formulations). Under exact arithmetic the BitNet b1.58 formulation and the absmean formulation of Eq. (6) coincide: substituting $\beta = 1/\gamma$ returns

$$\text{clip}(\text{round}(\mathbf{W}/\gamma),\ -1,\ 1) \cdot \gamma,$$

which is Eq. (6) exactly. The two therefore differ only in the association of the scale factor, and hence only in floating-point rounding. The two arms also receive the same standard initializer in this harness, so they are the same configuration measured twice rather than two independent recipes. We state this explicitly rather than presenting the recipes as substantively distinct: the value of retaining `bitnet158` as the pre-registered primary baseline is that the contrast is frozen to an externally specified recipe before the grid is executed, which removes the researcher degree of freedom of selecting, per scale, whichever ternary baseline happens to look weakest. The data-selected minimum over all ternary baselines is still reported, but only as a secondary sensitivity analysis and labelled as such.

3.12. Evaluation metrics

Table 4 fixes the name, symbol and unit of every quantity reported in this paper. These names are used verbatim in Sections 4 and 5; no synonym is introduced later.

3.12.1. Primary quality metric

The primary metric is held-out cross-entropy in nats per token on validation fold 0, averaged over 20 evaluation batches drawn from a fixed generator, with perplexity $\exp(v)$ reported as a derived quantity. Loss is measured under the run's own quantization state, so that the reported value describes the model as trained rather than a differently thresholded variant of it; the full-precision control is accordingly scored in full precision and provides the attainable-quality upper bound.

3.12.2. Sparsity

Final sparsity is the unweighted mean of $1-\rho$ over all BitLinear layers, evaluated through the same quantizer and the same live threshold scale in force at the end of training, and, critically, measured before the global quantization state is restored. Measuring at the canonical threshold instead would report a counterfactual for every arm whose mechanism is a shifted threshold, and would assign a spurious non-zero sparsity to the full-precision control.

3.12.3. Downstream evaluation

At the scales accessible under this budget, free generation is at chance, so downstream ability is read through the log-likelihood (multiple-choice) protocol. Consider a question with

Table 4: Reported quantities. The name in the first column is the only name used for that quantity in the Results and the Discussion.

Quantity	Symbol	Definition and unit
Held-out loss	$\mathcal{L}$	Cross-entropy on validation fold 0, nats per token. Held-out loss and held-out cross-entropy denote the same quantity.
Paired quality contrast	$\Delta\mathcal{L}$	$\mathcal{L}^{(\beta)} - \mathcal{L}^{(m)}$ in nats; positive favours method m.
Initial active fraction	ρ_0	Fraction of weights receiving a non-zero ternary code at step 0 (Definition 1). Not called active ratio.
Final sparsity	sp_T	$1 - \rho$ at the end of training, measured under the live threshold.
Retention	r	Slope of Eq. (32); dimensionless, $r = 1$ fully retained.
Steps to target	S_{target}	First recorded step at which the smoothed training trace reaches that scale's $\mathcal{L}_{\text{target}}$ (5.6539 at the primary scale, Eq. (30)); resolution 20 steps.
Paired speed contrast	ΔS	$S^{(\beta)}_{\text{target}} - S^{(m)}_{\text{target}}$; positive favours m.
Next-token accuracy	a_{nt}	Top-1 next-token accuracy on held-out data; internal transfer proxy.
Benchmark accuracy	acc, acc_norm	Token- and byte-normalized log-likelihood accuracy, Eq. (29).
Equivalence margin	Δ_{eq}	Data-derived TOST margin, Eq. (31), nats.
Family-wise level	α_H	Holm level, 0.05.
Per-hypothesis level	α_0	Standalone-test level, 0.0125; applies to preregistered tests outside the Holm family (H1 and H4).

context x and options $y^{(1)}, \ldots, y^{(m)}$, whose gold index is j^*, and let

$$s_j = \sum_t \log p\left(y_t^{(j)} \,\middle|\, x,\, y_{<t}^{(j)}\right) \tag{28}$$

denote the summed conditional log-probability of the continuation tokens of option j. Two length normalizations are reported:

$$\begin{aligned} \text{acc} &= \mathbf{1}\left[\arg\max_j \frac{s_j}{|y^{(j)}|_{\text{tok}}} = j^*\right], \\ \text{acc_norm} &= \mathbf{1}\left[\arg\max_j \frac{s_j}{|y^{(j)}|_{\text{bytes}}} = j^*\right], \end{aligned} \tag{29}$$

where $|\cdot|_{\text{tok}}$ and $|\cdot|_{\text{bytes}}$ are the continuation length in tokens and in UTF-8 bytes respectively, the latter being the normalization that removes option-length bias. One deviation from lm-evaluation-harness [54] should be recorded: there acc is the argmax of the unnormalized score s_j, whereas here it is normalized by continuation token length. Our acc_norm follows the harness definition, but our acc is therefore not directly comparable with published acc figures. We evaluate the validation splits of ARC-Easy and HellaSwag, restricted to the first 300 items of each for tractability, and report both metrics for both tasks. Tasks are cached once; if unavailable, downstream evaluation is skipped and flagged, never imputed.

Scoring is batched: all (item, option) pairs are flattened into a single work list and scored in padded groups of 16. Right padding is admissible under causal masking, since a pad token can influence only positions after itself and every scored position precedes the padding of its own row; a mask ensures pad positions contribute exactly zero. To control cost, downstream evaluation is performed on the first three reporting seeds of every (scale, method) cell, together with the first calibration seed where such a run is reused (the same seeds for every method, so the comparison remains paired), and missing entries are masked in the statistical analysis rather than imputed.

An internal transfer proxy, next-token top-1 accuracy on held-out data over 12 batches, is also recorded. A synthetic copy-task proxy used in preliminary work was removed: it tracks active-weight density mechanically and is not method-discriminating.

3.12.4. Efficiency: deployment model and simulation cost

We report two clearly separated efficiency quantities. The deployment model is primary and is a counting argument independent of the hardware used here: a ternary codebook carries $\log_2 3 = 1.585$ bits of information per weight, or 2 bits under naive packing, against 16 bits for a `bfloat16` weight, giving an information-theoretic compression of $16/1.585 = 10.1\times$ on the quantized (non-embedding) parameters, or $8\times$ under the naive 2-bit packing a simple kernel would use, with embeddings and the tied head retained in `bfloat16`. The simulation

cost (measured latency, throughput and peak memory of the straight-through ternary path relative to the full-precision control) is reported second and is explicitly labelled as a property of the research harness rather than of ternary inference. Under straight-through estimation the ternary path retains `bfloat16` master weights and adds per-forward work (normalization, activation quantization, weight quantization, rescaling) on top of the same `bfloat16` matrix multiplication; it is therefore expected to be slower than the control, and this says nothing about deployment. No ternary kernel is benchmarked in this work; the limitation is recorded in the manifest and restated in the Limitations.

3.13. Hypotheses and statistical analysis

3.13.1. Pre-registered hypotheses

The hypothesis family was frozen before the confirmatory grid was executed and is reproduced in Table 5. Hypotheses H1–H4 constitute the confirmatory family; H5–H8 and all secondary baseline contrasts are exploratory, are reported without family-wise correction and are labelled accordingly, so that the confirmatory claims are not diluted by exploratory multiplicity. Family-wise error control is applied by Holm–Bonferroni to the family of quality contrasts between each TAI variant and the frozen primary baseline across the confirmatory scales, which is where multiplicity accumulates. H1, H3 and H4 stand outside the Holm family: H1 and H4 are declared against the standalone level α_0, and H3 is reported without a family-wise correction. All three are to be read as such.

3.13.2. The H1 target loss

H1 concerns the number of optimizer steps needed to reach a common loss level, so that level must be fixed independently of the arms compared. It is taken from the standard-init calibration-seed trajectories as the median of their 60th-percentile smoothed loss, evaluated before the confirmatory grid was executed, and frozen. A single level would not be meaningful across the ladder, so one is calibrated by the same procedure at every scale; at the primary scale it is

$$\mathcal{L}_{\text{target}} \;=\; 5.6539 \text{ nats/token}, \tag{30}$$

with S_{target} denoting the first recorded step at which a run's smoothed training trace falls at or below its own scale's $\mathcal{L}_{\text{target}}$. Because the trace is recorded every 20 steps, S_{target} is quantized to that resolution, and differences smaller than 20 steps are not resolvable and are not interpreted.

3.13.3. Paired contrasts

All method contrasts are paired within seed: for a method m and the baseline β at a given scale, the per-seed difference is $\Delta_i = v_i^{(\beta)} - v_i^{(m)}$, so that $\Delta_i > 0$ favours m. We report the paired t-test as the primary inferential statistic, the Wilcoxon signed-rank test as distribution-free support, Cohen's d computed on the pooled standard deviation, and a nonparametric bootstrap confidence interval for $\bar{\Delta}$ obtained from 5,000 resamples of the difference vector at the 2.5th and 97.5th percentiles. Post-hoc power is deliberately not reported: computing power from the observed effect is circular, and the confidence interval conveys the same information without that fallacy. We note that the d values reported below are not comparable to the conventions of observational research and should not be read as such. Cohen's d is computed here on the pooled standard deviation of the two arms' held-out losses, and that denominator is small by construction: all runs share one corpus, budget, schedule and architecture and differ only in the seed and in the mechanism under test, so the spread it measures is seed-to-seed training noise alone (at `dim512`, 0.0192 nats, against 0.0270 for the standard deviation of the paired difference). Values of d in the high single digits therefore reflect the precision of the design rather than an implausibly large intervention. Where a confidence interval covers zero, the sign is declared unresolved rather than being reported as a null effect.

3.13.4. Equivalence testing

Non-inferiority is assessed by two one-sided tests (TOST) on the paired differences. Rather than fixing an arbitrary margin, the margin is derived from the data at each scale as a fixed fraction of the cost that quantization already incurs,

$$\Delta_{\text{eq}} \;=\; \max\Big(\lambda \,\big|\, \bar{v}^{(\beta)} - \bar{v}^{(\text{fp16})}\big|,\; 10^{-4}\Big), \qquad \lambda = 0.10, \tag{31}$$

so that equivalent means small relative to the ternary-versus-full-precision gap rather than a magic constant in nats; a fixed fallback of 0.02 nats is used only if the full-precision control is absent. Equivalence is declared when both one-sided p-values fall below 0.05. The convergence-conditional form of H2 is computed over the subset of seeds that are plateau-certified in both arms; if that subset is empty, the pre-registered test is reported as untestable rather than being silently replaced, and a clearly labelled companion test of the weaker token-equal claim is reported alongside it, over all paired seeds, with both recorded in the manifest.

3.13.5. Multiplicity control

Within the confirmatory family, p-values are adjusted by the Holm–Bonferroni step-down procedure: sorting $p_{(1)} \le \cdots \le p_{(m)}$, the k-th hypothesis is rejected if $p_{(k)} < \alpha_H/(m-k+1)$ and all preceding hypotheses were rejected, with family-wise level $\alpha_H = 0.05$. The family comprises one test per TAI variant per confirmatory scale, each being that variant's held-out-loss contrast against the frozen primary baseline.

Two distinct thresholds appear in this paper and are kept notationally separate throughout to prevent their conflation. $\alpha_H = 0.05$ is the family-wise level of the Holm procedure just described. $\alpha_0 = 0.0125$ is a separate, more stringent per-hypothesis level fixed in advance for pre-registered tests that stand outside the Holm family, in practice the H1 steps-to-target tests and the H4 variance test, and is the level against which those tests are declared. No test is evaluated against both.

3.13.6. Variance and stability

H4 is tested by Levene's test for equality of variances. Variance is computed over the pooled seed × fold matrix rather than over seeds alone: each trained model is scored on all $K = 5$

Table 5: Pre-registered hypotheses, primary test statistics and analysis set. Holm–Bonferroni correction is applied to the family of quality contrasts between TAI variants and the frozen primary baseline; the remaining confirmatory statistics are reported uncorrected.

	Statement	Primary test	Set
H1	At equal token budget, TAI attains a calibrated target loss in fewer optimizer steps than the pre-registered baseline.	Paired t-test on steps-to-target; Cohen's d.	confirmatory
H2	At the token budget, TAI is at least equivalent in quality to the pre-registered baseline.	Paired TOST with data-derived margin; paired t-test and bootstrap CI on the loss difference.	confirmatory
H3	Varying ρ_0 causally shifts the final learned sparsity.	Per-seed Spearman r_s over the ρ_0 grid; sign test over seeds; retention regression; placebo contrast.	confirmatory
H4	TAI does not inflate outcome variance.	Levene test on the pooled seed $\times$ fold loss matrix.	confirmatory
H5	There exists $\rho_0^\star$ placing TAI on or above the baseline quality-sparsity frontier.	Non-dominated frontier with bootstrap uncertainty over seeds.	exploratory
H6	The sign of the TAI effect is consistent across scale.	Per-scale sign test (primary); mixed-effects slope vs. $\log N$ (secondary).	exploratory
H7	The sign of the effect is preserved on the second block family.	Paired contrast on Family B.	exploratory
H8	Feeding $\rho_0^\star$ back into the grid does not degrade quality.	Paired contrast of `tai_star` vs. baseline.	exploratory

held-out sub-folds while resident in memory, so the reported spread encompasses data-sampling variance in addition to initialization and data-order variance, as the hypothesis states.

3.13.7. Causal sparsity analysis

The ρ_0 sweep is analysed within seed, exploiting the shared-base-noise construction of Remark 2. For each seed, Spearman's r_s is computed between ρ_0 and the final sparsity over the six-point grid; the per-seed coefficients are then aggregated by a sign test across the fifteen seeds. This is substantially more powerful than a single pooled correlation over all (ρ_0, seed) pairs, which can manufacture apparent monotonicity from between-seed offsets. The pooled coefficient is reported for continuity.

In addition to the ordinal test, we quantify the durability of the intervention by regressing the seed-mean final sparsity at each of the six grid points on the corresponding initial sparsity,

$$\mathrm{sp}_T \;=\; \mathrm{sp}_\infty + r \cdot \mathrm{sp}_0, \tag{32}$$

in which r is the retention coefficient, $r = 1$ denoting full retention and $r = 0$ complete erasure, and $\mathrm{sp}_\infty/(1 - r)$ is the implied fixed point of the training dynamics. Retention, rather than the mere existence of a shift, is the reported headline quantity for H3: whether the sparsity moves at all is trivially answered, and it says nothing about whether the intervention survives training. The placebo arm of Section 3.6 is subjected to the identical analysis and serves as the falsification instrument; a comparable retention slope in the placebo would establish that even the narrowed effect is an artefact.

3.13.8. Scale analysis

The primary readout for H6 is per-scale sign consistency: the number of ladder points at which the mean paired effect is positive, tested against a binomial null with $p = 0.5$. This is the claim a fixed budget can support honestly. As a secondary diagnostic, the paired effects are modelled as a function of $\log N$ with a random intercept per seed. Where the random-effects covariance is estimated at the boundary (which occurs when between-seed variance in the effect is small relative to between-scale variance, rendering the intercept unidentifiable), the singular fit is reported as such and the pre-specified fallback is used instead: ordinary least squares on the per-scale mean effects, in which the unit of analysis is the scale, since that is the unit H6 is about. Two clustered alternatives, with standard errors clustered on seed and on scale, are reported beside it as sensitivity analyses rather than as candidates to choose among, and all three enter the manifest. Where they disagree, the magnitude is reported as unresolved. A point estimate without an interval is not reported as a result. Any extrapolation beyond the tested range is labelled as prediction rather than evidence.

3.14. Ablations and confound controls

The following studies are run in addition to the confirmatory grid. Those marked recompute-only require no additional training and read stored trajectories.

Operating-point sweep and Pareto frontier: Two arms (`tai` and `gamma_offset`) are swept over $\rho_0 \in \{0.4, 0.5, 0.6, 0.7, 0.8, 0.9\}$ at the primary scale with the full fifteen seeds, and at the top confirmatory scale, giving a cross-scale test of whether the causal effect is scale-invariant. The non-dominated quality-sparsity frontier is traced for both arms with bootstrap uncertainty over seeds, and $\rho_0^\star$ is defined as the minimizer of validation loss subject to a sparsity floor, reported with a bootstrap confidence interval.

Operating-point feedback: $\rho_0^\star$ is fed back into the grid as the `tai_star` arm at the top scale and at the primary scale, and contrasted against the baseline under the same protocol as the fixed-ρ_0 arms (H8).

Cross-architecture transfer: Three arms (the pre-registered baseline `bitnet158`, the mechanism-matched control `gamma_offset`, and `tai_warmup`) are re-run on Family B (grouped-query attention with SwiGLU) at two representative scales, `dim256` and `dim384`, at the full 15 seeds (90 runs). Both contrasts against the baseline are reported, so the transfer question is put to the control as well as to the proposed method (H7). The remaining ladder points are defined for Family B but were not executed under the available budget.

Initialization versus threshold (2×2): A {full precision, ternary} $\times$ {standard init, TAI init} design at the primary scale, whose fourth cell is a dedicated arm `fp16_taiinit` (TAI initializer, quantization disabled), isolates whether the effect is a generic initialization benefit or specific to control of the ternary operating point. The pre-registered prediction under the latter account is that TAI initialization is indistinguishable from standard initialization in full precision, there being no threshold to control, while retaining its advantage under quantization. A significant full-precision advantage would indicate a generic component, which would be reported.

γ-warmup grid: A 3×3 sweep over $s_0 \in \{0.3, 0.4, 0.5\}$ and $\tau_w \in \{120, 240, 480\}$ at the primary scale on five seeds, testing whether the schedule contributes beyond TAI initialization alone and how sensitive it is.

TAI hyperparameter sensitivity: A sweep over the dormant magnitude δ and the jitter half-width a, which are genuine degrees of freedom of the construction, establishing the region in which Lemma 2 holds and confirming that ρ_0, and not δ, is the operative knob.

Learning-rate sensitivity: The key contrast is repeated across the learning-rate grid at the primary scale on five seeds, establishing that the conclusion is not an artefact of a single optimizer setting.

Depth-controlled scale trend: The scale slope is refitted on the depth-controlled ladder of Table 2, with depth held constant.

Plateau-criterion robustness (recompute-only): The plateau-verification rate and the H2 verdict are recomputed across $\varepsilon_p \in \{0.005, 0.01, 0.02\}$ and tail fractions {0.30, 0.40, 0.50}, which bracket the operating value, so the convergence claim does not hinge on one operationalization of converged. At the full budget the three fractions give tails of five, six and seven probes respectively, so the arm is not degenerate against the $\max(5, \cdot)$ floor.

Data-sampling variance decomposition (recompute-only): The seed $\times$ fold loss matrix is decomposed into seed and fold components, supporting the variance claim of H4 over both sources.

Theory verification (GPU-free in part): The σ-independence of Proposition 1 is verified by direct sampling and, at model level, by instantiating the network under each candidate initializer, with LSUV completed through its data-driven pass before measurement.

3.15. Reproducibility protocol

3.15.1. Seeding and comparability

The global seed is 1337. The fifteen reporting seeds, the three calibration seeds and the five ablation seeds are stated separately and are

$$\begin{aligned} \mathcal{S}_{\text{report}} &= \{1337,\ 2026,\ 7,\ 42,\ 2718,\ 9001,\ 13,\ 271,\\ &\qquad 99,\ 555,\ 8,\ 31415,\ 777,\ 2024,\ 11\},\\ \mathcal{S}_{\text{calib}} &= \{100001,\ 100002,\ 100003\},\\ \mathcal{S}_{\text{abl}} &= \{1337,\ 2026,\ 7,\ 42,\ 2718\}, \end{aligned} \tag{33}$$

with $\mathcal{S}_{\text{report}} \cap \mathcal{S}_{\text{calib}} = \varnothing$ by design. Within a run, three generators are seeded from the run seed and kept distinct by construction: the training-order generator (at the run seed), the initialization generator (run seed + 777), and the LSUV calibration-batch generator (run seed + 555). The separation guarantees that a data-driven initializer consumes no training randomness, so that every arm at a given seed observes exactly the same batch sequence. Evaluation draws are seeded from the global seed instead, so every arm at every seed is scored on one identical evaluation sequence.

3.15.2. Determinism

Two modes are supported and the active mode is recorded in the manifest. In bit-deterministic mode, attention is pinned to the mathematical backend, deterministic algorithms are enabled and autotuning is disabled, yielding bit-identical reruns. In throughput mode, used for the reported grid, flash and memory-efficient attention kernels, autotuning and TF32 matrix multiplication are permitted. We state precisely what this does and does not preserve: the identity of the initialization and of the batch sequence for a given seed is unaffected, since kernel selection alters only the floating-point reduction order within a kernel and not which batch a seed draws. Every arm at a given seed therefore starts from the same weights and reads the same data, and kernel nondeterminism enters as a perturbation common to all arms, which the paired design absorbs. We claim no more than

that: reduction-order differences propagate through training, so a rerun moves each point estimate within the run-to-run noise floor rather than leaving the aggregated numbers unchanged. What is not preserved is bit-identical reproduction of a single run. The manifest therefore reports statistically reproducible rather than bit-reproducible whenever throughput mode is active. Setting a single flag restores the stronger guarantee.

The compiled training step is used only for runs exceeding 600 optimizer steps, because compilation cost is incurred per run (each run constructs a fresh module, so the on-disk kernel cache does not amortize it) and exceeds the training time of a 300-step calibration run. The compiled and eager paths were verified during development to differ by less than the run-to-run noise floor induced by non-deterministic kernels. Section 5.10 contradicts that verification at the scales where the split falls between arms of a contrast, and the corrected statement is reported there: the path is a throughput decision with a measurable numerical consequence, and a corrected grid must pin it per arm. Model introspection, evaluation and sparsity measurement always operate on the uncompiled module.

3.15.3. Artefact persistence and resumption

Each completed run is persisted as a single JSON record keyed by (phase, scale, method, seed, ρ_0), written to a temporary file and moved into place atomically, so that an abrupt session termination cannot corrupt the store. Completed identifiers are skipped on subsequent invocations, making the entire multi-session grid idempotent and resumable; at worst, the single run in flight is lost and repeated. Each record stores the training and validation traces, the final validation loss and perplexity, the per-fold losses, the measured final sparsity, the plateau verdict and tail descent, the divergence flag and the uniform baseline, the downstream metrics, the wall-clock time inclusive of initialization, the learning rate actually used, and the threshold constant where applicable.

3.15.4. Manifest

A machine-readable manifest is emitted at the end of the grid, recording: the corpus provenance and whether the synthetic fallback was used; the tokenizer and vocabulary size; the software and hardware environment; the complete run configuration; the determinism mode; the measured LSUV degeneracy statistics; the realized ρ_0 distribution; the plateau-verification rate; the divergence count; the budget-sufficiency verdict and the licensed wording it entails; the retention coefficients for the treatment and placebo arms; the operative statistical verdicts with effect sizes and intervals; and a hash over the whole manifest, serialized with sorted keys, for provenance. Console output carrying the hypothesis verdicts is mirrored to persistent storage, since the grid spans several sessions and those verdicts are the primary artefact.

3.15.5. Software and hardware

All experiments were conducted on a single NVIDIA A100 (compute capability 8.0, 40 GB HBM2e). At install time the harness pins Transformers $\geq$ 4.49, Datasets $\geq$ 3.2, Accelerate $\geq$ 1.3, SciPy $\geq$ 1.14, statsmodels $\geq$ 0.14, pandas $\geq$ 2.2, Matplotlib $\geq$ 3.9, tokenizers $\geq$ 0.20 and pynvml $\geq$ 12.0. PyTorch, CUDA and NumPy are taken from the host image rather than pinned by the harness; the resolved versions are recorded in the manifest, and the reported grid was executed under Python 3.13.15, PyTorch 2.11.0 and CUDA 12.8. Where statsmodels is unavailable, the mixed-effects analysis degrades to a documented descriptive fallback rather than failing silently. The complete experimental harness, including the run store, the manifest generator and the figure and table export pipeline, is released with the paper.

3.16. Computational cost of the interventions

Because the paper recommends one intervention and declines to recommend another, the cost of each is stated explicitly.

Asymptotic cost: For a weight tensor of $n = d_{\text{out}} d_{\text{in}}$ entries, Algorithm 1 draws three uniform vectors, performs a constant number of elementwise operations and two reductions, and computes one verification pass; it is therefore $\Theta(n)$ in time and $\Theta(n)$ in auxiliary memory, executed once per tensor at step 0. The threshold-offset constant $c^\star(\rho_0)$ of Eq. (19) is obtained by 200 bisection steps on a scalar function, i.e. $O(1)$ work computed once per run and cached. The γ-warmup schedule of Eq. (17) evaluates one affine expression per optimizer step and writes a zero-dimensional tensor, again $O(1)$. None of the three interventions alters the asymptotic cost of the forward or backward pass, which remains that of the underlying BitLinear layer.

Measured cost: All initialization work is charged to the arm that incurs it: the per-run wall-clock timer starts before model construction, so the TAI construction and its verification pass are counted against the method rather than treated as free. Initialization is a single $\Theta(n)$ pass set against $\Theta(n)$ work repeated at each of the 1,221 optimizer steps, so its share of the total is negligible by construction rather than by separate measurement; the harness records the run total only. The arms are therefore separated in wall-clock only by the number of optimizer steps they require. The γ-warmup arm is consequently cheaper than the baseline in wall-clock terms at a matched target loss, since it reaches that target in fewer steps (Section 4.5); the token-equal protocol nevertheless runs every arm for the identical number of steps, so no arm receives a compute advantage in the reported comparisons.

Cost of the study: The complete experiment consumed 50.9 A100-hours over 3,016 persisted runs. The dominant single item is learning-rate calibration (1,462 runs at 300 steps each), a consequence of granting every (scale, method) cell its own $\eta^\star$ rather than sharing one rate; we regard this as a necessary cost of a fair comparison rather than overhead.

3.17. Deviations from the pre-registered protocol

In the interest of full disclosure, we record every deviation forced by measurement, together with the point in the experimental timeline at which it was made.

> *Convergence wording:* The budget was pre-registered as plateau-verified. The slope test of (26) certified plateau on

no run of the confirmatory grid; the validation trace continues to descend by approximately 0.2 nats over the final 40% of probes, uniformly across scales. The gate is functioning; the budget simply does not reach convergence. The protocol is therefore token-equal at a fixed budget, which is legitimate and applies identically to every arm, but H1 and H2 may not be worded as claims about converged quality. The budget itself was not changed: it is pre-registered, and moving it to chase a plateau would be precisely the post-hoc adjustment this design exists to prevent. This determination was recorded before the outcome statistics were computed.

H2 companion test: Because the convergence-conditional equivalence test is defined over plateau-certified pairs and that set is empty, the pre-registered test is reported as untestable. A separate, clearly labelled companion test of equivalence at the fixed token budget is reported over all paired seeds. The substitution is forced by a measurement rather than by an outcome, and it does not favour the proposed method.

H3 scope: The sweep establishes that the intervention is real, strictly ordered and highly reliable, but that the great majority of it is undone during training: the retention coefficient of (32) is small and the implied fixed point lies close to the value the locked baseline reaches unaided. The pre-registered strong form of H3, that the initial operating point shapes the final sparsity, is therefore not supported, and the reported claim is narrowed to the durable residue plus the existence of a strong attractor. This is a change in the reported quantity, not in the analysis: the sweep, the placebo arm and every test are unchanged.

Baseline breadth: `standard` and `lsuv` were initially described as two distinct locked-ρ_0 families, and an earlier draft of this methodology asserted on structural grounds that the LSUV calibration is degenerate under RMSNorm pre-normalization. The instrumentation of Remark 1 contradicts that assertion, and the corrected statement is reported in Section 4.2: the two arms share an operating point but are not the same intervention. No best-of-two contrast is taken over them in either case, and the mechanism-matched second family remains `gamma_offset`.

Compute-equal variant of H1: The wall-clock-equal variant of H1 requires re-running the entire confirmatory grid and is disabled by default under the stated budget. The token-equal formulation is the pre-registered primary; the compute-equal variant is available in the harness and can be enabled on request.

3.18. From protocol to evidence

The protocol specified above was fixed before the confirmatory grid was executed, and the deviations recorded in Section 3.17 are the complete list of what measurement subsequently forced. Section 4 reports the output of running it, grouped by the experiment that produces each hypothesis and using only the names and symbols fixed in Table 4. It opens by establishing that the grid executed as specified (scope, divergence, calibration), verifies the locked operating point on which the premise of the study rests, and then settles the single measurement on which the wording of every subsequent claim depends: whether the token budget reached convergence. That determination, and not the outcome of any hypothesis test, governs what the quality results are permitted to say, and it is therefore reported ahead of every hypothesis test. Section 5 supplies the mechanisms behind those findings and states the structural limitations of the design: the absence of a benchmarked ternary kernel, the token-equal rather than compute-optimal treatment of the upper ladder, and the restriction of all conclusions to the tested regime.

4. Results

Results are reported in the order set by the analysis plan of Section 3, which groups the hypotheses by the experiment that produces them rather than by hypothesis number. Section 4.1 establishes the scope and internal validity of the executed grid; Section 4.2 verifies the analytical premise; Section 4.3 determines the convergence status of the token budget and therefore the wording licensed for every quality claim that follows; Sections 4.4–4.7 report the confirmatory contrasts (H1, H2, H4); Sections 4.8–4.10 report the operating-point analysis, its pre-registered falsification, and the mechanism-attribution controls (H3, H5); Sections 4.11–4.14 report the external-validity analyses (H6, H7, H8) together with the mechanism-matched control at its default operating point; Sections 4.15–4.17 report downstream accuracy, the ablation suite and the efficiency model. Section 4.18 consolidates the outcome of every pre-registered hypothesis.

Throughout, the paired quality contrast is defined as

$$\Delta\mathcal{L} \;=\; \mathcal{L}^{(\beta)} - \mathcal{L}^{(m)}, \qquad \Delta S \;=\; S^{(\beta)}_{\text{target}} - S^{(m)}_{\text{target}}, \tag{34}$$

where β denotes the pre-registered primary baseline (`bitnet158`) and m the method under test, so that positive values favour the method. Losses are held-out cross-entropy in nats per token; sparsity is the mean fraction of ternary zeros over all BitLinear layers measured under the live threshold in force at the end of training. Unless stated otherwise, all intervals are 95% and all p-values are two-sided.

4.1. Scope, validity and internal consistency of the executed grid

The complete experiment comprises **3,016** persisted run records distributed over the phases listed in Table 6, consuming **50.9** A100-hours in aggregate. Of the 823 records in the `main` phase, 790 enter the confirmatory and exploratory analyses; the remaining 33 are the three calibration-seed runs per scale (3×11) that the protocol reuses as `standard` results (Section 3.9) and that are excluded from every reported contrast, since their seeds participated in learning-rate selection. All analysed `main`-phase runs use the full 1,221-step budget of

Table 6: Inventory of persisted run records by experimental phase. Learning-rate calibration runs are truncated to 300 steps; all others use the full 1,221-step budget.

Phase	Records	Purpose
`lrcal`	1,462	Per-arm learning-rate calibration (300 steps)
`main`	823	Confirmatory + exploratory scale ladder
`rho0`	360	ρ_0 sweep, two arms × two scales
`rho0neg`	90	H3 placebo (label-decoupled) arm
`arch2`	90	Cross-architecture family B
`wc`	60	Depth-controlled (width-only) ladder
`lr`	50	Learning-rate sensitivity ablation
`warmup`	45	γ-warmup (s_0, τ_w) grid
`taihp`	20	TAI (δ, a) hyperparameter ablation
`budget`	6	2× token-budget sufficiency check
`lrproxy`	5	Full-length validation of the calibration proxy
`fp16abl`	5	Full-precision arm with TAI initialization
Total	**3,016**	

Eq. (23). All runs were executed on a single NVIDIA A100-SXM4-40GB under Python 3.13.15, PyTorch 2.11.0+cu128 and CUDA 12.8. The opt-in `dim2048` ladder point was not enabled, so the executed ladder comprises the eleven scales of Table 1 up to `dim1536`; no result below depends on it.

Four internal-validity conditions specified in advance were met. (i) Corpus provenance. The real FineWeb-EDU corpus was used (48,000,000 GPT-2 BPE tokens, vocabulary 50,257); the synthetic fallback was not triggered, and the downstream benchmarks were therefore scored rather than suppressed. (ii) Divergence. Zero of the 790 analysed runs met the divergence criterion of Eq. (27) ($\mathcal{L} > 1.25 \log|\mathcal{V}| = 13.53$ nats, non-finite loss, or a late 1.5× rise in the validation trace); no run was excluded from any analysis on stability grounds. (iii) Learning-rate calibration. All 88 (scale, method) cells produced an interior optimum after adaptive grid extension; the boundary-optimum list is empty. The short-calibration proxy was validated at the primary scale, where the 300-step and full-length curves select the identical optimum ($\eta^\star = 3\times10^{-3}$) and are rank identical (Spearman = 1.000). (iv) Reproducibility class. The grid was executed in throughput mode (flash attention, TF32, chunked cross-entropy, compiled step), so the manifest records the run as statistically reproducible at seed level, identical initialization and data order per seed, rather than bit-reproducible.

One protocol defect is recorded here because it bears on the interpretation of five scales. The per-run wall-clock record resolves the arms at `dim512`, `dim640`, `dim1024` and `dim1280` into two groups differing by a factor of 1.9 to 2.6 in run time for arithmetically comparable work, indicating that the compiled execution path was obtained for some arms and not others. The primary scale `dim384` is affected in a narrower way, by a factor of 1.65: there the fast group contains `tai_star` alone, so every contrast among the remaining seven arms, the headline contrast included, is drawn within one execution group, and H8 is the single comparison that crosses groups. Across the six unaffected scales the largest spread is 1.28 (`dim192`). Section 5.10 diagnoses the cause, quantifies its effect, and shows that the primary result is unchanged when the four worst-affected scales are excluded.

The selected learning rates decline monotonically with scale, from 5×10^{-3} at the smallest ladder point to 1×10^{-3} at `dim1536`. One systematic asymmetry is recorded here because it bears on the interpretation of the full-precision control in Section 4.4: at five of the eleven ladder points (`dim512`, `dim768`, `dim896`, `dim1024`, `dim1536`) the `fp16_control` arm selects a strictly smaller $\eta^\star$ than the ternary arms, for example 5×10^{-4} versus 1×10^{-3} at `dim1536`; at the remaining six the selected rates coincide. The control's calibration optimum is interior at all eleven scales: at `dim1536` the adaptive extension added a 3×10^{-4} point, which is worse (6.114) than both 5×10^{-4} (5.966) and 1×10^{-3} (5.996), so the optimum is bracketed rather than truncated by the grid.

4.2. Verification of the locked ternary operating point

Proposition 1 predicts that the initial active fraction of an absmean ternary layer under any zero-mean Gaussian initializer is $\rho_0 = 0.6899$, independently of σ. Table 7 reports the two verification routes.

The empirical active fraction varies by at most 6×10^{-4} over two orders of magnitude in σ, and all four initializers, including LSUV after completion of its data-driven pass, land within 10^{-3} of the analytical value. The premise of the study is therefore confirmed: no standard initializer exercises any control over the initial operating point.

Two measurements settle the question left open by Remark 1 and qualify the baseline-family description. First, the LSUV calibration pass is not inert in this architecture: it performed 36 rescaling operations, exactly one per BitLinear layer, against a maximum absolute deviation of layer output variance from unity of 0.586 encountered at initialization. The structural argument that RMSNorm pre-normalization makes the calibration degenerate is therefore not borne out, which is why the protocol instrumented it rather than assuming it. Second, and consistent with this, `lsuv` attains a lower held-out loss than `standard` at nine of the eleven scales, by between 0.007 and 0.051 nats, and is the strongest non-TAI ternary arm at six of them (Table 9 and Fig. 2). The ordering reverses at the two largest ladder points, where `lsuv` is higher by 0.013 (`dim1280`) and 0.004 nats (`dim1536`); both differences fall within the implementation noise floor established in Section 5.10. The two arms are therefore indistinguishable in operating point ($\rho_0 = 0.689$ versus 0.690) but not in quality. That the calibration rescaled every layer and still left the operating point at 0.690 is itself a model-level demonstration of Proposition 1: an intervention that changed σ layer by layer moved ρ_0 by at most 10^{-3}.

4.3. Convergence status of the token budget

The plateau criterion of Eq. (26) certified convergence on **0 of 760** runs of the eight pre-registered arms, which are the runs the gate is evaluated on; the 30 `tai_star` runs enter the analysis after this pass and the remaining 33 are the excluded calibration-seed records. The median fitted tail descent over the final 40% of the validation trace is −0.225 nats, i.e. every arm is still improving measurably when the budget expires. Recomputation across the robustness grid (Table 8) shows this is not an

Table 7: Verification of σ-independence. Left: empirical initial active fraction of 10^6 Gaussian variates at five values of σ, under the canonical absmean threshold. Right: model-level initial active fraction at `dim384`, averaged over all 36 BitLinear layers, for four standard initializers. The analytical prediction of Eq. (10) is 0.6899. The pre-registered locked tolerance is $|\rho - 0.69| < 0.02$.

σ	Active fraction	Initializer	Active fraction	Verdict
0.02	0.6900	`standard`	0.689	locked
0.10	0.6896	`xavier`	0.689	locked
0.50	0.6902	`kaiming`	0.689	locked
1.00	0.6896	`lsuv`	0.690	locked
2.00	0.6900			
Analytical: 0.6899; spread 6×10^{-4}		All initializers within 10^{-3} of prediction		

Table 8: Plateau-certified fraction of the 790 main-grid runs that carry a validation trace, under the slope gate of Eq. (26), recomputed across the tolerance ε_p and the tail fraction. Recompute-only; no additional training.

$\varepsilon_p \setminus$ tail frac	0.30	0.40	0.50
0.005	0.00	0.00	0.00
0.010	0.00	0.00	0.00
0.020	0.03	0.00	0.00

artefact of the criterion's operationalization: the certified fraction is 0.00 in eight of nine (ε_p, frac) cells and 0.03 in the single most permissive cell.

The independent 2× budget check confirms the verdict directly. Doubling the token budget to 40 M at the primary scale improves held-out loss by 0.315 ± 0.009 nats for `bitnet158` ($n = 3$) and by 0.287 ± 0.001 nats for `tai_warmup` ($n = 3$), between 14 and 16 times the equivalence margin of 0.02 nats used in the check.

Under the decision rule fixed in advance (Section 3.10), two consequences follow and are applied without exception in the remainder of this section. First, the pre-registered form of H2 (an equivalence claim about converged quality, evaluated over seeds plateau-certified in both arms) is **untestable**, the qualifying set being empty; it is reported as unresolved rather than as a null result. Second, every quality statement below is a statement about performance at a fixed 20 M-token budget, and the word *converged* is not used of any result. The clearly labelled token-equal companion test defined in Section 3.13 is reported in its place.

4.4. Main grid: quality at a fixed token budget

Table 9 reports the complete arm set at the primary confirmatory scale. Figure 1 shows the corresponding loss trajectories and Fig. 2 the paired contrast against the pre-registered baseline at every ladder point.

Three orderings organize the results. The first two are reproduced at every ladder point (Fig. 2).

(i) `tai_warmup` **attains the lowest loss of any arm.** At the primary scale it improves on the pre-registered baseline by 0.168 nats and on the full-precision control by 0.235 nats.

(ii) `tai` **attains a higher loss than every locked-ρ_0 ternary baseline.** At the primary scale it is 0.080 nats above `bitnet158` and 0.116 nats above `lsuv`. The two variants of the proposed method therefore lie on opposite sides of the baseline, and `tai_star` (Section 4.14) tracks `tai`.

(iii) **The gradual-pruning baseline** `sparse_sched` **is the weakest ternary baseline at the primary scale** (5.1440 nats) while attaining by far the highest sparsity (48.8%), and `gamma_offset` at its grid-default $\rho_0 = 0.75$ sits between the locked baselines and `tai`.

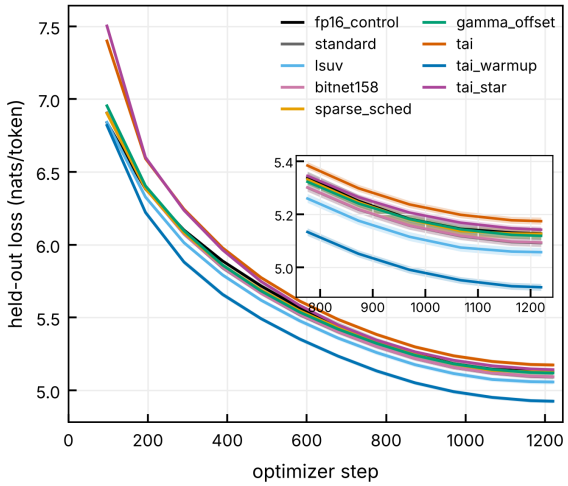


Figure 1: Held-out loss against optimization step for each arm at the primary confirmatory scale (`dim384`), averaged over 15 seeds at a matched 20 M-token budget. Shaded bands are bootstrap 95% confidence intervals.

Table 9: Complete method set at the primary confirmatory scale (`dim384`, 8.26 M non-embedding parameters, $n = 15$ seeds, 20 M-token budget). $\mathcal{L}$: held-out cross-entropy (nats/token); CI: half-width of the 95% confidence interval of the mean (Student t, $n = 15$); s: standard deviation across seeds; sp_T: final ternary sparsity; S_{target}: steps to reach the calibrated target loss of 5.6539 nats; a_{nt}: next-token top-1 accuracy. The pre-registered primary baseline is `bitnet158`.

Arm	$\mathcal{L}$	CI	s	sp_T	S_{target}	a_{nt}
`fp16_control`	5.1785	0.0248	0.0447	0.0%	564	0.216
`standard`	5.1094	0.0170	0.0307	32.1%	552	0.227
`lsuv`	5.0752	0.0185	0.0334	31.8%	524	0.230
`bitnet158`	5.1114	0.0161	0.0291	32.1%	555	0.227
`sparse_sched`	5.1440	0.0205	0.0370	48.8%	564	0.224
`gamma_offset`	5.1365	0.0143	0.0258	26.0%	560	0.224
`tai`	5.1914	0.0150	0.0271	30.0%	613	0.221
`tai_warmup`	**4.9437**	0.0150	0.0272	30.3%	**448**	**0.242**
`tai_star`	5.1919	0.0135	0.0244	28.4%	589	0.218

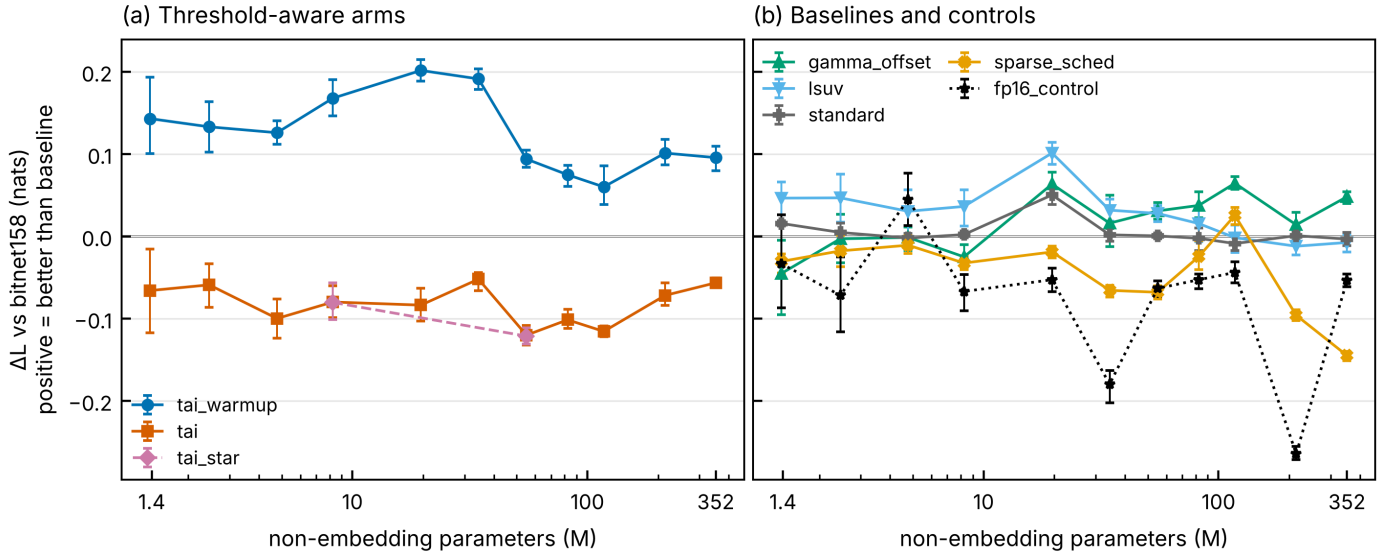


Figure 2: Paired quality contrast against the pre-registered baseline across the eleven-point ladder. Each point is $\Delta\mathcal{L} = \mathcal{L}(\texttt{bitnet158}) - \mathcal{L}(\text{arm})$, averaged over the seeds the two arms share at that scale, so positive values favour the arm; bars are bootstrap 95% confidence intervals over seeds ($n = 15$ at the four confirmatory scales, $n = 5$ elsewhere). Panel (a): the threshold-aware arms, with `tai_warmup` above the baseline at every scale and `tai` below it at every scale. Panel (b): the remaining baselines and the full-precision control. `tai_star` was executed at `dim384` and `dim768` only. Absolute losses at the primary scale are in Table 9; the differences are plotted here because they span 0.26 nats against absolute losses near 5.

A result that requires explicit statement is that the ternary arms do not sit above the full-precision control at this budget. That control is named `fp16_control` after its harness identifier but runs in `bfloat16` throughout (Table 3). Taking the strongest non-TAI ternary arm at each scale, we find that ternary training attains the lower loss at ten of the eleven ladder points, by between 0.080 and 0.278 nats; the sole exception is `dim320`, where `bitnet158` is 0.045 nats above `fp16_control`. Two properties of the protocol are recorded alongside this observation without further interpretation, which is deferred to Section 5: the budget does not reach convergence for any arm (Section 4.3), and the full-precision arm selects a strictly smaller calibrated learning rate than the ternary arms at five of the eleven scales, though its optimum is bracketed by the grid at all of them (Section 4.1).

4.5. *H1: convergence speed at a matched target loss*

Steps-to-target is measured against the calibrated target of 5.6539 nats, fixed before the grid on the disjoint calibration seeds. The results are reported in the ΔS column of Table 10.

`tai_warmup` reaches the target in fewer steps than the primary baseline at all eleven scales, with ΔS ranging from +28 steps (`dim1280`) to +148 steps (`dim512`). Every statistic in this subsection is computed on S_{target} and is therefore distinct from the held-out-loss columns of Table 10: Cohen's d on steps lies between +2.39 and +7.95, all eleven paired t-tests reject at $\alpha = 0.05$, and ten of eleven reject at the standalone level $\alpha_0 = 0.0125$ fixed in the design, the exception being `dim192` ($p = 0.0138$).

`tai` requires more steps than the baseline at all eleven scales, with ΔS between −48 and −100 and steps-based d between −1.52 and −4.47; ten of eleven contrasts are significant at $\alpha = 0.05$ (`dim192`: $p = 0.1248$).

H1 as pre-registered is therefore supported for the schedule variant and rejected in the opposite direction for the fixed-ρ_0 variant.

4.6. *H2: quality at a fixed token budget*

The pre-registered convergence-conditional test is untestable (Section 4.3). The token-equal companion test, defined over all paired seeds against the data-derived margin $\Delta_{\text{eq}} = 0.10\,|\bar{\mathcal{L}}^{(\beta)} - \bar{\mathcal{L}}^{(\text{fp16})}|$, is reported in Table 10.

The bootstrap interval excludes zero in all 24 cells, so the sign of $\Delta\mathcal{L}$ is resolved throughout. The one cell whose paired t-test does not reach $\alpha = 0.05$ is `tai` at `dim192` ($n = 5$, $p_t = 0.0922$), which is why its verdict is inconclusive. The companion test never returns equivalence: Δ_{eq} ranges from 0.0033 to 0.0264 nats while $|\Delta\mathcal{L}|$ ranges from 0.052 to 0.201 nats, i.e. every observed difference exceeds its margin by between 2.7× and 42.9×. The verdicts are accordingly categorical: `tai_warmup` is superior to the baseline at all eleven scales; `tai` is inferior at ten of eleven and inconclusive at the eleventh.

Holm–Bonferroni correction within the confirmatory held-out-loss family, at family-wise level $\alpha_H = 0.05$, rejects the null for both TAI variants at every scale except `tai` at `dim192`, which is retained. The Wilcoxon signed-rank test, computed wherever $n \geq 6$, supports the parametric verdict in every case ($p \leq 0.0009$).

The secondary sensitivity contrast against the data-selected minimum baseline attenuates the `tai_warmup` advantage, for example at `dim512`, $\Delta\mathcal{L}$ falls from +0.201 (versus `bitnet158`) to +0.100 (versus `lsuv`), and at `dim768` from +0.094 to +0.063 (versus `gamma_offset`). The sign survives at ten of the eleven

Table 10: Cross-scale paired contrasts against the pre-registered primary baseline `bitnet158`. $\Delta\mathcal{L}$ and ΔS follow Eq. (34) (positive favours the method). CI_{95} is the nonparametric bootstrap interval on $\Delta\mathcal{L}$ over 5,000 resamples; d is Cohen's d; p_t is the paired t-test p-value; p_{Lev} is Levene's test for H4; Δ_{eq} is the data-derived equivalence margin, one value per scale. The verdict column is the token-equal companion decision. Confirmatory scales are set in bold. n is the number of paired seeds.

Scale	Arm	n	$\Delta\mathcal{L}$	CI_{95}	d	p_t	ΔS	p_{Lev}	Δ_{eq}	Verdict
`dim192`	`tai`	5	−0.0661	[−0.117, −0.016]	−1.83	0.0922	−88	0.615	0.0033	inconclusive
	`tai_warmup`	5	+0.1428	[+0.101, +0.191]	+4.33	0.0051	+108	0.285		better
`dim256`	`tai`	15	−0.0591	[−0.086, −0.033]	−1.45	0.0008	−63	0.036	0.0072	worse
	`tai_warmup`	15	+0.1329	[+0.102, +0.164]	+2.88	$< 10^{-4}$	+101	0.447		better
`dim320`	`tai`	5	−0.1000	[−0.124, −0.076]	−4.79	0.0021	−100	0.996	0.0045	worse
	`tai_warmup`	5	+0.1257	[+0.111, +0.140]	+5.31	0.0001	+92	0.707		better
`dim384`	`tai`	15	−0.0800	[−0.099, −0.060]	−2.84	$< 10^{-4}$	−59	0.648	0.0067	worse
	`tai_warmup`	15	+0.1677	[+0.146, +0.190]	+5.95	$< 10^{-4}$	+107	0.889		better
	`tai_star`	15	−0.0805	[−0.102, −0.057]	−3.00	$< 10^{-4}$	−35	0.532		worse
`dim512`	`tai`	15	−0.0838	[−0.104, −0.064]	−3.59	$< 10^{-4}$	−71	0.175	0.0053	worse
	`tai_warmup`	15	+0.2014	[+0.188, +0.214]	+11.26	$< 10^{-4}$	+148	0.347		better
`dim640`	`tai`	5	−0.0522	[−0.066, −0.044]	−5.12	0.0016	−48	0.871	0.0180	worse
	`tai_warmup`	5	+0.1911	[+0.178, +0.204]	+13.01	$< 10^{-4}$	+104	0.633		better
`dim768`	`tai`	15	−0.1203	[−0.132, −0.109]	−6.35	$< 10^{-4}$	−95	0.957	0.0063	worse
	`tai_warmup`	15	+0.0937	[+0.084, +0.105]	+5.32	$< 10^{-4}$	+53	0.593		better
	`tai_star`	15	−0.1214	[−0.131, −0.111]	−7.23	$< 10^{-4}$	−95	0.461		worse
`dim896`	`tai`	5	−0.1016	[−0.112, −0.091]	−6.63	0.0001	−72	0.360	0.0053	worse
	`tai_warmup`	5	+0.0746	[+0.061, +0.086]	+6.38	0.0006	+40	0.819		better
`dim1024`	`tai`	5	−0.1157	[−0.122, −0.109]	−12.12	$< 10^{-4}$	−100	0.334	0.0044	worse
	`tai_warmup`	5	+0.0599	[+0.039, +0.085]	+3.48	0.0102	+32	0.621		better
`dim1280`	`tai`	5	−0.0720	[−0.084, −0.057]	−5.75	0.0008	−72	0.389	0.0264	worse
	`tai_warmup`	5	+0.1010	[+0.087, +0.116]	+8.99	0.0003	+28	0.163		better
`dim1536`	`tai`	5	−0.0566	[−0.062, −0.051]	−5.81	0.0001	−68	0.385	0.0054	worse
	`tai_warmup`	5	+0.0954	[+0.079, +0.110]	+8.27	0.0004	+36	0.955		better

Table 11: Variance decomposition for H4 at the primary scale (`dim384`, 15 seeds × 5 held-out folds). Mean $\mathcal{L}$ is the grand mean over all five folds and therefore differs from the fold-0 value of Table 9. Levene's test on the pooled matrix is the pre-registered statistic.

Arm	Mean $\mathcal{L}$	s across seeds	s across folds	s pooled
`tai`	5.2113	0.0272	0.0328	0.0397
`bitnet158`	5.1324	0.0309	0.0339	0.0429

Levene, pooled seed × fold: $W = 0.554$, $p = 0.458$
Levene, seed only: $W = 0.343$, $p = 0.563$

scales and reverses at `dim1024`, where `gamma_offset` is the strongest baseline and $\Delta\mathcal{L} = -0.0044$. This contrast is exploratory and carries no family-wise correction.

4.7. H4: outcome variance

Levene's test finds no evidence that either TAI variant inflates variance relative to the baseline. Across the 24 cells of Table 10, p_{Lev} falls below 0.05 in exactly one (`dim256`/`tai`, $p = 0.036$). That cell does not bear on H4 in the direction the hypothesis concerns: the TAI standard deviation there is less than half the baseline's (0.0237 versus 0.0525), so what the test detects is a reduction in spread, not inflation. It also does not survive the pre-registered per-hypothesis level $\alpha_0 = 0.0125$.

The pre-registered form of H4 is evaluated on the pooled seed × fold matrix, in which every trained model is scored on all $K = 5$ held-out sub-folds. Table 11 reports the decomposition at the primary scale.

H4 is supported: there is no evidence of variance inflation ($p = 0.458$, clearing both the conventional 0.05 and the pre-registered $\alpha_0 = 0.0125$), and this holds whether or not data-sampling spread is included. A secondary observation from the decomposition is that the dominant variance component is the evaluation fold rather than the seed (mean s across folds 0.0334 versus across seeds 0.0290), for both arms.

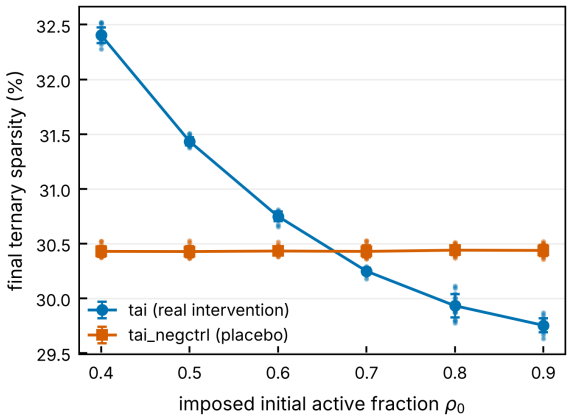


Figure 3: Final ternary sparsity, in per cent, as a function of the imposed initial active fraction ρ_0 at the primary scale (`dim384`), for the `tai` arm and for the ρ_0-labelled placebo `tai_negctrl`. Every one of the 15 seeds per point is shown, with the mean and one standard deviation. The placebo holds the realized threshold partition fixed at $\rho_0 = 0.69$ while varying only the label, so any dependence it exhibits is measurement artefact.

4.8. H3: control of the ternary operating point and its retention

4.8.1. Ordinal control

The rank association between ρ_0 and final sparsity is essentially perfect and negative, as predicted (Fig. 3): a larger initial active fraction implies fewer terminal zeros. At the primary scale the pooled Spearman coefficient is −0.983 ($p = 4.3 \times 10^{-66}$) and the mean per-seed coefficient is −0.996, 95% CI [−1.000, −0.989] over 15 seeds, with 93% of seeds strictly monotone. The intervention is far above the nuisance floor:

the across-ρ_0 range of final sparsity is 0.0265 against a within-ρ_0 across-seed standard deviation of 0.00059, a signal-to-noise ratio of 44.9×. The control is statistically indistinguishable across scale (`dim384` mean -0.996 ± 0.015 versus `dim768` -1.000 ± 0.000; Welch $t = +1.00$, $p = 0.334$).

4.8.2. Retention: how much of the intervention survives training

The ordinal result establishes that the ordering survives; it does not establish that the magnitude does. Regressing final on initial sparsity per Eq. (32) gives the retention coefficients of Table 12, which is the principal quantitative result of this section.

The two mechanisms behave qualitatively differently. For the TAI arm, a 50-point swing in initial sparsity produces a final swing of 2.65 points at `dim384` and 3.55 points at `dim768`: retention is 0.052–0.069, so approximately 93–95% of the imposed intervention is erased during training. The residual is nonetheless perfectly ordered and highly reproducible, the within-cell standard deviation being $3\text{–}11 \times 10^{-4}$. The implied fixed point is 0.305 at `dim384` and 0.300 at `dim768`; for reference, the `standard` arm reaches 0.321 and 0.318 at those scales without any intervention.

For the γ-offset arm, retention is 1.003 and 1.006 with $R^2 = 1.000$ at both scales: the final operating point tracks the target one-for-one across the entire 50-point range (Table 13). This arm reaches its operating point by holding the threshold scale at the constant $c^\star(\rho_0)$ of Eq. (19), whose calibrated values over the grid are 2.110, 1.691, 1.314, 0.966, 0.635 and 0.315 for $\rho_0 = 0.4, \ldots, 0.9$.

H3 as pre-registered, that the initial operating point causally shapes the final learned sparsity, is therefore **not supported in its strong form for the initialization mechanism**, and is supported almost exactly for the threshold-offset mechanism. The operative statement for the initialization variant is a bounded residual effect with the retention fraction reported.

4.8.3. Pre-registered falsification

The placebo arm `tai_negctrl`, which varies the ρ_0 label while holding the realized partition fixed at 0.69 (Fig. 4), produces a final sparsity of 0.3043–0.3044 across all six labels ($n = 15$ per label): a range of 1.3×10^{-4}, against 2.65×10^{-2} for the real intervention, a factor of 212. The placebo's mean per-seed Spearman coefficient is +0.135, generated entirely by fourth-decimal noise and of opposite sign to the real effect (−0.996). The pre-registered sign test records the real coefficient as exceeding the placebo coefficient in 0 of 15 seeds ($p = 6.1 \times 10^{-5}$); because the test was specified as a one-sided comparison of signed coefficients while the true effect is negative, this statistic is reported here for completeness and the magnitude comparison above is the interpretable falsification result. The placebo is flat, as predicted, and the sparsity effect is therefore not an artefact of the analysis.

4.9. H5: the quality-sparsity frontier

Table 13 reports the full sweep for both arms at both scales.

For the initialization mechanism the loss decreases monotonically in ρ_0 over the whole grid, from 5.2519 at $\rho_0 = 0.4$ to 5.1555 at $\rho_0 = 0.9$ (`dim384`), while final sparsity varies by only 2.6 points; the frontier is consequently near-degenerate, and all six points are formally non-dominated only because the sparsity differences, though tiny, are strictly ordered. Under the pre-registered selection rule, minimum loss subject to $\mathrm{sp}_T \geq 30\%$, the optimum is $\rho_0^\star = 0.70$ ($\mathcal{L} = 5.1875$, $\mathrm{sp}_T = 30.2\%$), with bootstrap selection probabilities $P(0.60) = 0.237$, $P(0.70) = 0.758$ and ≤ 0.003 elsewhere.

For the threshold-offset mechanism the loss is U-shaped with an interior optimum, and the sparsity axis spans 50 points (Fig. 5). At `dim384` the minimum is at $\rho_0 = 0.70$ ($\mathcal{L} = 5.0724$ at 31.1% sparsity); at `dim768` it is at $\rho_0 = 0.80$ ($\mathcal{L} = 4.8968$ at 20.7% sparsity), where the loss is 0.011 nats below the same arm's $\rho_0 = 0.70$ point and 0.020 nats below the `dim768` `bitnet158` baseline (4.9167) at two-thirds of its sparsity.

One point associated with this arm does not lie on the curve its neighbours describe, and is recorded here. The main grid runs `gamma_offset` at the default $\rho_0 = 0.75$, which the sweep grid does not contain; at `dim384` that point attains 5.1365, above both $\rho_0 = 0.70$ (5.0724) and $\rho_0 = 0.80$ (5.1000), by some five standard errors of the 15-seed mean. It is not a configuration error: the realized sparsity interpolates its two neighbours to within 10^{-4}, and the calibrated learning rate and the threshold constant $c^\star(0.75)$ are the values the protocol specifies. The sweep and the main grid do, however, fall on opposite sides of the execution-path split of Section 4.1. We record the discrepancy rather than resolve it. Contrasts drawn within a single phase are unaffected, including the main-grid comparison of this arm against the baseline in Section 4.12; what the discrepancy rules out is treating the sweep and the main grid as interchangeable measurements of the same arm.

H5 is therefore supported for the threshold-offset mechanism, which traces a genuine frontier, and is uninformative for the initialization mechanism, whose sparsity axis is effectively fixed.

4.10. Mechanism attribution

4.10.1. Initialization versus threshold offset

Paired over all 90 (ρ_0, seed) cells at the primary scale, the initialization mechanism attains a loss 0.0610 nats higher than the threshold-offset mechanism, 95% CI [+0.0474, +0.0742], paired t-test $p = 1.06 \times 10^{-13}$ (Fig. 6). Both arms control the operating point ordinally with essentially equal reliability (mean per-seed Spearman −1.000 for `gamma_offset`, −0.996 for `tai`), but only the threshold-offset arm retains the magnitude (Table 12). Of the three pre-registered readings of this contrast (Section 3.5), the outcome is the third: the simpler mechanism wins, on both quality and retention. This contrast is exploratory and carries no family-wise correction.

4.10.2. The 2×2 initialization × precision ablation

The {full precision, ternary} × {standard init, TAI init} design at the primary scale ($n = 5$ seeds) is assembled from three arms already in the grid (`fp16_control`, `standard` and `tai`) plus

Table 12: Retention of the operating-point intervention, $\mathrm{sp}_T = \mathrm{sp}_\infty + r\,\mathrm{sp}_0$, over the six-point ρ_0 grid at 15 seeds per point. A 50-percentage-point swing in initial sparsity is imposed in every cell. r: retention coefficient; final swing: the resulting range of final sparsity.

Scale	Arm	r	R^2	Final swing	Implied fixed point
`dim384`	`tai`	0.052	0.931	2.65 pt	0.305
`dim384`	`gamma_offset`	1.003	1.000	50.09 pt	n/a
`dim768`	`tai`	0.069	0.959	3.55 pt	0.300
`dim768`	`gamma_offset`	1.006	1.000	50.33 pt	n/a

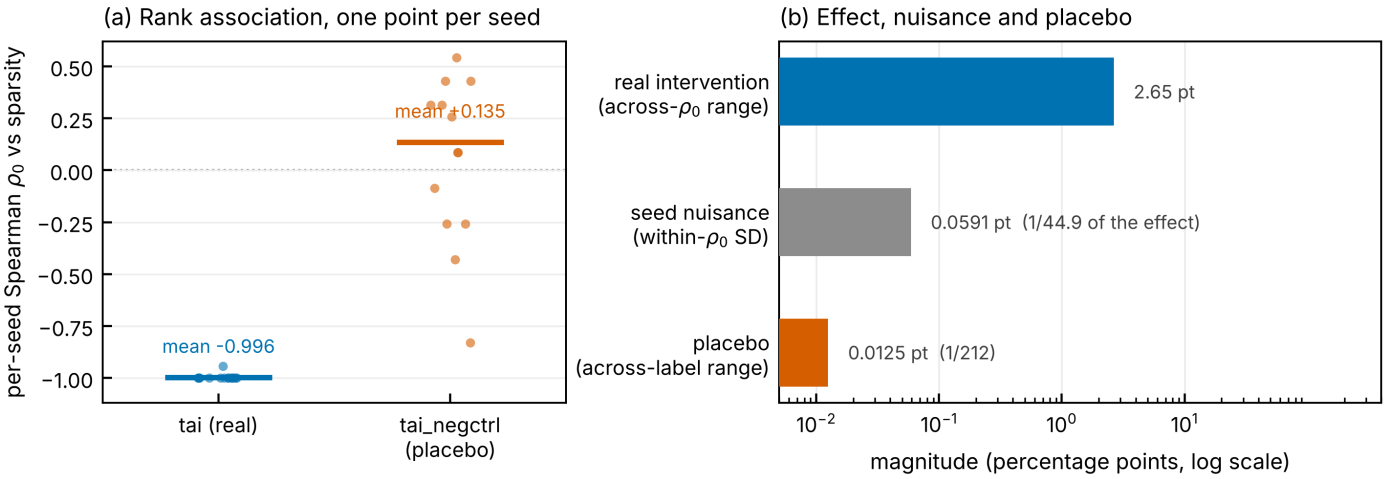


Figure 4: Pre-registered falsification instrument for H3; the sweep itself is Fig. 3. Panel (a): the per-seed Spearman coefficient between the imposed ρ_0 label and final sparsity, one point per seed, for the real intervention and for the placebo `tai_negctrl`. Panel (b): the across-ρ_0 effect, the within-ρ_0 seed nuisance and the placebo's across-label range on one logarithmic axis, so that both ratios quoted in the text are legible at once. The signed sign test of the pre-registration (0 of 15, $p = 6.1 \times 10^{-5}$) is not plotted, for the reason given in the text.

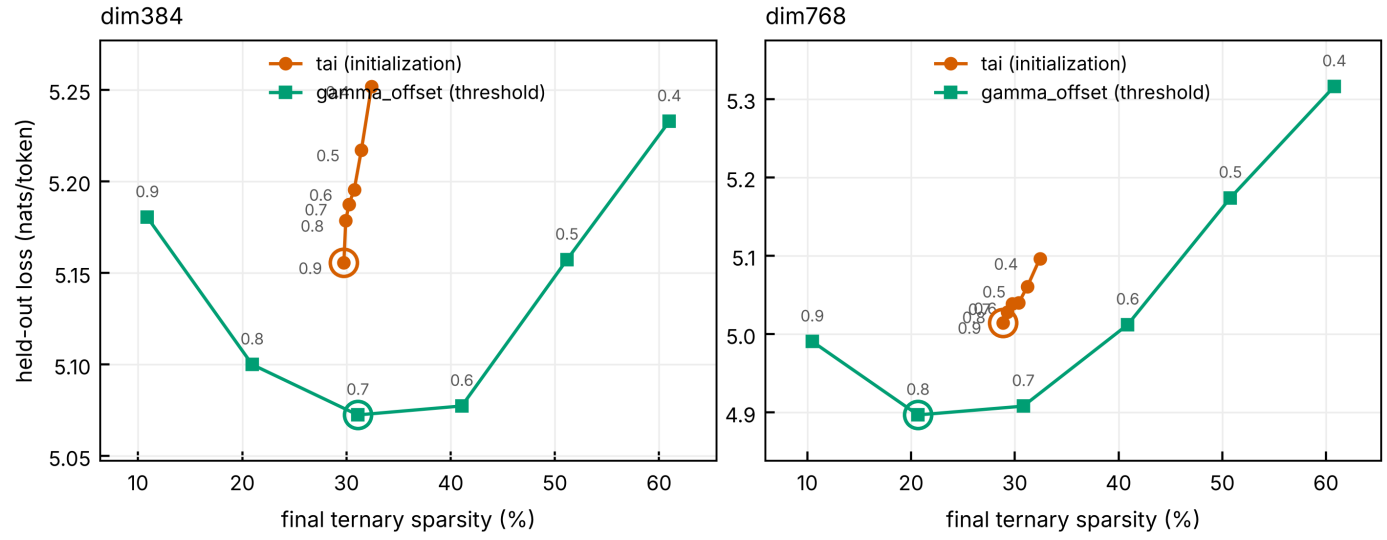


Figure 5: Held-out loss, in nats per token, against final ternary sparsity, in per cent, across the ρ_0 sweep, for both mechanisms and at both swept scales. Each point is one operating point, labelled with its ρ_0 and averaged over 15 seeds; the ringed marker is each arm's minimum. The threshold-offset arm traces a U-shaped frontier spanning 50 points of sparsity with an interior optimum, while the initialization arm's sparsity axis spans under four points, rendering its frontier near-degenerate.

one arm run specifically for it, `fp16_taiinit`, which applies the TAI initializer with quantization disabled. The design yields the initialization effect $\mathcal{L}^{\mathrm{std}} - \mathcal{L}^{\mathrm{TAI}}$ reported in Table 14 and shown in Fig. 7.

The TAI initialization distribution carries a penalty of 0.086 nats in full precision and 0.090 nats under ternary quantization. The two effects are of the same sign, of nearly equal magnitude, and both significant at $\alpha = 0.05$; the difference between them, 0.0038 nats, is an order of magnitude smaller than either, and the paired interaction contrast does not resolve a difference

Table 13: Operating-point sweep. Held-out loss (mean ± SD over 15 seeds) and final ternary sparsity for the initialization mechanism (`tai`) and the threshold-offset mechanism (`gamma_offset`) at two scales. The minimum-loss point of each column is set in bold.

	dim384				dim768			
	`tai`		`gamma_offset`		`tai`		`gamma_offset`	
ρ_0	$\mathcal{L}$	sp_T	$\mathcal{L}$	sp_T	$\mathcal{L}$	sp_T	$\mathcal{L}$	sp_T
0.4	5.2519	32.4%	5.2328	61.0%	5.0963	32.4%	5.3160	60.8%
0.5	5.2171	31.4%	5.1572	51.2%	5.0609	31.2%	5.1739	50.8%
0.6	5.1955	30.7%	5.0774	41.1%	5.0397	30.4%	5.0120	40.9%
0.7	5.1875	30.2%	**5.0724**	31.1%	5.0386	29.8%	4.9080	30.8%
0.8	5.1786	29.9%	5.1000	21.0%	5.0282	29.3%	**4.8968**	20.7%
0.9	**5.1555**	29.8%	5.1804	10.9%	**5.0146**	28.9%	4.9905	10.5%

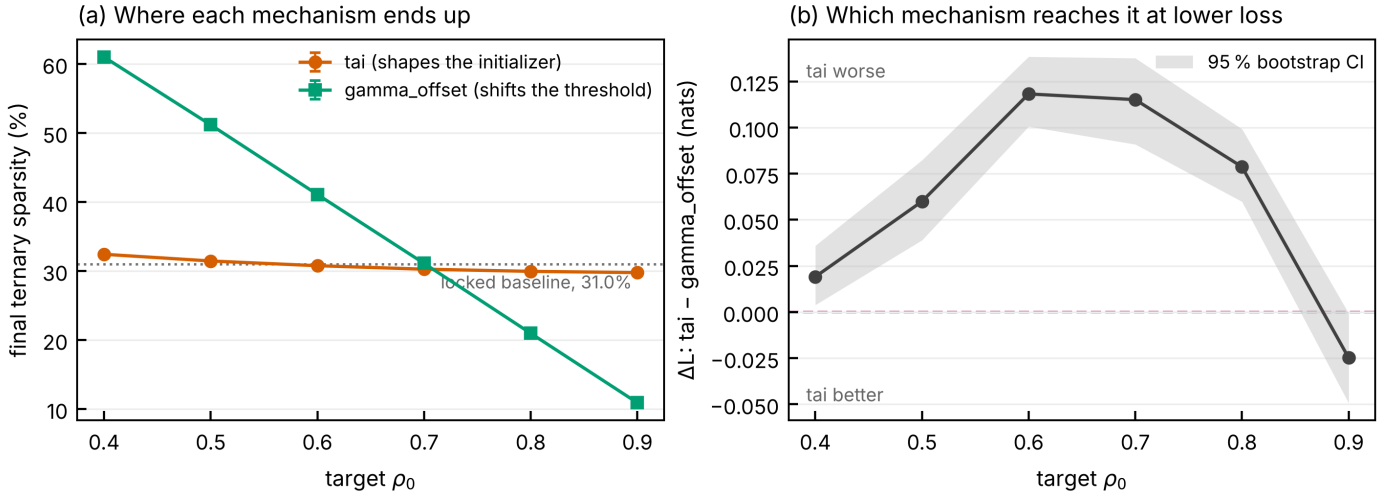


Figure 6: Panel (a): the *final* operating point each mechanism reaches as a function of the target ρ_0, with the analytically locked sparsity of Proposition 1 marked. Panel (b): the paired difference in held-out loss, in nats per token, between the initialization and threshold-offset mechanisms at matched ρ_0, with a bootstrap 95% confidence interval over the 15 seeds; an interval covering zero would indicate that the two are matched.

Table 14: Initialization × precision ablation at the primary scale ($n = 5$ seeds). The tabulated quantity is $\mathcal{L}^{\text{std-init}} - \mathcal{L}^{\text{TAI-init}}$; a negative value indicates that the standard initializer attains the lower loss.

Precision	Initialization effect (nats)	Paired t-test p
Full precision (`bfloat16`)	−0.0860	0.0086
Ternary (absmean)	−0.0898	0.0090
Difference between precisions	−0.0038	0.869

($t = +0.18$, $p = 0.869$, bootstrap 95% CI $[-0.035, +0.040]$); at $n = 5$ that interval is too wide to exclude an interaction of half the size of either main effect, so what is established is the absence of a detectable difference rather than equality. The pre-registered discriminating pattern for attributing the effect to operating-point control, a null effect in full precision alongside a positive effect under quantization, is not observed: neither leg holds, since the full-precision effect is significant and the quantized effect runs opposite to the predicted direction.

4.11. H6: behaviour across the scale ladder

The primary readout is per-scale sign consistency. The effect $\Delta\mathcal{L}$ for `tai_warmup` is positive at 11 of 11 ladder points spanning 1.4 M to 352.3 M non-embedding parameters (binomial $p = 0.0010$; Fig. 8), with per-scale means listed in Table 10. The mean effect over scales is +0.1260 nats. Against the data-selected minimum baseline, the exploratory sensitivity of Section 4.6, the mean effect is +0.0818 nats and the sign reverses at one scale.

The secondary slope diagnostic is reported with its estimator-dependence made explicit, because the pre-registered mixed-effects model with a random intercept per seed failed with a singular random-effects covariance, the between-seed standard deviation of the effect being small relative to the between-scale standard deviation. Table 15 reports the three fallback estimators.

All estimators agree on the point estimate (≈ -0.0126 nats per natural log-unit of parameters) and on the sign: the effect

Table 15: Slope of the `tai_warmup` effect against $\log N$ (non-embedding parameters). The pre-registered mixed-effects estimator did not converge. Clustering by seed treats scale-level variation as replicated within seed and is anticonservative for a scale-level question; it is reported for completeness only.

Estimator	Unit	Slope	95% CI	p
Scale-mean OLS	scale ($n = 11$)	-0.0127	$[-0.0291, +0.0037]$	0.114
Cluster-robust by scale	scale ($n = 11$)	-0.0126	$[-0.0236, -0.0016]$	0.024
Cluster-robust by seed	seed ($n = 15$)	-0.0126	$[-0.0181, -0.0071]$	7.9×10^{-6}
Mixed-effects (pre-registered)	failed: singular random-effects covariance			
Depth-controlled ladder	scale ($n = 4$)	-0.0553	$[-0.0701, -0.0392]$	n/a

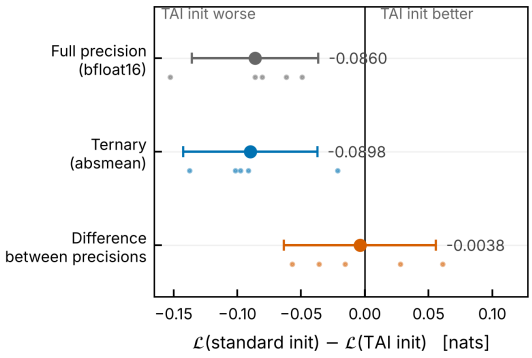


Figure 7: The {full precision, ternary} × {standard init, TAI init} ablation at the primary scale ($n = 5$ seeds). Each row is the paired initialization effect $\mathcal{L}^{\text{std-init}} - \mathcal{L}^{\text{TAI-init}}$, shown as the five per-seed values, their mean and its 95% confidence interval; negative values mean the TAI initializer attains the higher loss. The penalty is of the same size under both precisions, and the difference between them is indistinguishable from zero ($p = 0.869$), which is the signature of a generically weaker initializer rather than of failed operating-point control.

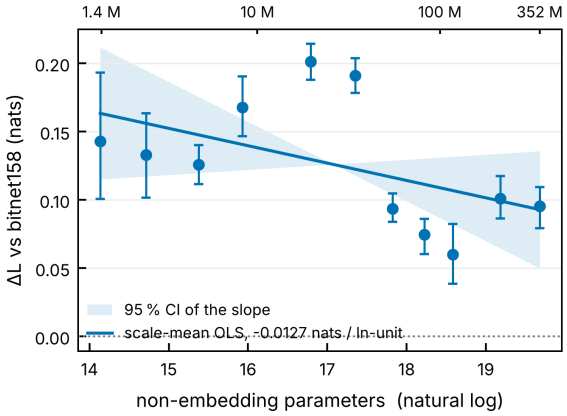


Figure 8: The paired contrast of `tai_warmup` against the pre-registered baseline at each ladder point, with bootstrap 95% confidence intervals over seeds, and the primary scale-mean OLS slope of Section 4.11 with its confidence interval. The slope is in nats per natural log-unit of non-embedding parameters, and its interval spans zero. The depth-controlled ladder is shown in Fig. 9.

narrows as scale grows. The scale-level interval, which is the appropriate unit for a scale-level question, spans zero under OLS and excludes zero marginally under cluster-robust inference. The operative statement is accordingly that the direction is consistent across scale while the magnitude is not resolved as scale-varying. The corresponding slopes for the auxiliary outcomes, fitted with the same scale-mean estimator against the same baseline, are -16.47 steps per log-unit for steps-to-

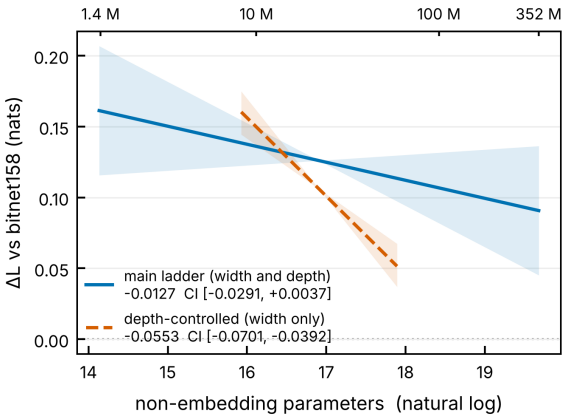


Figure 9: Scaling slope estimated with width and depth varying together, against the depth-controlled ladder in which depth is held fixed at $L = 6$. Both slopes are in nats per natural log-unit of non-embedding parameters, and each band is the 95% confidence interval of its slope. The depth-controlled fit is the steeper of the two and its interval excludes zero; the main-ladder interval does not.

target (95% CI $[-27.49, -5.44]$) and -0.0056 for the sparsity gap (95% CI $[-0.0093, -0.0019]$). Extrapolations from these fits are labelled as predictions and are not treated as evidence.

One confound has to be ruled out before the slope can be read as a statement about scale at all, since width and depth co-vary along the principal ladder. The depth-controlled ladder (Table 16, Fig. 9), in which depth is fixed at $L = 6$ and only width varies, reproduces the sign of the main-ladder slope with a steeper magnitude (-0.0553, 95% CI $[-0.0701, -0.0392]$). The narrowing trend is therefore not an artefact of depth co-varying with width along the principal ladder.

4.12. The mechanism-matched control at its default operating point

Sections 4.8 and 4.10 compare `gamma_offset` against `tai` at matched ρ_0. Because the arm also appears in the main grid at the single default $\rho_0 = 0.75$, its contrast against the pre-registered baseline is reported separately here; the two questions have different answers and conflating them would overstate the control.

Against `bitnet158` at the default operating point, `gamma_offset` attains the lower held-out loss at 7 of the 11 ladder points (binomial $p = 0.55$). The sign is scale-dependent and changes once: the arm is worse at the four smallest scales (significantly so at `dim384`, $\Delta\mathcal{L} = -0.0251$, $p = 7.0 \times 10^{-3}$) and better at the seven largest, significantly

Table 16: Depth-controlled (width-only) ladder, $L = 6$ fixed, $n = 5$ seeds. Held-out loss in nats per token, mean ± SD; $\Delta\mathcal{L}$ is the paired contrast of `tai_warmup` against `bitnet158`, following Eq. (34).

Scale	`standard`	`bitnet158`	`tai_warmup`	$\Delta\mathcal{L}$
`wc384`	5.1397 ± 0.0298	5.1402 ± 0.0304	4.9710 ± 0.0196	+0.169
`wc512`	4.9683 ± 0.0205	4.9684 ± 0.0173	4.8487 ± 0.0283	+0.120
`wc768`	4.8489 ± 0.0141	4.9196 ± 0.0142	4.7603 ± 0.0138	+0.159
`wc1024`	4.9382 ± 0.0198	4.9375 ± 0.0265	4.8831 ± 0.0107	+0.054

at five of them, reaching $\Delta\mathcal{L} = +0.0643$ ($p = 1.4 \times 10^{-4}$) at `dim1024` and +0.0480 ($p = 3.1 \times 10^{-4}$) at `dim1536`. A single fixed threshold constant is therefore not uniformly preferable to the absmean convention at this budget; the sweep of Table 13 shows that its advantage requires ρ_0 to be tuned, and that the tuned optimum itself moves with scale (0.70 at `dim384`, 0.80 at `dim768`).

4.13. H7: transfer to a second architecture family

Three arms were re-run on block family B (grouped-query attention with SwiGLU feed-forward) at two scales, 15 seeds each. For `tai_warmup` the sign of the effect is preserved at both: $\Delta\mathcal{L} = +0.0794$ ($p = 1.9 \times 10^{-4}$, $n = 15$) at `dim256` and $\Delta\mathcal{L} = +0.0347$ ($p = 8.3 \times 10^{-4}$, $n = 15$) at `dim384`.

For `gamma_offset` at its default $\rho_0 = 0.75$ the sign is not preserved: the arm is significantly worse than the baseline on family B at both scales, $\Delta\mathcal{L} = -0.0424$ ($p = 3.7 \times 10^{-3}$) at `dim256` and $\Delta\mathcal{L} = -0.0873$ ($p = 2.2 \times 10^{-8}$) at `dim384`, the latter being the largest deficit recorded for that arm anywhere in the study. Table 17 reports both contrasts. No ρ_0 sweep was executed on family B, so it is not known whether a re-tuned threshold constant would recover the deficit; the retention property of the threshold mechanism was likewise measured only on family A. Aggregated over the 13 independent (architecture × scale) cells for which a sign is available, the effect favours `tai_warmup` in all 13 against the pre-registered baseline (binomial $p = 2.4 \times 10^{-4}$); against the data-selected minimum baseline it is 12 of 13, the discordant cell being `dim1024` on family A. Magnitudes for `tai_warmup` on family B are smaller than the corresponding family-A values (+0.133 and +0.168 at the same two scales).

4.14. H8: feedback of the sweep-optimal operating point

Re-running the grid with $\rho_0^\star = 0.70$ as the `tai_star` arm (30 runs at `dim384` and `dim768`) does not recover the deficit of the fixed-ρ_0 initialization variant. At `dim384`, `tai_star` attains 5.1919 nats against 5.1914 for `tai` at $\rho_0 = 0.75$, i.e. $\Delta\mathcal{L} = -0.0805$ against the baseline; at `dim768` it attains 5.0382 against 5.0370, $\Delta\mathcal{L} = -0.1214$.

The comparison is not a pure change of ρ_0. Per-arm calibration (Section 3.9.3) gives `tai_star` an $\eta^\star$ of 2×10^{-3} at `dim384` against 3×10^{-3} for `tai`, which is why its final sparsity (28.4%) falls below the 30.2% the sweep records at the same $\rho_0 = 0.70$, rather than above `tai`'s 30.0% as the monotone relation of Section 4.8 would predict. At `dim768` the two arms calibrate to the same rate and their sparsities agree (29.8% versus 29.5%). This is also the one contrast in the study drawn across the execution-path split of Section 4.1, `tai_star` falling in the fast group at `dim384` and the baseline in the slow one; at `dim768` both arms lie in the same group and the deficit is unchanged, which is the comparison the conclusion rests on.

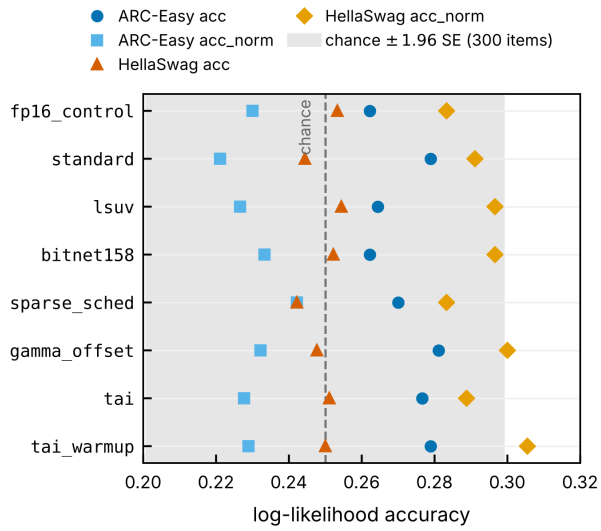


Figure 10: Zero-shot accuracy on ARC-Easy and HellaSwag under log-likelihood scoring, by arm, at the primary scale on the same 300 scored items as Table 18. Both token-normalized (acc) and byte-normalized (acc_norm) metrics are shown. The shaded band is the random-choice rate plus or minus 1.96 sampling standard errors for a 300-item subset; every observation falls inside it.

Both contrasts are significant, and for both the null is rejected under Holm–Bonferroni within the confirmatory family. The steps-to-target deficit is reduced at `dim384` ($\Delta S = -35$ for `tai_star` versus −59 for `tai`) and unchanged at `dim768` (−95 in both). H8 is therefore not supported: feeding back the sweep-optimal operating point neither degrades nor materially improves the fixed-ρ_0 arm.

4.15. Downstream evaluation

Table 18 reports the four benchmark metrics at the primary scale, and Fig. 10 shows them across the arm set.

No benchmark contrast is significant at any scale after correction, and the observed differences ($|\Delta| \le 0.025$) are of the order of the sampling noise of a 300-item subset, $\sqrt{p(1-p)/300} \approx 0.025$ at the random-choice rate. Absolute accuracies lie between 0.21 and 0.31 across all arms, tasks and metrics, i.e. at or marginally above the random-choice baseline. Aggregated over the 88 benchmark contrasts computed across the ladder, three

Table 17: Cross-architecture contrasts on block family B (GQA + SwiGLU), paired against `bitnet158` over 15 seeds. $\Delta\mathcal{L}$ follows Eq. (34); positive favours the arm. Family-A values at the same scales are reproduced from Table 10 and Section 4.12 for comparison.

Scale	Arm	$\Delta\mathcal{L}$ (family B)	p_t	$\Delta\mathcal{L}$ (family A)
`dim256`	`tai_warmup`	+0.0794	1.9×10^{-4}	+0.1329
	`gamma_offset`	−0.0424	3.7×10^{-3}	−0.0032
`dim384`	`tai_warmup`	+0.0347	8.3×10^{-4}	+0.1677
	`gamma_offset`	−0.0873	2.2×10^{-8}	−0.0251

Table 18: Downstream log-likelihood accuracy at the primary scale, scored on the first three seeds of each cell so that the comparison remains paired (300 items per task). Δ is the paired difference against `bitnet158`; p_t is the paired t-test p-value. Random-choice baselines are approximately 0.25 for both tasks.

Task / metric	Arm	Arm acc.	Baseline acc.	Δ	p_t
ARC-Easy (acc)	`tai`	0.277	0.262	+0.014	0.423
	`tai_warmup`	0.279	0.262	+0.017	0.138
ARC-Easy (acc_norm)	`tai`	0.228	0.233	−0.006	0.797
	`tai_warmup`	0.229	0.233	−0.004	0.673
HellaSwag (acc)	`tai`	0.251	0.252	−0.001	0.921
	`tai_warmup`	0.250	0.252	−0.002	0.754
HellaSwag (acc_norm)	`tai`	0.289	0.297	−0.008	0.465
	`tai_warmup`	0.306	0.297	+0.009	0.490

reach nominal $p < 0.05$, which is consistent with the number expected by chance.

The internal transfer proxy is the sole downstream measure that discriminates between arms. Next-token top-1 accuracy at the primary scale is 0.242 for `tai_warmup` against 0.227 for the baseline ($p < 10^{-4}$) and 0.221 for `tai` ($p < 10^{-4}$); the ordering of these three arms matches their held-out loss ordering at every scale.

Table 19: γ-warmup ablation at the primary scale (`dim384`): mean held-out loss in nats per token over 5 seeds, as a function of the warmup start s_0 and the duration τ_w in optimizer steps.

$s_0 \backslash \tau_w$	120	240	480
0.3	5.0536	5.1249	5.1564
0.4	**5.0437**	5.0514	5.1130
0.5	5.1553	5.1568	5.1838

4.16. Ablations

4.16.1. Realization map of the TAI construction

The separation condition of Lemma 2 was probed over a 5×3 grid in the dormant magnitude δ and jitter half-width a at target $\rho_0 = 0.75$. In 11 of 15 cells the condition holds and the target is realized to $|\rho_{\text{real}} - \rho_0| \leq 1.3 \times 10^{-4}$, independently of (δ, a); in the remaining four cells ($\delta = 0.45$ at every a, and $\delta = 0.35$ with $a = 0.30$) the assertion fires and the target is not realizable. The boundary is thus located empirically and matches the analytical prediction, and the operating defaults $(\delta, a) = (0.15, 0.15)$ sit in the interior of the valid region.

4.16.2. Quality sensitivity to the dormant magnitude

Although ρ_{real} is invariant to δ within the valid region, final quality is not. At fixed $\rho_0 = 0.75$ and $n = 5$ seeds, mean held-out loss falls monotonically from 5.2304 ± 0.0466 at $\delta = 0.05$ to 5.1048 ± 0.0202 at $\delta = 0.35$, a spread of 0.126 nats (one-way ANOVA $F = 12.11$, $p = 2.2 \times 10^{-4}$; exploratory, outside the Holm family). The construction therefore exposes a second effective degree of freedom whose influence on quality is comparable in magnitude to the ρ_0 effect itself.

4.16.3. γ-warmup schedule grid

The 3×3 grid over warmup start s_0 and duration τ_w at the primary scale ($n = 5$ seeds) is reported in Table 19. The best cell is $(s_0, \tau_w) = (0.4, 120)$ at 5.0437 nats and the worst is $(0.5, 480)$ at 5.1838; loss increases monotonically in τ_w at every s_0, and $s_0 = 0.5$ is dominated by $s_0 = 0.4$ at every τ_w. The grid-default configuration used in the main grid, $(0.4, 240)$, is the second-best cell (5.0514).

4.16.4. Learning-rate sensitivity

Across the five-point learning-rate grid at the primary scale (Table 20), `tai_warmup` attains the lower loss at four of five settings; the ordering reverses only at the smallest rate (5×10^{-4}), where both arms are far from their optima. At the baseline's own optimum (3×10^{-3}), where the comparison is least favourable to `tai_warmup`, the advantage is 0.218 nats, and it is larger still at 5×10^{-3} (0.250). The reported conclusion is therefore not an artefact of a single optimizer setting.

4.17. Efficiency

Table 21 separates the theoretical deployment model from the measured simulation cost, in the order specified by the protocol.

The simulated ternary path is 1.67× slower than the full-precision control at identical peak memory, which is the expected direction: under straight-through estimation the ternary path executes the same `bfloat16` matrix multiplication and additionally performs normalization, activation quantization,

Table 20: Learning-rate sensitivity at the primary scale (`dim384`, $n = 5$ ablation seeds). Entries are mean held-out loss $\mathcal{L}$ in nats/token; $\Delta\mathcal{L}$ is the paired contrast against `standard`, the comparison arm of this ablation, so positive favours `tai_warmup`. The baseline's minimum over the grid (5.1196 at 3×10^{-3}) is interior and therefore a genuine envelope; `tai_warmup` falls below it at the three largest rates, which excludes the global effective-learning-rate account of the schedule (Section 5.3).

Peak LR	`standard`	`tai_warmup`	$\Delta\mathcal{L}$
5×10^{-4}	5.7185	5.8034	−0.085
1×10^{-3}	5.4115	5.3928	+0.019
2×10^{-3}	5.1893	5.0409	+0.148
3×10^{-3}	5.1196	4.9019	+0.218
5×10^{-3}	5.1891	4.9390	+0.250

weight quantization and rescaling, while retaining `bfloat16` master weights. This quantity is charged honestly to the method in the compute-equal accounting and carries no deployment implication in either direction.

4.18. Summary of pre-registered outcomes

Table 22 consolidates the outcome of each pre-registered hypothesis under the wording licensed by Section 4.3.

Two auxiliary controls complete the picture. The mechanism-matched contrast places the quality difference in `gamma_offset`'s favour at 0.0610 nats, 95% CI $[+0.0474, +0.0742]$, over 90 paired cells at matched ρ_0, though, as Sections 4.12 and 4.13 record, that arm's advantage over the baseline is scale-dependent on family A and reverses on family B. The 2×2 ablation finds the TAI initialization distribution to carry an almost identical penalty in full precision (−0.086 nats) and under quantization (−0.090 nats), so the pre-registered pattern that would attribute the effect to ternary operating-point control is not observed.

Taken together, the results separate cleanly into three groups: what the initialization intervention does (little, and durably almost nothing), what the threshold intervention does (everything the initialization was supposed to do, with caveats of its own), and what annealing the threshold does (the largest and most consistent effect in the study). Section 5 takes each group in turn and asks why the quantizer's own definition makes this separation almost inevitable.

5. Discussion

5.1. What the experiment establishes, and what it does not

It is worth stating plainly, and early, that this study did not confirm the hypothesis it was designed to confirm. The pre-registered claim was that shaping the initial weight-magnitude distribution so as to place a ternary network at a chosen operating point ρ_0 would improve both convergence speed and final quality, and would causally determine the sparsity the network eventually learns. Of these, the first is supported only for the schedule variant, the second is reversed for the fixed-ρ_0 variant, and the third is supported for a mechanism that is not the proposed one. The value of the work lies in the sharpness with which the design separates these outcomes, and we have organised the discussion around that separation rather than around a defence of the original claim.

Three findings survive scrutiny, and they are of unequal importance.

First, and least contentious: the ternary operating point is analytically locked and the lock is real. For any zero-mean initializer, the absmean rule places the network at $\rho_0 = 0.6899$ irrespective of the initialization scale, and the measurement confirms this across two orders of magnitude in σ, over which the empirical active fraction varies by 6×10^{-4}, and across four initializer families, all of which land within 10^{-3} of the analytical value (Table 7). This is a small analytical observation, but it has a practical consequence that the quantization literature has not made explicit: in an absmean ternary network, initialization scale is not a lever on sparsity, and any study that varies σ hoping to trade quality against density is varying nothing.

Second, and most important: the two mechanisms that can move the operating point behave completely differently under training. The threshold offset `gamma_offset` retains its target essentially perfectly ($r = 1.003$ and 1.006 at the two tested scales, $R^2 = 1.000$); an initialization intervention of identical nominal magnitude is almost entirely erased ($r = 0.052$ and 0.069). A 50-percentage-point manipulation of initial sparsity leaves 50 points of final sparsity in the first case and between 2.6 and 3.6 points in the second. We regard this contrast, rather than the performance of any single arm, as the central empirical result of the paper, and Section 5.2 is devoted to why it arises.

Third: annealing the threshold works, and does so consistently. The γ-warmup arm `tai_warmup` attains a lower held-out loss $\mathcal{L}$ and reaches its scale's pre-registered target loss ($\mathcal{L}_{\text{target}} = 5.6539$ at the primary scale) in fewer steps than the frozen primary baseline `bitnet158` at 11 of 11 scales spanning 1.4 M–352 M non-embedding parameters, at all 13 architecture × scale cells, and at four of five learning rates. The effect is not fragile with respect to seed, fold, architecture family, depth, or optimizer setting. Whether it is fragile with respect to training budget is the principal open question, and we treat it at length in Section 5.7.

Against these, three pre-registered claims failed, and the reasons are instructive rather than merely negative: the fixed-ρ_0 initializer `tai` is a worse initializer in full precision as well as under quantization (Section 5.4); the mechanism-matched control `gamma_offset` outperforms it at matched ρ_0 (Section 5.4); and the feedback of the sweep-optimal operating point, `tai_star`, had almost no room to change anything, for reasons that are a consequence of the retention result itself (Section 5.11).

One qualification must be carried through everything that follows, because it bounds the strongest reframing these data will support. That `gamma_offset` beats `tai` at matched ρ_0 is robust (+0.0610 nats, 95% CI $[+0.0474, +0.0742]$, 90 paired cells). That `gamma_offset` beats the baseline is not a general fact: at its default $\rho_0 = 0.75$ it does so at only seven of eleven scales on family A, with the sign turning over mid-ladder, and it is significantly worse than the baseline on family B at both tested scales (Section 4.13). The durability of the threshold mechanism and

Table 21: Efficiency. Panel (A) is a counting argument over the ternary codebook at the largest *declared* configuration, `dim2048`, which was not executed (Section 4.1); the compression ratios do not depend on the parameter count, so only the absolute sizes are specific to it. It is independent of the hardware used here. Panel (B) is the measured cost of simulating ternary arithmetic in `bfloat16` under straight-through estimation; it is a property of the research harness and is not a ternary-inference measurement. No ternary kernel was benchmarked in this work.

(A) Deployment model (theoretical, primary)	
Quantized parameters	782.91 M
Half-precision size	1493.28 MB
Ternary size, information-theoretic ($\log_2 3 = 1.585$ bits/weight)	147.92 MB
Ternary size, 2-bit packed	186.66 MB
Compression vs. half precision (information-theoretic)	10.09×
Compression vs. half precision (2-bit packed)	8.00×

(B) Simulation cost (measured, secondary)		
	`fp16_control`	ternary (simulated)
Latency	430.9 ms	718.9 ms
Throughput	38,025 tok/s	22,792 tok/s
Peak memory	3.336 GB	3.336 GB
Forward sparsity	0.0%	25.0%
Latency ratio		1.67×

Table 22: Outcome of every pre-registered hypothesis. Token-equal indicates that the claim is restricted to performance at the fixed 20 M-token budget, convergence not having been established for any arm.

Hyp.	Outcome	Operative statement
H1	Split	Supported for `tai_warmup` (11/11 scales, ΔS up to +148 steps); rejected in the opposite direction for `tai` (11/11 scales slower).
H2	Untestable (pre-reg.)	No run reached plateau (0/760); the convergence-conditional equivalence test has an empty qualifying set. The token-equal companion returns `tai_warmup` superior at 11/11 scales and `tai` inferior at 10/11; equivalence is never returned.
H3	Not supported (strong form)	Ordinal control is near-perfect (per-seed Spearman −0.996, SNR 44.9×) and the placebo is flat, but retention for the initialization mechanism is 0.052–0.069: 93–95% of the intervention is erased, and final sparsity converges to an attractor near 0.30. The threshold-offset mechanism retains 1.003–1.006.
H4	Supported	No evidence of variance inflation on the pooled seed × fold matrix ($W = 0.554$, $p = 0.458$); 23/24 per-cell Levene tests are non-significant and the one exception favours lower TAI variance.
H5	Supported for the threshold mechanism	The threshold-offset arm traces a U-shaped frontier spanning 50 points of sparsity with an interior optimum; the initialization arm's sparsity axis spans 2.6 points, rendering its frontier near-degenerate. $\rho_0^\star = 0.70$ (bootstrap $P = 0.758$).
H6	Direction supported; magnitude unresolved	Sign positive at 11/11 scales (binomial $p = 0.0010$) over 1.4 M–352 M parameters. Slope −0.0127 per log-unit under the scale-level OLS estimator, whose interval spans zero. The depth-controlled ladder reproduces the sign.
H7	Supported for `tai_warmup`; not for `gamma_offset`	Sign preserved for `tai_warmup` on family B at both tested scales ($p \leq 8.3 \times 10^{-4}$); all 13 architecture × scale cells consistent against the pre-registered baseline (binomial $p = 2.4 \times 10^{-4}$), 12/13 against the data-selected minimum. `gamma_offset` at its default ρ_0 is significantly worse than the baseline on family B at both scales (Table 17).
H8	Not supported	`tai_star` at $\rho_0^\star = 0.70$ tracks `tai` to within 0.0013 nats at both tested scales and remains below the baseline.

the competitiveness of any particular threshold constant are separate claims, and only the first is established here.

5.2. *Why the initialization intervention is erased: the absmean attractor*

The retention asymmetry has a straightforward explanation, and articulating it is the main theoretical contribution we can defend.

The absmean threshold is not an external constant. It is a functional of the weight matrix itself,

$$\gamma(\mathbf{W}; c) \;=\; c\,\mathbb{E}|w|, \tag{35}$$

so the criterion by which a weight is judged large enough to survive moves whenever the weight distribution moves. An initialization intervention perturbs the numerator and the denominator of the same ratio: TAI shapes $|w|$ into two modes, but $\mathbb{E}|w|$ is computed from those same modes, and the surviving fraction is determined only by where the modes sit relative to their own mean. Gradient descent then acts on the weights, and as it does so it drags γ along with them. The threshold is, in

this precise sense, adaptive; nothing in the training dynamics preserves the initial partition, and the system relaxes to whatever active fraction the optimizer's own steady-state magnitude distribution implies.

The measurements are consistent with exactly this picture. The fitted fixed point of the retention regression is 0.305 at `dim384` and 0.300 at `dim768`, and the untouched `standard` arm lands at 0.321 and 0.318 respectively. All four straddle the analytical initialization value $1 - \rho_0 = 0.310$ within 0.011, the fitted attractors falling just below it and the untouched arm just above. In other words, the value ≈ 0.31 that Proposition 1 identifies as a property of initialization turns out to be approximately where the optimizer goes anyway. This reframes the premise of the paper in a way we did not anticipate: the locked operating point is not merely a shared default that every zero-mean initializer happens to inherit; it is close to an attractor of ternary training under this quantizer.

A threshold offset escapes this because it changes the constant c, which no gradient touches. The survival condition $|w| \geq \frac{c}{2}\mathbb{E}|w|$ is homogeneous of degree zero in the weights, so rescaling the weights cannot undo a change in c; only a change in the shape of the magnitude distribution could, and the observed $R^2 = 1.000$ indicates that shape change is negligible over the tested range. The consequence is a clean prediction that we did not test and that a follow-up study should: any quantizer whose threshold is a fixed multiple of a distribution statistic of the current weights will exhibit low retention of initialization interventions, while any quantizer whose threshold is constant in the weights will retain them. Trained ternary quantization [6] falls on the first side of this line despite learning its per-layer scaling factors, because its threshold is a fixed multiple of $\max|w|$ and therefore moves with the weights exactly as the absmean rule does. The result reported here is therefore not about TAI specifically but about the class of self-referential thresholds to which absmean and BitNet b1.58 [4] belong.

There is an important edge case. Retention was measured over $\rho_0 \in [0.4, 0.9]$, i.e. initial sparsity in $[0.10, 0.60]$. We do not know whether the attractor persists at extreme operating points, at, say, 90% sparsity, where the surviving weights are few enough that the mean absolute weight is dominated by the dormant mass and the feedback in Eq. (35) may behave differently. The claim is bounded to the tested range.

5.3. Timing rather than distribution, and an honest confound

The single largest effect in the study is not an initialization effect at all. It is the γ-warmup schedule of `tai_warmup`, which lowers the threshold early, raising the active fraction above its locked value, and relaxes the threshold to the nominal $c = 1$ over τ_w steps. This arm improves on `bitnet158` by 0.060 to 0.201 nats depending on scale, whereas the fixed-ρ_0 initializer `tai` falls short of it by 0.052 to 0.120 nats. Same target operating point, opposite outcome; the difference is when the network is required to be there.

The ablation is consistent with a transient-density reading. Loss increases monotonically in τ_w at every warmup start, and the best configuration is the one with the shortest warmup at an intermediate start, $(s_0, \tau_w) = (0.4, 120)$; the least aggressive start, $s_0 = 0.5$, is the worst at every duration. That is, the benefit comes from a brief early period in which more weights carry non-zero ternary codes, and therefore more weights receive a non-zero contribution through the straight-through estimator [17], not from a prolonged departure from the nominal threshold. The natural mechanistic story is that early ternary training is gradient-starved: roughly 31% of weights are quantized to zero at initialization, and a temporarily denser forward pass lets the network establish useful structure before the threshold tightens.

We must, however, flag a confound that a referee will raise and that our data cannot fully dispel. Scaling γ down by s_0 scales the magnitude of the quantized weights by the same factor, which scales the forward activations and, through the straight-through path, the effective gradient magnitude of every BitLinear layer. A γ-warmup is therefore not only a sparsity schedule; it is also, in part, an implicit rescaling of the network's effective learning rate during exactly the period when the explicit learning-rate warmup is active. The coincidence is uncomfortably exact: the best warmup duration found by ablation, $\tau_w = 120$ steps, is precisely the length of the learning-rate warmup fixed in the protocol. We do not think this fully explains the effect. The learning-rate sensitivity ablation shows the advantage persisting, and indeed growing, at the upper four of the five peak learning rates tested, reversing only at 5×10^{-4} where both arms are far from their optima; an implicit-learning-rate account would not naturally predict that pattern. The same ablation excludes the simplest form of the confound outright.

The exclusion is an envelope argument. If γ-warmup acted purely as a multiplier on the effective learning rate, then every outcome it can reach would also be reachable by the baseline at some other learning rate: the set of achievable losses would be a subset of the baseline's learning-rate curve, and the schedule could not beat that curve's minimum. It does. At the primary scale the baseline's loss over the five-point grid is 5.7185, 5.4115, 5.1893, 5.1196 and 5.1891 nats, an interior minimum of 5.1196 that is bracketed on both sides and therefore a genuine envelope rather than a truncated one. The γ-warmup arm falls below that envelope at three of the five learning rates (by 0.079 nats at 2×10^{-3}, 0.218 at 3×10^{-3} and 0.181 at 5×10^{-3}), reaching 4.9019 nats at its own optimum. No reparameterization of a single global learning rate can produce a loss the baseline cannot attain at any learning rate, so the account of γ-warmup as an implicit learning-rate schedule is refuted in its global form.

What the argument does not exclude is a gradient-scale effect that is time-varying or layer-dependent, since such an effect is not equivalent to any single global rate. Settling that requires the control we specify in Section 5.15: a matched arm holding the threshold at $c = 1$ while rescaling the BitLinear output by the same $g_{\text{warm}}(t)$, reproducing the gradient-scale trajectory with no change to the sparsity schedule.

A second observation belongs here because it is easy to overread. The gradual-pruning baseline `sparse_sched`, which drives the threshold in the opposite direction, is the weakest ternary baseline at the primary scale. It is tempting to present this as a clean mirror image (annealing down helps, annealing up hurts), but the comparison is confounded: the gradual-

pruning arm terminates at $c = 1.6$ and 48.8% sparsity, whereas γ-warmup terminates at $c = 1$ and 30.3%. Its deficit therefore conflates schedule direction with terminal operating point, and at least part of it is simply the quality cost of ending 18 points sparser. The honest statement is that we did not test annealing direction with the terminal point held fixed.

5.4. Attribution: the contribution is about γ, not about **W**

Two controls, both pre-registered and both designed so that the proposed method could lose, agree that the initialization distribution is not where the benefit lives.

The 2×2 initialization $\times$ precision ablation is the more decisive of the two. Had TAI's advantage derived from control of the ternary operating point, the initializer should have been neutral in full precision, where there is no threshold to control, and beneficial under quantization. The measured pattern is neither: the TAI distribution costs 0.086 nats in full precision and 0.090 nats under ternary quantization, effects of the same sign, nearly equal magnitude, and comparable significance, differing by 0.0038 nats, an order of magnitude less than either. The parsimonious reading is that the two-point magnitude law is simply a worse initializer, in the ordinary sense in which initializers are better or worse [8, 9], and that its cost is paid identically whether or not the weights are subsequently ternarized.

The dormant-magnitude ablation supports this and supplies a mechanism. Within the region where the separation condition of Lemma 2 holds, the realized ρ_0 is exactly on target and completely independent of the dormant magnitude δ, yet quality is not: loss falls monotonically from 5.2304 at $\delta = 0.05$ to 5.1048 at $\delta = 0.35$, a spread of 0.126 nats, which is comparable to the entire ρ_0 effect over the sweep. Since increasing δ toward the active magnitude α makes the distribution progressively less bimodal, quality improves monotonically as the TAI construction is deformed toward a unimodal distribution, that is, toward the standard initializer it was meant to replace. Two facts follow that the paper must state rather than bury: TAI possesses a second, quality-relevant degree of freedom that is not the operating point, so the framing of ρ_0 as the knob is incomplete; and the direction in which that knob helps points away from the method.

The mechanism-matched control closes the argument from the other side. Paired over 90 (ρ_0, seed) cells, `gamma_offset` attains a loss 0.0610 nats lower than `tai`, 95% CI $[+0.0474, +0.0742]$, and it retains the operating point essentially perfectly while `tai` does not. Of the three readings we pre-registered for this contrast, the outcome is the one in which the simpler mechanism wins on both axes on which the two were designed to be compared.

The finding must not be over-read. It concerns two ways of reaching the same operating point, and does not establish that a constant threshold offset is a better ternary recipe than the absmean convention. The evidence there is mixed and in one place negative: at its default ρ_0, `gamma_offset` beats `bitnet158` at seven of eleven family-A scales with the sign turning over mid-ladder (Section 4.12), and it is significantly worse on family B at both tested scales, by -0.042 and -0.087 nats (Section 4.13). Two readings are available and these data do not separate them: either the optimal constant is architecture-dependent (which is plausible, since the sweep already shows it to be scale-dependent, moving from $\rho_0^\star = 0.70$ at `dim384` to 0.80 at `dim768`) or the permanent rescaling of the quantized weight magnitudes, and hence of the effective gradient scale, that a fixed $c \neq 1$ imposes interacts badly with the gated feed-forward block of family B. No ρ_0 sweep was executed on family B, so this is an open question rather than a resolved one. What survives unqualified is the narrower and more durable claim: if an operating point is to be held, the threshold holds it and the initializer does not.

The contribution is therefore best stated as follows: the ternary operating point is analytically locked under every standard initializer; it can be unlocked, but only a threshold intervention unlocks it durably; and annealing that threshold, rather than displacing the initial weight distribution, is what improves training. The initializer is the vehicle by which we discovered this, and reporting its failure is what makes the threshold result credible.

5.5. Relation to prior work

We relate the results to four lines of work: the ternary quantizers whose thresholds we reparameterize, the initialization literature whose framework turns out to be silent in this regime, the sparsity-schedule literature our warmup result inverts, and the methodological literature the design follows.

Ternary weight networks [5] introduced the symmetric threshold $\Delta = 0.7\,\mathbb{E}|w|$ as a heuristic approximation to the ℓ_2-optimal ternary projection, and trained ternary quantization [6] made the two non-zero levels learnable. BitNet [3] brought absmean scaling to large-scale language-model pre-training in the binary setting, and BitNet b1.58 [4] carried it to the ternary one as the absmean threshold rule, i.e. $c = 1$ in our parameterization. Our Proposition 1 supplies the map between these conventions that the literature has, to our knowledge, left implicit: the threshold scale c and the initial active fraction are related by $\rho(c) = 2[1 - \Phi(\frac{c}{2}\sqrt{2/\pi})]$, so a stated threshold constant is a stated operating point.

Evaluating that map at the classical TWN value gives the closest point of contact with prior art. Substituting $c = 0.7$ gives $\rho_0 = 0.780$, i.e. 22.0% initial sparsity, a value we compute here from Eq. (9) rather than measure. The optimum located empirically by our threshold-offset sweep at `dim768` lies at $\rho_0 = 0.80$ ($c^\star = 0.635$, 20.7% final sparsity), and at `dim384` at $\rho_0 = 0.70$ ($c^\star = 0.966$). The decade-old TWN heuristic thus sits between our two measured optima and within two percentage points of the better one. We read this as convergent evidence rather than as a claim of priority: an approximation derived analytically for a different purpose and a sparsity-quality frontier traced empirically under a language-modelling objective identify nearly the same operating region, and the absmean convention $c = 1$ used by the current generation of 1.58-bit models sits measurably off it in the direction of excess sparsity.

The initialization literature is concerned almost exclusively with variance propagation [8, 9, 11]. Our results identify a regime in which that entire framework is silent by construction: because the absmean code is scale-invariant (Lemma 1),

every variance-based prescription lands on the same operating point. A referee may reasonably ask whether this makes initialization irrelevant in ternary networks. The data say no, and say so in an unexpected direction: `lsuv`, whose calibration pass we measured performing exactly one rescaling operation per BitLinear layer against a maximum initial variance deviation of 0.586, attains a lower loss than the standard initializer at nine of the eleven scales, by 0.007 to 0.051 nats, and is the strongest non-TAI ternary arm at six of them, while sitting at precisely the same operating point ($\rho_0 = 0.690$ versus 0.689). The two exceptions are the largest ladder points, where the difference reverses by 0.013 and 0.004 nats, within the noise floor of Section 5.10. Initialization therefore still matters in ternary training; it simply does not matter through the operating point. We note that this contradicts a claim made in an earlier draft of our own methodology, that LSUV is mathematically degenerate under RMSNorm pre-normalization, and we have corrected it against the measurement.

Gradual magnitude pruning [33] establishes that sparsity is best increased slowly rather than imposed at once, and the lottery-ticket line of work [39] that the identity of the surviving weights is established early. Our γ-warmup result is a schedule result of the same family, but with the direction inverted: in ternary pre-training under a fixed budget, the productive schedule starts denser than the nominal operating point and tightens, rather than starting dense in the unquantized sense and pruning toward a sparse target. The retention finding also bears on the lottery-ticket framing: since the surviving set at the end of training is almost independent of the surviving set at initialization ($r \approx 0.05$–0.07), the ticket in an absmean ternary network is not drawn at initialization in any strong sense.

The design follows the line of work arguing that machine-learning comparisons require variance accounting, pre-registration of the primary contrast, and equivalence rather than significance testing when the claim is non-inferiority [41–46]. Our variance decomposition contributes a small empirical datum to that literature: at the primary scale, the standard deviation across evaluation folds (0.0334) exceeds the standard deviation across seeds (0.0290) for both arms. Benchmarks that report seed variance alone, as is standard, therefore understate the total uncertainty of a held-out language-modelling number, in our case by roughly a factor of two in variance terms (Table 11).

5.6. Scale behaviour: what the trend licenses

The direction of the effect is stable across the ladder: positive at 11 of 11 scales, binomial $p = 0.0010$, over more than two orders of magnitude in non-embedding parameters. The magnitude is not stable, and the trend is downward. All three fallback estimators agree on a slope of approximately -0.0126 nats per natural log-unit of parameters; the scale-level ordinary-least-squares interval spans zero ($[-0.0291, +0.0037]$, $p = 0.114$), the cluster-robust scale-level interval excludes it marginally, and the seed-clustered interval, which treats scale-level variation as replicated within seed and is anticonservative for this question, excludes it decisively. We report the scale-level intervals as primary precisely because the seed-clustered one would flatter the result.

The depth-controlled ladder matters here. Because the principal ladder co-varies width and depth, a narrowing trend could have been a depth artefact; refitting with depth fixed at six layers reproduces the sign with a steeper slope (-0.0553, 95% CI $[-0.0701, -0.0392]$). The narrowing is thus a property of scale rather than of the particular width-depth schedule.

The obvious referee question—does this survive to the scales at which 1.58-bit models are actually deployed?—cannot be answered by this experiment, and we decline to let the extrapolation answer it. Carried forward naively, the fitted slope predicts an effect of +0.064 nats one decade above the tested range and +0.034 nats two decades above; the latter falls inside the 0.03–0.05 nat cross-path implementation noise floor documented in Section 5.10. Two considerations cut in opposite directions. Against the pessimistic reading, the per-scale effects are not monotone in scale (they dip over the first three scales from +0.143 at 1.4 M, peak at +0.201 at 19.6 M, decline to their minimum at `dim1024`, and recover at the two largest points to +0.101 and +0.095), so a monotone log-linear fit is a crude summary of a non-monotone series. In its favour, the largest ladder points are the furthest from any sensible compute-optimal token allocation [55, 56]: at 352 M parameters, a 20 M-token budget is roughly two orders of magnitude below the Chinchilla-style allocation, so those points measure very early training of a large model rather than the training of a large model. Neither consideration is decisive, and the operative statement remains the one recorded in the manifest: direction consistent across scale, magnitude not resolved as scale-varying.

5.7. The principal threat to validity: an unconverged budget

No run in this study converged. The pre-registered plateau gate of Eq. (26) (a slope test on the held-out validation trace, not a level test on the training loss) certified 0 of 760 runs, the median fitted tail descent was -0.225 nats, and the independent $2\times$ budget check found 0.315 and 0.287 nats of remaining improvement for the two arms tested, between 14 and 16 times the equivalence margin. This is the weakest point of the paper and we treat it as such rather than as a technicality.

The consequence for wording has already been applied throughout: the pre-registered equivalence hypothesis is reported as untestable rather than as a null result, and no claim in the Results section uses the word *converged*. The deeper consequence is interpretive. An initialization or schedule intervention is, on any reasonable prior, expected to matter most during the transient and to matter less once the optimizer has had time to forget where it started. The entire study lives inside that transient. It is therefore possible that the γ-warmup advantage is a budget-limited effect that would shrink or vanish under longer training, and the design cannot exclude this.

What the design can do is bound the erosion rate, and the budget check happens to supply the necessary numbers. The comparison is taken within run, as Section 3.10 specifies: both quantities are read off the same three long runs, at the probe nearest the short-budget step count and at the end, so that no

cross-run seed noise enters. On that basis the advantage is 0.1269 nats at 20 M tokens. Doubling to 40 M improves the baseline by 0.3153 nats and the γ-warmup arm by 0.2865 nats, so the advantage falls to 0.0981 nats, an erosion of 22.7% per doubling. Extrapolated linearly in log tokens, the advantage would reach zero after approximately four further doublings, at roughly 0.4 billion tokens. We present this projection with three explicit caveats and we do not build any claim upon it: it rests on $n = 3$ seeds and a single doubling; erosion is very unlikely to be linear in log tokens; and one doubling gives no information about curvature. Its only legitimate use is to establish an order of magnitude: the advantage is not a few-hundred-step artefact, since it survives a doubling with 83% of its magnitude intact, but neither is there evidence that it would survive to a compute-optimal budget.

We note in passing that the direction of the erosion is itself informative: the baseline gains more from the additional budget than the γ-warmup arm does, which is what one expects if the schedule accelerates early progress rather than raising the attainable ceiling. That reading is consistent with the transient-density mechanism proposed in Section 5.3 and inconsistent with a claim that TAI or its schedule changes what the network can ultimately represent. Neither the paper nor its title should make the latter claim.

5.8. The anomaly: ternary training below full precision

At ten of the eleven scales, the strongest non-TAI ternary arm attains a lower held-out loss than the full-precision control `fp16_control`, by between 0.080 and 0.278 nats; including the TAI arms the control is beaten at all eleven, by up to 0.371 nats. This inverts the ordering the quantization literature would predict [57]. We cannot fully explain it. Four accounts are available; the data discriminate among some of them.

Regularization under a short budget, the leading account: Straight-through ternarization injects a large, structured perturbation into the forward pass, which at a budget two orders of magnitude short of convergence may act as an effective regularizer, in the manner long observed for BinaryConnect and its successors [14]. With the tuning explanation largely excluded below, this is the account we regard as most likely. It predicts that the ordering should revert with budget. Our 2× check did not include the full-precision arm, so this is untested and constitutes the most valuable single follow-up experiment.

An unfair schedule for the control, largely excluded. The natural suspicion is that the protocol is tuned for the ternary arms and handicaps the control. The calibration data do not support it. The control's learning-rate optimum is interior at all eleven scales, so it is bracketed rather than truncated by the grid; at `dim1536` the adaptive extension added a 3×10^{-4} point that is worse than both of its neighbours, confirming 5×10^{-4} as a genuine minimum. Each arm additionally receives its own $\eta^\star$ selected on seeds disjoint from those reported. What remains shared across arms is warmup length, weight decay and schedule shape, so a residual disadvantage arising from those cannot be excluded; but the simplest version of this objection, that the control was never given its best learning rate, is refuted by the measurement.

Not a normalization artefact: One tempting explanation can be excluded: BitLinear applies RMSNorm to its input in both modes, so the full-precision control receives the identical normalization structure and does not differ from the ternary arms in that respect.

Not a divergence artefact. No run met the divergence criterion of Eq. (27), and the control's seed-to-seed standard deviations are of the same order as the ternary arms', so the control is not being dragged down by unstable seeds.

The consequence for the paper is a restriction that we state explicitly: **this study does not measure the cost of ternary quantization, and its full-precision arm should not be read as a quantization-cost baseline.** The control's legitimate role here is narrower: it supplies the denominator for the data-derived equivalence margin and confirms that no arm is catastrophically broken. Every comparison on which a claim rests is ternary-versus-ternary.

5.9. Why the downstream benchmarks do not discriminate

The log-likelihood benchmarks separate no pair of arms at any scale, and the reason is arithmetic rather than mysterious. The largest arm-versus-baseline difference the benchmarks are asked to resolve is 0.201 nats at `dim512`, on models whose absolute loss is near 5.0 nats, a perplexity of roughly 150 against a uniform baseline of 50,257. At that level of language modelling the calibrated preference among four plausible continuations is almost uninformative, and every arm sits between 0.21 and 0.31 accuracy against a chance baseline of 0.25. A 300-item subset carries a binomial standard error of approximately 0.025, which exceeds every observed difference; the benchmarks are therefore not merely insensitive to the intervention but incapable of resolving differences of the size it produces. That three of 88 contrasts reach nominal $p < 0.05$ is what chance predicts.

The corollary matters for how the paper should be read, and we state it plainly: this study demonstrates an effect on language-modelling loss and convergence speed and supplies no evidence of an effect on downstream capability. The only transfer measure that tracks the loss ordering is the internal next-token proxy a_{nt}, which is not independent of the training objective and cannot supply that evidence either. Establishing downstream consequences requires scales at which the benchmarks themselves become informative, which the budget of this study does not reach.

5.10. An empirical noise floor, and which effects are near it

The design furnishes an unplanned but useful instrument for bounding implementation noise. As noted in the Methods, the BitNet b1.58 formulation and the absmean formulation are algebraically identical under exact arithmetic: substituting $\beta = 1/\gamma$ into $\text{clip}(\text{round}(\beta\mathbf{W}), -1, 1)/\beta$ returns Eq. (6) exactly. The two arms differ only in the order in which the scale factor is applied and, in this harness, share the same standard initializer. Any measured difference between them is therefore attributable to floating-point association compounded through 1,221 optimization steps under non-deterministic kernels, not to method.

Measured per seed, the mean absolute difference between these two nominally identical arms is 0.005 to 0.019 nats at

ten of the eleven scales, with a worst-case per-seed difference of 0.036 nats. At `dim512` it is far larger and systematic: a mean paired difference of −0.050 nats with a standard deviation of 0.023 over 15 seeds, and a maximum per-seed difference of 0.082 nats. At `dim192` the difference is also systematic though smaller (−0.016 ± 0.004 over five seeds).

5.10.1. The cause: a non-uniform execution path

The `dim512` separation is too large and too systematic to be accumulated rounding, and we traced it. The per-run wall-clock record, stored for every run, resolves the arms at each scale into two sharply separated groups. At `dim512` the arms `standard`, `lsuv`, `gamma_offset`, `tai` and `tai_warmup` complete in a median of 134–140 seconds, while `bitnet158` and `sparse_sched` require 250–251 seconds for arithmetically comparable work, a ratio of 1.9. The grouping is read over the seven ternary arms; `fp16_control` is excluded throughout because it bypasses the quantizer and so does not perform comparable work. The same bimodality appears at `dim640`, `dim1024` and `dim1280`, with ratios of 2.0, 2.2 and 2.6. Narrower instances appear at `dim384`, where the ratio is 1.65 and the fast group contains `tai_star` alone, and at `dim192` (1.28); across the remaining five scales the spread is at most 1.15. The grouping tracks the quantizer branch taken inside the fused step, and its magnitude is that of compiled versus eager execution: at the four scales analysed here the compiled path was obtained for some arms and not for others.

This supplies the mechanism, because compiled and eager execution differ in floating-point reduction order, which over 1,221 steps of a chaotic optimization amplifies into a systematic offset. The prediction it makes is sharp: `standard` and `bitnet158` should diverge when those two arms fall on opposite sides of the split, and agree otherwise. They fall on opposite sides at `dim512` and `dim1024` only; at `dim640` and `dim1280` the split separates the TAI arms from the baselines, leaving `standard` and `bitnet158` together. In aggregate the prediction holds: the measured mean absolute gap between them is 0.0295 nats at the two scales where their paths differ and 0.0037 nats at the nine where they coincide, a factor of eight. Cell by cell it is weaker. The aggregate is carried by `dim512` (−0.050, $p < 10^{-4}$); at `dim1024` the two arms do not separate detectably (+0.009, $p = 0.17$), and at `dim192`, where their paths coincide, they do separate (−0.016, $p = 0.001$). Execution path is therefore one source of systematic offset between nominally identical arms, but not the only one.

5.10.2. Consequences for the reported effects

We report this as a protocol defect rather than a curiosity: at four of eleven scales the arms split into two execution groups by a factor of 1.9 or more, and at all four (`dim512`, `dim640`, `dim1024`, `dim1280`) the primary contrast is measured across that boundary. The two narrower splits noted above, at `dim384` and `dim192`, leave `bitnet158` and `tai_warmup` in the same group and so do not affect it. Two things follow.

First, the effective reproducibility floor under throughput mode is of order 0.004 nats when arms share a path and rises to 0.03–0.05 nats when they do not. The smallest reported effects must be read against the appropriate figure: the γ-warmup advantage of +0.060 nats at `dim1024` sits at roughly twice the cross-path floor, while +0.075 at `dim896`, a scale with no split, sits at roughly twenty times the same-path floor.

Second, and decisively for the paper's conclusion, the primary result does not depend on the affected scales. Restricting attention to the seven scales at which the two arms of the primary contrast shared one execution path, the γ-warmup advantage is positive at 7 of 7 (sign test $p = 0.016$) with a mean of +0.119 nats and a range of +0.075 to +0.168; the four split scales give a mean of +0.138. The effect is therefore neither created nor materially inflated by the defect, and the sign-consistency claim of Section 4.11 survives the exclusion of every scale the defect touches. A corrected grid should nonetheless pin the execution path per arm, and we specify this in Section 5.15.

5.11. A critique of our own design

Three design decisions look weaker after the fact than they did before, and naming them is cheaper than being told.

The operating-point feedback hypothesis had no room to move. H8 fed the sweep-optimal $\rho_0^\star = 0.70$ back into the grid against the default $\rho_0 = 0.75$. But the retention coefficient of the initialization arm is 0.052: a 0.05 change in ρ_0 therefore changes final sparsity by roughly 0.3 percentage points. The observed outcome, `tai_star` tracking `tai` to within 0.0013 nats at both scales, was close to guaranteed by the retention result, and H8 as specified was not a meaningful test. It would have been meaningful had it been applied to the threshold-offset arm, where the same ρ_0 change moves the final operating point by five percentage points.

The Pareto constraint excluded the better operating point: The pre-registered selection rule required $\mathrm{sp}_T \geq 30\%$. For the initialization arm this constraint is vacuous, since the attractor pins every point between 29.8% and 32.4%. For the threshold-offset arm it is binding, and it excludes the arm's actual optimum at `dim768` ($\rho_0 = 0.80$, 20.7% sparsity, 4.8968 nats, better than the constrained $\rho_0 = 0.70$ point by 0.011 nats and better than the pre-registered baseline by 0.020 nats at two-thirds of its sparsity). The floor was chosen to prevent the degenerate outcome of simply choosing the densest model; in retrospect it encodes an assumption about the desirable operating region that the experiment was meant to test. We report both the constrained and unconstrained optima.

The falsification statistic was specified with the wrong sign. The `tai_negctrl` sign test was written as a one-sided comparison of signed Spearman coefficients, real versus placebo, on the implicit expectation of a positive effect. The true effect is negative (−0.996) and the placebo's noise-driven coefficient is positive (+0.135), so the test returns 0 of 15 wins with $p = 6.1 \times 10^{-5}$, a number that is meaningless as stated. The falsification nevertheless succeeds decisively on the interpretable comparison: the placebo's final-sparsity range across labels is 1.3×10^{-4} against 2.65×10^{-2} for the real intervention, a factor of 212. We report the magnitude comparison as the operative result and the mis-specified statistic as a recorded error. That

we can do so without altering any conclusion is a direct benefit of having frozen the analysis in advance; had the placebo been analysed post hoc, the temptation to re-specify would have been considerable.

5.12. Threats to validity

Table 23 consolidates the objections we consider most likely to be raised, the evidence in the study that bears on each, and the residual risk that remains unaddressed.

5.13. Practical implications

For practitioners training ternary language models, four recommendations follow from the measurements. The first three are ordered by the strength of the evidence behind them; the fourth is a measurement practice rather than an empirical finding.

Treat the threshold constant as a hyperparameter, not a convention. The absmean choice $c = 1$ is an operating-point choice equivalent to $\rho_0 = 0.690$, and Eq. (19) gives the closed-form map from any desired ρ_0 to the constant that realizes it. The sweep was executed at two scales, and it places the optimum at $\rho_0 = 0.70$ ($c^\star = 0.966$) at `dim384` and at $\rho_0 = 0.80$ ($c^\star = 0.635$) at `dim768`. Only the second separates from the convention: the first is the grid point nearest $\rho_0 = 0.690$ and is not distinguishable from it at a grid resolution of 0.1. The two bracket the classical TWN constant $c = 0.7$ [5], which the same map places at $\rho_0 = 0.780$.

If sparsity must be controlled, control it at the threshold. Shaping the initialization retains 5–7% of the imposed shift; a threshold offset retains essentially all of it ($r = 1.003$–1.006), at lower loss than the initialization route, with a two-line implementation and no additional hyperparameter beyond c itself. The recommendation concerns control, not a default: our evidence does not support replacing the absmean convention with a fixed $c \neq 1$ unconditionally, since the best constant moves with scale and, on the second architecture family tested, the default constant was worse than the convention (Section 5.4). Tune c when a particular density is required; do not adopt one blindly.

Consider a brief threshold anneal. At a fixed budget, the γ-warmup arm is ahead of `bitnet158` by 0.060 to 0.201 nats at every one of the eleven scales in the main grid, where it runs the grid default $(s_0, \tau_w) = (0.4, 240)$. The 3×3 schedule ablation was executed at the primary scale only; there a shorter ramp, $(0.4, 120)$, is better still, and loss rises monotonically in τ_w at every start. The caveat of Section 5.3 applies: part of this may be an effective learning-rate effect, in which case the practical recommendation survives even though the mechanistic explanation changes.

Measure sparsity through the live forward path. The protocol evaluates sparsity under the threshold scale actually in force at the end of training (Section 3.12), because any other choice reports a quantity the model never computed. An arm whose mechanism is a threshold shift, read by an instrument hard-coded to $c = 1$, is mismeasured by the size of that shift, and the failure is silent: the reported numbers remain plausible.

For the research community, the transferable methodological point is the retention coefficient itself. Claims of the form "intervention X controls final sparsity" are routinely supported by a monotone trend, and a monotone trend is what we observed: a mean per-seed Spearman coefficient of -0.996, essentially perfect ordering, and a pooled coefficient of -0.983 at $p = 4.3 \times 10^{-66}$. Had we stopped there, we would have reported H3 as strongly confirmed. The retention regression converts the same data into the statement that 93–95% of the intervention is erased. We would encourage the reporting of a retention or durability statistic alongside any correlational sparsity-control claim.

5.14. Limitations

The following limitations are structural and bound every claim in this paper.

No ternary kernel was benchmarked: The compression figures are an arithmetic counting argument; the measured latency and throughput numbers are the cost of simulating ternary arithmetic in `bfloat16` under straight-through estimation and carry no deployment implication in either direction.

No run converged: All quality statements are token-equal at 20 M tokens. Convergence behaviour is unmeasured beyond a single doubling on three seeds at one scale.

Scale is bounded: The ladder stops at 352 M non-embedding parameters, and the larger ladder points are trained roughly two orders of magnitude below any compute-optimal token allocation, so they measure early training of large models rather than trained large models.

One corpus, one tokenizer, one language: FineWeb-EDU with the GPT-2 byte-pair encoder. Nothing here speaks to multilingual, code, or domain-specific pre-training.

Downstream evaluation does not discriminate: All arms sit at or near chance on both benchmarks; the only discriminating transfer measure is an internal next-token proxy, which is not independent of the training objective.

The full-precision control is not a clean quantization-cost baseline: The reasons are set out in Section 5.8.

The pre-registered primary baseline is algebraically equivalent to the standard arm: The design therefore contains one fewer independent ternary recipe than intended. Its retained value is procedural, freezing the primary contrast before execution, rather than evidential breadth. The empirical near-coincidence of the two arms at ten of eleven scales is consistent with this equivalence and was used above as a noise instrument.

Reproducibility is statistical, not bit-level: The grid ran in throughput mode; identical initialization and data order per seed are guaranteed, bit-identical reruns are not.

Table 23: Anticipated objections, the evidence bearing on each, and the residual risk after that evidence is taken into account. A residual risk of *low* means the design contains a measurement that constrains the objection; it does not mean the objection is eliminated.

Objection	Evidence in this study	Residual risk
The effect is an artefact of an unconverged budget.	2× budget check: advantage retains 77% of its magnitude after one doubling; erosion 22.7% per doubling.	**High.** One doubling, $n = 3$. No evidence bearing on a compute-optimal budget.
The effect is an implicit learning-rate schedule, not a sparsity schedule.	Advantage persists and grows over the upper four of five peak learning rates, a five-fold range, reversing only at the smallest; per-arm learning rates calibrated on disjoint seeds.	**Moderate.** The decisive control (gradient-scale schedule with fixed threshold) was not run.
The baseline is weak or badly tuned.	Primary baseline frozen before the grid; every arm receives its own $\eta^\star$; secondary contrast against the data-selected minimum baseline preserves the sign at ten of eleven scales.	**Moderate.** That contrast attenuates the advantage by 14 to 50% (e.g. +0.201 → +0.100 at `dim512`) and reverses it at `dim1024`.
The sparsity control is an analysis artefact.	Placebo arm flat to 1.3×10^{-4} against 2.65×10^{-2}; SNR 44.9×; sparsity measured under the live threshold.	**Low.**
The results are seed-lucky.	$n = 15$ at confirmatory scales; bootstrap CIs; Holm correction; pooled seed × fold variance; zero diverged runs.	**Low** at confirmatory scales, **moderate** at the $n = 5$ exploratory scales.
Effects are within implementation noise.	Two algebraically equivalent arms bound the floor at ≈ 0.01 nats typically, 0.05 worst case.	**Moderate** for the two smallest effects; **low** for the rest.
The threshold result does not transfer.	Retention, frontier and optimum measured on family A; family B tested at the default constant only, where the arm underperforms.	**Moderate.** A family-B sweep was not run; the negative result is reported rather than resolved.
Downstream ability is unmeasured.	ARC-Easy and HellaSwag scored under log-likelihood with both normalizations.	**High.** All arms are at or near chance; the benchmarks do not discriminate at this scale.
Efficiency claims are unsupported.	Deployment figures presented as an explicit counting argument; measured figures labelled as simulation overhead.	**Acknowledged, not mitigated.** No ternary kernel was benchmarked.
Does it hold at deployment scale?	11/11 scales to 352 M; depth-controlled replication; cross-architecture replication.	**High.** Slope is negative; extrapolation is labelled prediction.

The consequence is larger than a rerun tolerance: as Section 5.10 documents, kernel selection was not pinned per arm, and where two arms of a contrast fell on opposite sides of the compiled/eager split the reproducibility floor rises from about 0.004 to 0.03–0.05 nats.

The threshold-offset arm was swept on one architecture family only: Its retention property, its frontier and its optimal constant were all measured on family A. On family B only the default constant was tested, and it underperformed the baseline; whether a re-tuned constant recovers this is unknown.

Single hardware platform, single software stack: All 3,016 runs executed on one NVIDIA A100-SXM4-40GB under one Python, PyTorch and CUDA combination, with one random-number implementation. Nothing here separates an effect of the method from an interaction with that stack.

5.15. Future work

Seven experiments would materially strengthen or overturn the conclusions, and we list them in the order in which we would run them.

(i) Disentangle the schedule from the gradient scale. Compare γ-warmup against an arm that holds the threshold at $c = 1$ while scaling the BitLinear output by the same $g_{\text{warm}}(t)$. If the second arm reproduces the advantage, the effect is an effective

learning-rate schedule and the paper's mechanistic claim must change. It is also cheap—one additional arm at the primary scale—and it is the single most informative missing control.

(ii) Extend the budget until the advantage resolves. Run the primary contrast at 4× and 8× tokens on the full seed set, including the full-precision control. This simultaneously tests the erosion projection of Section 5.7 and the regularization account of the anomaly in Section 5.8.

(iii) Sweep the threshold constant directly, at scale. The threshold-offset arm, not the initializer, is the method that survived. It deserves the treatment the initializer received: a full sweep of c at the confirmatory scales, jointly with the anneal schedule, since the two interact by construction.

(iv) Sweep the threshold constant on the second architecture family. The one arm that fails to transfer across architectures is `gamma_offset` at its default constant, which is significantly worse than the baseline on family B at both scales tested. Since the optimal constant is already known to move with scale, the minimal experiment is a ρ_0 sweep on family B at `dim384`, the scale carrying the larger deficit ($\Delta\mathcal{L} = -0.0873$). It would distinguish an architecture-dependent optimum from an interaction between the threshold offset and gated feed-forward blocks, and the two carry different implications for the recommendation above.

(v) Pin the execution path per arm. The wall-clock evidence in Section 5.10 shows that at four of eleven scales the compiled path was available to some arms and not others by a factor of 1.9 or more in run time, with two narrower instances besides. A corrected grid should either compile every arm or none, and should assert the chosen state at run entry rather than inferring it afterwards from timings. The cost is negligible and it removes the only confound in this study that is a property of the harness rather than of the method.

(vi) Test the retention prediction across quantizer families. The mechanism proposed in Section 5.2 is a claim about a class rather than about one quantizer: any threshold defined as a statistic of the current weights should retain an initialization intervention poorly, while a threshold constant in the weights, or learned independently of them, should retain it. Trained ternary quantization [6] is the sharpest test of the first half, since it learns its per-layer scaling factors and yet keeps a threshold proportional to $\max|w|$; soft-threshold reparameterization [7] supplies the contrasting case.

(vii) Benchmark a real ternary kernel. Until this is done, every efficiency statement in this literature, ours included, remains an arithmetic model rather than a measurement.

5.16. Concluding remarks

The study set out to show that where ternary training begins determines where it ends, and found approximately the opposite: under an absmean threshold, training returns to a final sparsity near 0.30 from wherever it is placed, and the initialization that placed it there is a mildly worse initializer in full precision as well. What survived is a different result. The threshold constant, not the weight distribution, is the durable control surface, and the conventional absmean value is not self-evidently the right setting of it: at the larger of the two scales swept, the optimum lies a full grid step away from the convention and within two percentage points of the operating point that a decade-old heuristic identifies on entirely different grounds, while at the smaller it is not distinguishable from the convention. Whether a different constant should become the new default is not settled by these data: its advantage over the convention is scale-dependent on one architecture family and reversed on the other. What is settled, and what holds across eleven scales, two architecture families, a depth-controlled ladder and the upper four of five learning rates, is that briefly relaxing the threshold early in training improves budget-matched quality and convergence speed.

We are able to say which of these claims is which only because the hypothesis family, the primary baseline, the equivalence margin and the falsification instrument were frozen before the grid was executed, and because the design included two arms built for the proposed method to lose. Both did their work: the mechanism-matched control defeated it outright, and the placebo established that the operating-point effect is not an artefact of the analysis. A design in which the central hypothesis cannot fail produces no information about the world; the value of this experiment is largely the information contained in the ways it failed.

6. Conclusion and Future Work

6.1. Core takeaways

Three findings survive the controls built to defeat them.

(i) The ternary operating point is analytically locked, and the threshold constant is its coordinate. Under the absmean rule the initial active fraction is $\rho_0 = 0.6899$ for every zero-mean initializer, independently of scale, confirmed across two decades in σ, over which the measured active fraction varies by 6×10^{-4}, and across four initializer families, each landing within 10^{-3} of the prediction. Standard initialization theory, being a theory of second moments, is therefore silent about the variable that determines where ternary training begins. The closed-form bijection between threshold constant and operating point (Eqs. (9) and (19)) supplies the missing coordinate, and places the classical threshold of Li et al. [5] at $\rho_0 = 0.780$ and the absmean convention of Ma et al. [4] at 0.690.

(ii) Only a threshold intervention is durable. A fifty-percentage-point manipulation of initial sparsity leaves fifty points of final sparsity when imposed at the threshold (retention 1.003–1.006, $R^2 = 1.000$) and two to four points when imposed at the initializer (0.052–0.069). Both mechanisms produce a near-perfect rank association, so the ordinal evidence this literature customarily reports would have supported a strong causal claim that the retention coefficient refutes. The cause is structural: the absmean threshold is a functional of the weights it judges, so gradient descent carries it along with them, and training returns the network to a final sparsity near 0.30 regardless of where it starts.

(iii) The effect lies in the timing of the threshold, not the weight distribution. Annealing the threshold briefly at the start of training improves held-out loss by 0.060 to 0.201 nats and

reduces steps-to-target at all eleven scales, at all thirteen architecture × scale cells against the pre-registered baseline and twelve of thirteen against the data-selected minimum, on a depth-controlled ladder, and at the upper four of five learning rates. The proposed initializer, by contrast, is outperformed by its own mechanism-matched control by 0.0610 nats (95% CI [+0.0474, +0.0742], 90 paired cells) and carries an almost identical penalty in full precision (−0.086 nats) and under quantization (−0.090), the signature of a generically weaker initializer rather than of failed operating-point control.

6.2. Scope and limitations

The conclusions are bounded by the protocol that produced them, and four limits are structural rather than incidental.

No run converged. The plateau gate certified 0 of 760 runs and the 2× budget check found 0.29–0.32 nats of remaining improvement. All quality claims are token-equal at 20 M tokens; the advantage erodes by 22.7% per token doubling, so it may not persist to a compute-optimal budget.

Scale is bounded at 352 M non-embedding parameters, which are trained some two orders of magnitude below a Chinchilla-style token allocation at that size. The larger ladder points measure early training of large models, not trained large models, which is why the scale readout is sign consistency rather than magnitude.

No ternary kernel was benchmarked. The compression figures are a counting argument; the measured latencies are the cost of simulating ternary arithmetic under straight-through estimation and carry no deployment implication.

Two further caveats attach to the arms themselves. The threshold-offset arm was swept on one architecture family only, and at its default constant it underperforms the baseline on the second family, so its frontier and its optimal constant are family A results. Separately, and as a defect of the harness rather than of the method, the execution path was not uniform across arms, and at four of eleven scales the primary contrast was measured across that boundary (Section 5.10); the effect survives excluding all four, but a corrected grid should pin the path per arm.

Two further boundaries are worth stating plainly. The downstream benchmarks sit at chance at these scales and discriminate between no pair of arms, so this work demonstrates an effect on language-modelling loss and convergence speed and supplies no evidence of an effect on downstream capability. And the γ-warmup result excludes the global effective-learning-rate account (Section 5.3) but not a time-varying or layer-dependent one.

6.3. Future directions

Three experiments would most directly strengthen or overturn these results.

The first would settle the mechanism. Lowering the threshold also scales the quantized weight magnitudes, and with them the effective gradient scale reaching each layer. The envelope argument of Section 5.3 excludes a global learning-rate equivalence, but not a time-varying or layer-dependent one. One additional arm settles it: hold $c = 1$ and rescale the BitLinear output by the same $g_{\text{warm}}(t)$, reproducing the gradient-scale trajectory with no change to sparsity.

The second would settle whether the advantage is a short-budget phenomenon. Running the primary contrast at four and eight times the token budget, with the full-precision reference included, would test simultaneously whether the annealing advantage survives toward convergence, whether the measured 22.7%-per-doubling erosion is linear in log tokens, and whether the anomaly by which ternary training undercuts full precision at ten of eleven scales is the short-budget regularization effect we identify as most likely.

The third would establish how far the retention result reaches. The mechanism we propose is a claim about a class, not about one quantizer: any threshold defined as a statistic of the current weights should exhibit low retention, while thresholds constant in the weights or learned independently of them, as in soft-threshold reparameterization [7], should retain it; trained ternary quantization [6] falls on the first side despite learning its scaling factors, and is therefore the sharpest test. Measuring the retention coefficient across these families would convert a mechanism inferred from one quantizer into a general property of quantization-aware training, and identify which existing methods already possess the durable control ours does not.

6.4. Closing remark

The hypothesis this study was built to confirm did not survive it. What replaced it is narrower and better supported: the threshold, not the initial weight distribution, is where ternary training can be steered, and the gain lies in relaxing that threshold briefly rather than in placing the network at any particular operating point to begin with. Drawing that distinction at all was a property of the design rather than of the data. The commitments were frozen before execution and two arms were included so that the proposed method could lose, and one of them duly defeated it. The complete run store, provenance manifest and analysis pipeline are released, so that every table and figure regenerates without a GPU and the reader can verify not only what we found but what we had committed to finding.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Haydar Kilic: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Project administration.

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Data availability

The complete research artefact is released with this paper: the per-run checkpoint store (3,016 JSON records), the provenance manifest, the tidy per-run and per-cell result tables, the analysis and figure-generation pipeline, and the mirrored execution logs. Every table and figure in this manuscript can be regenerated from the run store without a GPU. The corpus is the publicly available FineWeb-EDU sample tokenized with the public GPT-2 byte-pair encoder; no proprietary or personal data were used.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Anthropic's Claude in order to edit prose for clarity and consistency, to format tables, figures and algorithm environments in LaTeX, and to check reported values and cross-references against the pipeline's stored artefacts. All experiments were designed and executed by the author. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Author biography

Haydar Kilic received the B.Sc. degree in mathematics from Kırıkkale University, Türkiye, in 2006, the M.Sc. degree in computer engineering from Case Western Reserve University, Cleveland, OH, USA, in 2013, and the Ph.D. degree in electronics and computer engineering from Bilecik Şeyh Edebali University, Türkiye, in 2022. From 2014 to 2024 he was a Lecturer with Bilecik Şeyh Edebali University. He is currently an Assistant Professor with the Department of Artificial Intelligence Engineering, Ostim Technical University, Ankara, Türkiye. His research interests include image processing, geometric methods in computer vision, optimization methods in artificial intelligence, and efficient deep learning.

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